

Projection Craft

S Essays

Books 0–1

Ambiguity-Resolved Source Edition

Reconciled against the canonical definitions, errata, and scope rules established in the project thread.

Editorial Resolution Notice

This edition preserves the numbering, theorem order, and canonical operator identities of the stable S0–S1 corpus. It resolves only ambiguities already identified in the project thread. No physical interpretation is promoted to theorem status.

1. **Carrier convention.** Analytic calculations use the lifted coordinate $x \in \mathbb{R}$. Periodic phase statements use the quotient circle $\mathbb{T} = \mathbb{R}/(2\pi\mathbb{Z})$. The notation $x \equiv x + 2\pi$ means equality of quotient classes, not literal equality in $\mathbb{R}$.
2. **Dihedral convention.** D_4 denotes the order-eight symmetry group of the four-quadrant phase circle. Some references denote the same group by D_8 .
3. **Boundary terminology.** Closing a single open quadrant to $[b_k, b_{k+1}]$ is a legitimate componentwise compactification. The countable disjoint union $\widehat{\mathcal{D}} = \bigsqcup_k \overline{Q}_k$ is instead called the *split-boundary extension domain*; it is not asserted to be compact as a whole.
4. **Validity-domain logic.** In S1.II, the declared domain $\Omega_{\mathcal{P}}$ is separated from the maximal operational admissibility envelope. Lawfulness requires inclusion in that envelope; equality means the declared domain is maximal. This removes the former circularity caused by using the instance-domain guard to define the domain against which that same guard was tested.
5. **Naming.** *Projection Craft* is the project title, *S Essays* is the corpus title, and *R-operator* remains the established name of the mathematical operator family. These names do not designate competing document series.

A complete change ledger is included in `AMBIGUITY_RESOLUTION.md` in the source package.

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Book 0: Formal Core

S0.I

The Pythagorean Foundation of the Bounded Phase Operators

Book 0 - Formal Core

IDENTIFICATION DIVISION

PROGRAM-ID

R-THEORY-PYTHAGOREAN-CORE

PURPOSE

To derive the exact algebraic properties of the R-operators from the Pythagorean identity

$$\sin^2 x + \cos^2 x = 1,$$

together with standard definitions of the reciprocal and quotient trigonometric functions.

NON-PURPOSE

This division does not establish:

- physical conservation of energy, mass, or information;
- uniqueness of the R-operator family;
- empirical applicability;
- replacement of exponential or logarithmic laws;
- a law of temporal evolution.

Those require additional arguments beyond the Pythagorean foundation.

ENVIRONMENT DIVISION

Definition E.1 — Phase variable

Analytic calculations use a lifted phase coordinate

$$x \in \mathbb{R}$$

measured in radians. The operator family is 2π -periodic. When one revolution is identified, the phase circle is

$$\mathbb{T} = \mathbb{R}/(2\pi\mathbb{Z}),$$

and $[x] \in \mathbb{T}$ denotes the quotient class of x . Thus

$$x \equiv x + 2\pi$$

means equality in $\mathbb{T}$, not literal equality of real numbers. All analytic domains, derivatives, and one-sided limits in Book 0 are taken on the lifted real line unless the quotient circle is named explicitly.

Definition E.2 — Admissible domain

Define

$$\mathcal{D} = \{x \in \mathbb{R} : \sin x \neq 0 \text{ and } \cos x \neq 0\}.$$

Equivalently,

$$\mathcal{D} = \mathbb{R} \setminus \left\{ \frac{k\pi}{2} : k \in \mathbb{Z} \right\}.$$

All expressions below are evaluated only for $x \in \mathcal{D}$.

Definition E.3 — Unit-circle coordinates

Let

$$s = \sin x, \quad c = \cos x.$$

The point

$$P(x) = (c, s)$$

lies on the unit circle.

AXIOM DIVISION

Inherited Theorem A.1 — Pythagorean theorem

For a right triangle with legs a, b and hypotenuse h ,

$$a^2 + b^2 = h^2.$$

Corollary A.2 — Unit-circle Pythagorean identity

Since the radius of the unit circle is 1,

$$\boxed{s^2 + c^2 = 1}.$$

Equivalently,

$$\boxed{\sin^2 x + \cos^2 x = 1}.$$

This identity is the foundational algebraic relation used throughout the remainder of the proof.

DATA DIVISION

Definition D.1 — Absolute reciprocal functions

Define

$$M = |\sec x| = \frac{1}{|c|},$$

and

$$N = |\csc x| = \frac{1}{|s|}.$$

Both satisfy

$$M > 0, \quad N > 0.$$

Definition D.2 — Primitive displayed operators

Define Sininverex:

$$\text{srx}(x) = N + \cot x.$$

Define Sinexpon:

$$\text{sxp}(x) = \frac{1}{\text{srx}(x)}.$$

Define Cosinexpon:

$$\text{cxp}(x) = M + \tan x.$$

Define Cosininverex:

$$\text{crx}(x) = \frac{1}{\text{cxp}(x)}.$$

Definition D.3 — UNA operators

Define Unainverex:

$$\text{urx}(x) = \text{srx}(x) - \text{crx}(x).$$

Define Unaexpon:

$$\text{uxp}(x) = \text{cxp}(x) - \text{sxp}(x).$$

This sign convention is fixed throughout the calculus.

Definition D.4 — Saw operators

Define

$$\text{saw}_r(x) = \frac{1}{\text{urx}(x)},$$

and

$$\text{saw}_x(x) = \frac{1}{\text{uxp}(x)}.$$

Definition D.5 — Flatwave

Define

$$\text{Flatwave}(x) = \text{saw}_r(x) + \text{saw}_x(x).$$

Thus,

$$\text{Flatwave}(x) = \frac{1}{\text{urx}(x)} + \frac{1}{\text{uxp}(x)}.$$

PROCEDURE DIVISION

SECTION 1 — Reciprocal Trigonometric Identities

Lemma 1.1 — Secant–tangent Pythagorean identity

For every $x \in \mathcal{D}$,

$$\boxed{M^2 - \tan^2 x = 1}.$$

Proof. Since

$$M = |\sec x|,$$

we have

$$M^2 = \sec^2 x.$$

Using the definitions of secant and tangent,

$$\sec^2 x - \tan^2 x = \frac{1}{c^2} - \frac{s^2}{c^2}.$$

Therefore,

$$\sec^2 x - \tan^2 x = \frac{1 - s^2}{c^2}.$$

By the Pythagorean identity,

$$1 - s^2 = c^2.$$

Hence,

$$\sec^2 x - \tan^2 x = \frac{c^2}{c^2} = 1.$$

Therefore,

$$\boxed{M^2 - \tan^2 x = 1}.$$

□

Lemma 1.2 — Cosecant–cotangent Pythagorean identity

For every $x \in \mathcal{D}$,

$$\boxed{N^2 - \cot^2 x = 1}.$$

Proof. Since

$$N = |\csc x|,$$

we have

$$N^2 = \csc^2 x.$$

Thus,

$$\csc^2 x - \cot^2 x = \frac{1}{s^2} - \frac{c^2}{s^2} = \frac{1 - c^2}{s^2}.$$

By the Pythagorean identity,

$$1 - c^2 = s^2.$$

Therefore,

$$\csc^2 x - \cot^2 x = \frac{s^2}{s^2} = 1.$$

Hence,

$$\boxed{N^2 - \cot^2 x = 1}.$$

□

Corollary 1.3 — Strict positivity of the conjugate factors

Because

$$M^2 - \tan^2 x = 1 > 0$$

and $M > 0$,

$$M > |\tan x|.$$

Therefore,

$$M + \tan x > 0, \quad M - \tan x > 0.$$

Likewise,

$$N > |\cot x|,$$

so

$$N + \cot x > 0, \quad N - \cot x > 0.$$

No denominator introduced below vanishes on $\mathcal{D}$.

SECTION 2 — Exact Reciprocal Conjugates

Theorem 2.1 — Cosine-family reciprocal conjugate

For every $x \in \mathcal{D}$,

$$\boxed{\frac{1}{M + \tan x} = M - \tan x}.$$

Proof. From Lemma 1.1,

$$M^2 - \tan^2 x = 1.$$

Factoring the difference of squares gives

$$(M + \tan x)(M - \tan x) = 1.$$

Since $M + \tan x \neq 0$, division gives

$$\frac{1}{M + \tan x} = M - \tan x.$$

□

Theorem 2.2 — Sine-family reciprocal conjugate

For every $x \in \mathcal{D}$,

$$\boxed{\frac{1}{N + \cot x} = N - \cot x}.$$

Proof. From Lemma 1.2,

$$N^2 - \cot^2 x = 1.$$

Therefore,

$$(N + \cot x)(N - \cot x) = 1.$$

Since $N + \cot x \neq 0$,

$$\frac{1}{N + \cot x} = N - \cot x.$$

□

Corollary 2.3 — Explicit forms of the reciprocal operators

The reciprocal R-operators may be written as

$$\boxed{\text{crx}(x) = M - \tan x},$$

and

$$\boxed{\text{sxp}(x) = N - \cot x}.$$

Corollary 2.4 — Exact reciprocal pairing

$$\boxed{\text{exp}(x) \text{crx}(x) = 1},$$

and

$$\boxed{\text{srx}(x) \text{sxp}(x) = 1}.$$

Corollary 2.5 — Positivity of all four displayed operators

For every $x \in \mathcal{D}$,

$$\boxed{\text{srx}, \text{sxp}, \text{exp}, \text{crx} > 0}.$$

The primitive operators carry magnitude but no negative sign. Signed direction first appears in the UNA differences.

SECTION 3 — Sign Variables and Branch Forms

Definition 3.1 — Quadrant signs

Define

$$\alpha = \operatorname{sgn}(s), \quad \beta = \operatorname{sgn}(c).$$

Since $x \in \mathcal{D}$,

$$\alpha, \beta \in \{-1, +1\}.$$

Because

$$|s| = \alpha s, \quad |c| = \beta c,$$

we obtain

$$N = \frac{\alpha}{s}, \quad M = \frac{\beta}{c}.$$

Lemma 3.2 — Signed rational forms

The primitive operators satisfy

$$\boxed{\operatorname{srx}(x) = \frac{\alpha + c}{s}},$$

$$\boxed{\operatorname{sxp}(x) = \frac{\alpha - c}{s}},$$

$$\boxed{\operatorname{cxp}(x) = \frac{\beta + s}{c}},$$

and

$$\boxed{\operatorname{ctx}(x) = \frac{\beta - s}{c}}.$$

Proof. For example,

$$\operatorname{srx} = N + \cot x = \frac{\alpha}{s} + \frac{c}{s} = \frac{\alpha + c}{s}.$$

The other formulas follow identically.

□

SECTION 4 — UNA Structure

Definition 4.1 — Auxiliary branch quantity

Define

$$p = \alpha c - \beta s.$$

Lemma 4.2 — Exact UNA branch forms

For every $x \in \mathcal{D}$,

$$\boxed{\text{urx}(x) = \frac{1+p}{sc}},$$

and

$$\boxed{\text{uxp}(x) = \frac{1-p}{sc}}.$$

Proof. Using the signed rational forms,

$$\text{urx} = \text{srx} - \text{crx}.$$

Therefore,

$$\text{urx} = \frac{\alpha + c}{s} - \frac{\beta - s}{c}.$$

Using the common denominator sc ,

$$\text{urx} = \frac{c(\alpha + c) - s(\beta - s)}{sc}.$$

Expanding,

$$\text{urx} = \frac{\alpha c + c^2 - \beta s + s^2}{sc}.$$

Since

$$s^2 + c^2 = 1,$$

we obtain

$$\text{urx} = \frac{1 + \alpha c - \beta s}{sc}.$$

Thus,

$$\text{urx} = \frac{1 + p}{sc}.$$

Similarly,

$$\text{uxp} = \text{cxp} - \text{sxp},$$

so

$$\text{uxp} = \frac{\beta + s}{c} - \frac{\alpha - c}{s}.$$

Combining denominators gives

$$\text{uxp} = \frac{\beta s + s^2 - \alpha c + c^2}{sc}.$$

Hence,

$$\text{uxp} = \frac{1 + \beta s - \alpha c}{sc} = \frac{1 - p}{sc}.$$

□

Lemma 4.3 — Bound on the auxiliary quantity

For every $x \in \mathcal{D}$,

$$\boxed{|p| < 1}.$$

Proof. Squaring p ,

$$p^2 = (\alpha c - \beta s)^2.$$

Expanding,

$$p^2 = \alpha^2 c^2 + \beta^2 s^2 - 2\alpha\beta sc.$$

Since

$$\alpha^2 = \beta^2 = 1,$$

and

$$s^2 + c^2 = 1,$$

we have

$$p^2 = 1 - 2\alpha\beta sc.$$

But

$$\alpha\beta sc = |sc|.$$

Therefore,

$$p^2 = 1 - 2|sc|.$$

Since $x \in \mathcal{D}$,

$$|sc| > 0.$$

Thus,

$$p^2 < 1,$$

and therefore

$$|p| < 1.$$

□

Corollary 4.4 — Positivity of the UNA numerators

Because $|p| < 1$,

$$1 + p > 0, \quad 1 - p > 0.$$

Theorem 4.5 — Common UNA sign

For every $x \in \mathcal{D}$,

$$\boxed{\operatorname{sgn}(\operatorname{ur}x) = \operatorname{sgn}(\operatorname{ux}p) = \operatorname{sgn}(sc)}.$$

Since

$$\sin 2x = 2sc,$$

we also have

$$\boxed{\operatorname{sgn}(\operatorname{urx}) = \operatorname{sgn}(\operatorname{uxp}) = \operatorname{sgn}(\sin 2x)}.$$

Proof. Both UNA numerators $1 + p$ and $1 - p$ are positive. Their common denominator is sc . Therefore, both UNA operators have the sign of sc .

□

SECTION 5 — Double-Angle Closure

Theorem 5.1 — UNA sum identity

For every $x \in \mathcal{D}$,

$$\boxed{\operatorname{urx}(x) + \operatorname{uxp}(x) = \frac{4}{\sin 2x}}.$$

Proof. From Lemma 4.2,

$$\operatorname{urx} + \operatorname{uxp} = \frac{1+p}{sc} + \frac{1-p}{sc}.$$

Therefore,

$$\operatorname{urx} + \operatorname{uxp} = \frac{2}{sc}.$$

Since

$$\sin 2x = 2sc,$$

we obtain

$$\frac{2}{sc} = \frac{4}{\sin 2x}.$$

Thus,

$$\boxed{\operatorname{urx} + \operatorname{uxp} = \frac{4}{\sin 2x}}.$$

□

Corollary 5.2 — Inverse UNA sum identity

Because the UNA sum is nonzero on $\mathcal{D}$,

$$\boxed{\frac{1}{\text{urx}(x) + \text{uxp}(x)} = \frac{\sin 2x}{4}}.$$

This is the exact double-angle closure identity.

SECTION 6 — UNA Product

Theorem 6.1 — UNA product identity

For every $x \in \mathcal{D}$,

$$\boxed{\text{urx}(x) \text{uxp}(x) = \frac{2}{|sc|}}.$$

Equivalently,

$$\boxed{\text{urx}(x) \text{uxp}(x) = \frac{4}{|\sin 2x|}}.$$

Proof. Using Lemma 4.2,

$$\text{urx} \text{uxp} = \frac{(1+p)(1-p)}{s^2 c^2}.$$

Thus,

$$\text{urx} \text{uxp} = \frac{1-p^2}{s^2 c^2}.$$

From Lemma 4.3,

$$p^2 = 1 - 2|sc|.$$

Therefore,

$$1 - p^2 = 2|sc|.$$

Hence,

$$\text{urx} \text{uxp} = \frac{2|sc|}{s^2 c^2}.$$

Since

$$s^2 c^2 = |sc|^2,$$

we obtain

$$\text{urx} \text{ uxp} = \frac{2}{|sc|}.$$

Finally,

$$|\sin 2x| = 2|sc|,$$

so

$$\frac{2}{|sc|} = \frac{4}{|\sin 2x|}.$$

□

SECTION 7 — Flatwave

Theorem 7.1 — Exact Flatwave theorem

For every $x \in \mathcal{D}$,

$$\boxed{\text{Flatwave}(x) = \text{sgn}(\sin 2x)}.$$

Proof. By definition,

$$\text{Flatwave} = \frac{1}{\text{urx}} + \frac{1}{\text{uxp}}.$$

Combining fractions,

$$\text{Flatwave} = \frac{\text{urx} + \text{uxp}}{\text{urx} \text{ uxp}}.$$

By Theorems 5.1 and 6.1,

$$\text{Flatwave} = \frac{2/(sc)}{2/|sc|}.$$

Therefore,

$$\text{Flatwave} = \frac{|sc|}{sc}.$$

But

$$\frac{|sc|}{sc} = \text{sgn}(sc).$$

Since

$$\text{sgn}(sc) = \text{sgn}(\sin 2x),$$

we obtain

$$\boxed{\text{Flatwave}(x) = \text{sgn}(\sin 2x)}.$$

□

Corollary 7.2 — Exact Flatwave magnitude

$$\boxed{|\text{Flatwave}(x)| = 1}.$$

Corollary 7.3 — Quadrant values

$$\text{Flatwave}(x) = \begin{cases} +1, & x \text{ lies in Quadrant I or III,} \\ -1, & x \text{ lies in Quadrant II or IV.} \end{cases}$$

Corollary 7.4 — Flatwave derivative

On every open quadrant,

$$\boxed{\frac{d}{dx} \text{Flatwave}(x) = 0}.$$

Flatwave is undefined at the excluded quadrant boundaries.

SECTION 8 — Saw Partition

Theorem 8.1 — Signed saw partition

For every $x \in \mathcal{D}$,

$$\boxed{\text{saw}_r(x) + \text{saw}_x(x) = \text{sgn}(\sin 2x)}.$$

This follows directly from the definition of Flatwave and Theorem 7.1.

Corollary 8.2 — Unsigned saw partition

For every $x \in \mathcal{D}$,

$$\boxed{|\text{saw}_r(x)| + |\text{saw}_x(x)| = 1}.$$

Proof. By Theorem 4.5, urx and uxp have the same sign. Their reciprocals therefore also have the same sign.

Consequently,

$$|\text{saw}_r + \text{saw}_x| = |\text{saw}_r| + |\text{saw}_x|.$$

But by Theorem 8.1,

$$|\text{saw}_r + \text{saw}_x| = |\text{sgn}(\sin 2x)| = 1.$$

Therefore,

$$\boxed{|\text{saw}_r| + |\text{saw}_x| = 1}.$$

This establishes an exact signed and unsigned partition of unity within the operator system.

It does not yet establish a physical conservation law.

□

SECTION 9 — Half-Angle Representation

Theorem 9.1 — Quadrant-I half-angle forms

For

$$0 < x < \frac{\pi}{2},$$

we have

$$\boxed{\text{srx}(x) = \cot \frac{x}{2}},$$

$$\boxed{\text{sxp}(x) = \tan \frac{x}{2}},$$

$$\boxed{\text{cxp}(x) = \tan \left(\frac{\pi}{4} + \frac{x}{2} \right)},$$

and

$$\boxed{\text{crx}(x) = \tan \left(\frac{\pi}{4} - \frac{x}{2} \right)}.$$

Proof. In Quadrant I,

$$|\csc x| = \csc x, \quad |\sec x| = \sec x.$$

The standard half-angle identities give

$$\csc x + \cot x = \frac{1 + \cos x}{\sin x} = \cot \frac{x}{2},$$

and

$$\csc x - \cot x = \frac{1 - \cos x}{\sin x} = \tan \frac{x}{2}.$$

Similarly,

$$\sec x + \tan x = \frac{1 + \sin x}{\cos x} = \tan \left(\frac{\pi}{4} + \frac{x}{2} \right),$$

and

$$\sec x - \tan x = \frac{1 - \sin x}{\cos x} = \tan \left(\frac{\pi}{4} - \frac{x}{2} \right).$$

□

SECTION 10 — Octant Dominance

Theorem 10.1 — Dominance in the first octant

For

$$0 < x < \frac{\pi}{4},$$

$$\boxed{\text{srx}(x) > \text{cxp}(x), \text{sxp}(x), \text{crx}(x)}.$$

Proof. In this interval,

$$c > s > 0.$$

Compare srx and exp :

$$\text{srx} = \frac{1+c}{s}, \quad \text{exp} = \frac{1+s}{c}.$$

Because $sc > 0$,

$$\text{srx} > \text{exp}$$

if and only if

$$c(1+c) > s(1+s).$$

Subtracting the right side from the left gives

$$c + c^2 - s - s^2 = (c - s) + (c^2 - s^2).$$

Factoring,

$$(c - s) + (c - s)(c + s) = (c - s)(1 + c + s).$$

Both factors are positive. Therefore,

$$\text{srx} > \text{exp}.$$

Also,

$$\text{srx} > 1,$$

while

$$\text{sxp} < 1$$

because they are reciprocal.

Likewise,

$$\text{crx} < 1$$

in Quadrant I.

Therefore, srx is the unique dominant operator in the first octant.

□

Theorem 10.2 — Dominance in the second octant

For

$$\frac{\pi}{4} < x < \frac{\pi}{2},$$

$$\boxed{\text{exp}(x) > \text{srx}(x), \text{sxp}(x), \text{crx}(x)}.$$

The proof is identical, with $s > c > 0$.

Lemma 10.3 — Quarter-turn operator permutation

For every admissible x ,

$$\text{srx}\left(x + \frac{\pi}{2}\right) = \text{crx}(x),$$

$$\text{exp}\left(x + \frac{\pi}{2}\right) = \text{sxp}(x),$$

$$\text{crx}\left(x + \frac{\pi}{2}\right) = \text{srx}(x),$$

and

$$\text{sxp}\left(x + \frac{\pi}{2}\right) = \text{exp}(x).$$

Theorem 10.4 — Complete octant dominance cycle

Using Theorems 10.1–10.2 and the quarter-turn permutation,

$$\boxed{\text{srx} \rightarrow \text{exp} \rightarrow \text{crx} \rightarrow \text{sxp} \rightarrow \text{srx}}$$

is the unique magnitude-dominance cycle over successive open octants.

Explicitly:

Octant	Unique dominant operator
$(0, \pi/4)$	srx
$(\pi/4, \pi/2)$	exp
$(\pi/2, 3\pi/4)$	crx
$(3\pi/4, \pi)$	sxp
$(\pi, 5\pi/4)$	srx
$(5\pi/4, 3\pi/2)$	exp
$(3\pi/2, 7\pi/4)$	crx
$(7\pi/4, 2\pi)$	sxp

At odd multiples of $\pi/4$, adjacent dominant operators tie. Those points are admissible but are not unique-winner points.

CLOSURE DIVISION

Starting from the Pythagorean identity alone, together with ordinary trigonometric definitions, we have rigorously established:

1. Exact reciprocal conjugates:

$$\text{crx} = M - \tan x, \quad \text{sxp} = N - \cot x.$$

1. Positivity of all four primitive displayed operators.
2. Exact reciprocal pairing.
3. Exact signed forms of the UNA operators.
4. Common UNA sign:

$$\text{sgn}(\text{urx}) = \text{sgn}(\text{uxp}) = \text{sgn}(\sin 2x).$$

1. The double-angle closure identity:

$$\text{urx} + \text{uxp} = \frac{4}{\sin 2x}.$$

1. The UNA product identity:

$$\text{urx} \text{uxp} = \frac{4}{|\sin 2x|}.$$

1. The exact Flatwave theorem:

$$\text{Flatwave} = \text{sgn}(\sin 2x).$$

1. The signed and unsigned saw partitions:

$$\text{saw}_r + \text{saw}_x = \text{sgn}(\sin 2x),$$

$$|\text{saw}_r| + |\text{saw}_x| = 1.$$

1. The complete octant dominance cycle:

$$\text{srx} \rightarrow \text{cxp} \rightarrow \text{crx} \rightarrow \text{sxp}.$$

These results constitute a rigorous Pythagorean core for the bounded phase operator calculus.

They establish the operator geometry.

They do not yet establish its physical interpretation.

S0.II

The Canonical Generator and Quadrant-Local Closure

Book 0 - Formal Core

Abstract

This essay determines the algebraic size of the bounded phase operator calculus established in S0.I. On each open quadrant, let

$$z(x) = \text{sr}\mathbf{x}(x).$$

Using the exact Flatwave theorem, the second primary envelope is shown to be a Möbius transformation of z :

$$\text{exp}(x) = \frac{z(x) + \varepsilon}{\varepsilon z(x) - 1},$$

where

$$\varepsilon = \text{sgn}(\sin 2x)$$

is constant on the quadrant. The reciprocal operators, UNA operators, saw responses, and Flatwave then reduce to rational functions of the single generator z .

The phase itself also admits a rational readout:

$$\sin 2x = \frac{4z(z^2 - 1)}{(z^2 + 1)^2}.$$

The generator obeys the autonomous differential equation

$$\frac{dz}{dx} = -\frac{1 + z^2}{2}.$$

Consequently, every rational expression in z remains rational in z after quadrant-local differentiation. This establishes a precise closure result: the calculus is not a finite algebra of named functions, but a one-generator differential field on each admissible quadrant.

No claim of global analytic continuation, physical interpretation, or uniqueness of the chosen generator is made.

0. Scope and Proof Standard

0.1 Scope

This essay proves:

1. the rational dependence of all named operators on one generator;
2. the exact Möbius relation between the two primary envelopes;
3. canonical rational forms for the UNA and saw operators;
4. a rational representation of $\sin 2x$;
5. the differential equation satisfied by each primitive operator;
6. quadrant-local closure under differentiation.

0.2 Non-scope

This essay does not prove:

- that srx is the only possible generator;
- that the named operator list is closed under arbitrary operations;
- that every rational function of the generator is physically meaningful;
- that the calculus supplies a law of temporal evolution;
- that differentiation with respect to phase is differentiation with respect to time.

0.3 Inheritance

All definitions and results of S0.I are inherited.

In particular, for every admissible x ,

$$\text{srx} > 0, \quad \text{cxp} > 0,$$

$$\text{sxp} = \frac{1}{\text{srx}}, \quad \text{crx} = \frac{1}{\text{cxp}},$$

and

$$\text{Flatwave}(x) = \text{sgn}(\sin 2x).$$

The admissible domain remains

$$\mathcal{D} = \mathbb{R} \setminus \left\{ \frac{k\pi}{2} : k \in \mathbb{Z} \right\}.$$

1. Quadrant-Local Setting

Definition 1.1 — Open quadrants

For each integer k , define

$$Q_k = \left(\frac{k\pi}{2}, \frac{(k+1)\pi}{2} \right).$$

Every Q_k is a connected component of $\mathcal{D}$.

Definition 1.2 — Quadrant sign

On a fixed quadrant Q_k , define

$$\varepsilon_k = \operatorname{sgn}(\sin 2x).$$

Because $\sin 2x$ has no zero inside Q_k , ε_k is constant throughout the quadrant.

For readability, write simply

$$\varepsilon = \varepsilon_k$$

when a quadrant has been fixed.

Thus,

$$\varepsilon \in \{-1, +1\}.$$

Definition 1.3 — Canonical generator

Define

$$\boxed{z(x) = \operatorname{srx}(x)}.$$

Since $\operatorname{srx} > 0$,

$$z(x) > 0$$

throughout every admissible quadrant.

Definition 1.4 — Second primary envelope

For temporary convenience, write

$$w(x) = \text{csp}(x).$$

The objective of the following sections is to eliminate w as an independent algebraic quantity.

2. Rational Form of Flatwave**Lemma 2.1 — Rational saw forms**

For every $x \in \mathcal{D}$,

$$\text{saw}_r = \frac{w}{zw - 1},$$

and

$$\text{saw}_x = \frac{z}{zw - 1}.$$

Proof. By definition,

$$\text{urx} = \text{srx} - \text{crx}.$$

Using

$$\text{srx} = z \quad \text{and} \quad \text{crx} = \frac{1}{w},$$

we have

$$\text{urx} = z - \frac{1}{w}.$$

Combining the terms,

$$\text{urx} = \frac{zw - 1}{w}.$$

Therefore,

$$\text{saw}_r = \frac{1}{\text{urx}} = \frac{w}{zw - 1}.$$

Similarly,

$$\text{uxp} = w - \frac{1}{z} = \frac{zw - 1}{z}.$$

Hence,

$$\text{saw}_x = \frac{1}{\text{uxp}} = \frac{z}{zw - 1}.$$

□

Corollary 2.2 — Rational Flatwave form

$$\boxed{\text{Flatwave} = \frac{z + w}{zw - 1}}.$$

Proof. Adding the two saw forms gives

$$\text{Flatwave} = \frac{w}{zw - 1} + \frac{z}{zw - 1}.$$

Therefore,

$$\text{Flatwave} = \frac{z + w}{zw - 1}.$$

□

3. The Möbius Reduction

Theorem 3.1 — Primary-envelope dependence

On every open quadrant,

$$\boxed{w = \frac{z + \varepsilon}{\varepsilon z - 1}}.$$

Equivalently,

$$\boxed{\text{cpx}(x) = \frac{\text{srx}(x) + \varepsilon}{\varepsilon \text{srx}(x) - 1}}.$$

Proof. By the Flatwave theorem of S0.I,

$$\text{Flatwave} = \varepsilon.$$

By Corollary 2.2,

$$\frac{z+w}{zw-1} = \varepsilon.$$

Multiplying by $zw-1$,

$$z+w = \varepsilon(zw-1).$$

Expanding the right-hand side,

$$z+w = \varepsilon zw - \varepsilon.$$

Collect the terms containing w :

$$w - \varepsilon zw = -\varepsilon - z.$$

Factor w :

$$w(1 - \varepsilon z) = -(z + \varepsilon).$$

Multiplying numerator and denominator by -1 ,

$$w = \frac{z + \varepsilon}{\varepsilon z - 1}.$$

It remains to verify that the denominator is nonzero.

If $\varepsilon = -1$, then

$$\varepsilon z - 1 = -z - 1 < 0,$$

because $z > 0$.

If $\varepsilon = +1$, the denominator could vanish only if $z = 1$. But $z = 1$ occurs only at an excluded quadrant boundary. Therefore,

$$\varepsilon z - 1 \neq 0$$

on every open quadrant.

Hence,

$$\boxed{w = \frac{z + \varepsilon}{\varepsilon z - 1}}.$$

□

Corollary 3.2 — Reciprocal forms

The two reciprocal operators are

$$\boxed{\text{sxp} = \frac{1}{z}},$$

and

$$\boxed{\text{crx} = \frac{\varepsilon z - 1}{z + \varepsilon}}.$$

Proof. The first identity follows from the definition of sxp .

For the second,

$$\text{crx} = \frac{1}{\text{exp}} = \frac{1}{w}.$$

Taking the reciprocal of Theorem 3.1 gives

$$\text{crx} = \frac{\varepsilon z - 1}{z + \varepsilon}.$$

□

Corollary 3.3 — One-generator sufficiency

On a fixed quadrant, every primitive displayed operator is a rational function of z .

Explicitly,

$$\text{srx} = z,$$

$$\text{sxp} = \frac{1}{z},$$

$$\text{exp} = \frac{z + \varepsilon}{\varepsilon z - 1},$$

and

$$\text{crx} = \frac{\varepsilon z - 1}{z + \varepsilon}.$$

Thus the four displayed operators do not constitute four algebraically independent generators.

4. Generator Range and Direction Sign

Theorem 4.1 — Generator range

On every quadrant with $\varepsilon = +1$,

$$\boxed{z > 1}.$$

On every quadrant with $\varepsilon = -1$,

$$\boxed{0 < z < 1}.$$

Proof. From S0.I, on an upper sine half-cycle,

$$z = \cot \frac{x}{2},$$

while on the corresponding lower half-cycle,

$$z = -\tan \frac{x}{2}.$$

In Quadrants I and III, $\sin 2x > 0$, so $\varepsilon = +1$. On both such quadrants, the corresponding positive half-angle expression takes values greater than 1.

In Quadrants II and IV, $\sin 2x < 0$, so $\varepsilon = -1$. On both such quadrants, the corresponding positive half-angle expression lies strictly between 0 and 1.

The value $z = 1$ occurs only at odd multiples of $\pi/2$, which are excluded from $\mathcal{D}$.

Therefore,

$$\varepsilon = +1 \implies z > 1,$$

and

$$\varepsilon = -1 \implies 0 < z < 1.$$

□

Corollary 4.2 — Direction sign from the generator

$$\boxed{\varepsilon = \operatorname{sgn}(z - 1)}.$$

Since $z > 0$,

$$\boxed{\varepsilon = \operatorname{sgn}(z^2 - 1)}.$$

The Flatwave sign therefore need not be supplied independently once the generator value is known.

5. Canonical UNA Forms

Theorem 5.1 — Canonical Unainverex form

$$\boxed{\text{urx} = \frac{z^2 + 1}{z + \varepsilon}}.$$

Proof. By definition,

$$\text{urx} = \text{srx} - \text{crx}.$$

Using the canonical expressions,

$$\text{urx} = z - \frac{\varepsilon z - 1}{z + \varepsilon}.$$

Place the terms over a common denominator:

$$\text{urx} = \frac{z(z + \varepsilon) - (\varepsilon z - 1)}{z + \varepsilon}.$$

Expand the numerator:

$$z(z + \varepsilon) - (\varepsilon z - 1) = z^2 + \varepsilon z - \varepsilon z + 1.$$

Therefore,

$$\text{urx} = \frac{z^2 + 1}{z + \varepsilon}.$$

□

Theorem 5.2 — Canonical Unaexpon form

$$\boxed{\text{uxp} = \frac{z^2 + 1}{z(\varepsilon z - 1)}}.$$

Proof. By definition,

$$\text{uxp} = \text{cxp} - \text{sxp}.$$

Substitute the canonical expressions:

$$\text{uxp} = \frac{z + \varepsilon}{\varepsilon z - 1} - \frac{1}{z}.$$

Use the common denominator

$$z(\varepsilon z - 1) :$$

$$\text{uxp} = \frac{z(z + \varepsilon) - (\varepsilon z - 1)}{z(\varepsilon z - 1)}.$$

The numerator is again

$$z^2 + \varepsilon z - \varepsilon z + 1 = z^2 + 1.$$

Thus,

$$\text{uxp} = \frac{z^2 + 1}{z(\varepsilon z - 1)}.$$

□

Corollary 5.3 — UNA signs

Because

$$z^2 + 1 > 0,$$

the signs are determined entirely by the denominators.

Theorem 4.1 therefore gives

$$\text{sgn}(\text{urx}) = \text{sgn}(\text{uxp}) = \varepsilon.$$

This recovers the common-sign theorem of S0.I in canonical-generator form.

6. Canonical Saw Forms

Theorem 6.1 — Canonical inverex saw

$$\boxed{\text{saw}_r = \frac{z + \varepsilon}{z^2 + 1}}.$$

Proof. By definition,

$$\text{saw}_r = \frac{1}{\text{urx}}.$$

Using Theorem 5.1,

$$\text{saw}_r = \frac{z + \varepsilon}{z^2 + 1}.$$

□

Theorem 6.2 — Canonical expon saw

$$\boxed{\text{saw}_x = \frac{z(\varepsilon z - 1)}{z^2 + 1}}.$$

Proof. By definition,

$$\text{saw}_x = \frac{1}{\text{uxp}}.$$

Taking the reciprocal of Theorem 5.2 gives

$$\text{saw}_x = \frac{z(\varepsilon z - 1)}{z^2 + 1}.$$

□

Corollary 6.3 — Flatwave reduction

Adding the two canonical saw forms,

$$\text{Flatwave} = \frac{z + \varepsilon}{z^2 + 1} + \frac{z(\varepsilon z - 1)}{z^2 + 1}.$$

The numerator becomes

$$z + \varepsilon + \varepsilon z^2 - z.$$

Therefore,

$$\text{Flatwave} = \frac{\varepsilon(z^2 + 1)}{z^2 + 1}.$$

Hence,

$$\boxed{\text{Flatwave} = \varepsilon}.$$

The exact Flatwave theorem is therefore built directly into the canonical forms.

7. Rational Phase Readout

Theorem 7.1 — Double-angle phase in generator form

$$\boxed{\sin 2x = \frac{4z(z^2 - 1)}{(z^2 + 1)^2}}.$$

Proof. From S0.I,

$$\text{urx} + \text{uxp} = \frac{4}{\sin 2x}.$$

Using Theorems 5.1 and 5.2,

$$\text{urx} + \text{uxp} = \frac{z^2 + 1}{z + \varepsilon} + \frac{z^2 + 1}{z(\varepsilon z - 1)}.$$

Factor $z^2 + 1$:

$$\text{urx} + \text{uxp} = (z^2 + 1) \left[\frac{1}{z + \varepsilon} + \frac{1}{z(\varepsilon z - 1)} \right].$$

Combine the fractions:

$$\text{urx} + \text{uxp} = (z^2 + 1) \frac{z(\varepsilon z - 1) + (z + \varepsilon)}{z(z + \varepsilon)(\varepsilon z - 1)}.$$

Simplify the numerator:

$$z(\varepsilon z - 1) + (z + \varepsilon) = \varepsilon z^2 - z + z + \varepsilon.$$

Thus,

$$z(\varepsilon z - 1) + (z + \varepsilon) = \varepsilon(z^2 + 1).$$

Next simplify

$$(z + \varepsilon)(\varepsilon z - 1).$$

Expanding,

$$(z + \varepsilon)(\varepsilon z - 1) = \varepsilon z^2 - z + \varepsilon^2 z - \varepsilon.$$

Since $\varepsilon^2 = 1$,

$$(z + \varepsilon)(\varepsilon z - 1) = \varepsilon z^2 - z + z - \varepsilon = \varepsilon(z^2 - 1).$$

Therefore,

$$\text{urx} + \text{uxp} = (z^2 + 1) \frac{\varepsilon(z^2 + 1)}{\varepsilon z(z^2 - 1)}.$$

Cancel ε :

$$\text{urx} + \text{uxp} = \frac{(z^2 + 1)^2}{z(z^2 - 1)}.$$

Since this also equals $4/\sin 2x$,

$$\frac{4}{\sin 2x} = \frac{(z^2 + 1)^2}{z(z^2 - 1)}.$$

Taking reciprocals,

$$\frac{\sin 2x}{4} = \frac{z(z^2 - 1)}{(z^2 + 1)^2}.$$

Therefore,

$$\boxed{\sin 2x = \frac{4z(z^2 - 1)}{(z^2 + 1)^2}}.$$

□

Corollary 7.2 — Flatwave sign

Because $z > 0$ and $(z^2 + 1)^2 > 0$,

$$\text{sgn}(\sin 2x) = \text{sgn}(z^2 - 1).$$

Therefore,

$$\text{Flatwave} = \text{sgn}(z^2 - 1).$$

Corollary 7.3 — Boundary identification

The excluded quadrant boundaries correspond to:

$$z \rightarrow \infty, \quad z \rightarrow 0, \quad \text{or} \quad z \rightarrow 1,$$

depending on the side of approach.

The value $z = 1$ is never attained on an admissible open quadrant.

8. Differential Law of the Generator

Analytic Convention 8.1

Ordinary real differentiation rules are inherited on each open quadrant, including the chain rule and the derivatives

$$\frac{d}{du} \tan u = \sec^2 u,$$

and

$$\frac{d}{du} \cot u = -\csc^2 u.$$

No differentiation across an excluded boundary is asserted.

Theorem 8.2 — Generator differential equation

On every open quadrant,

$$\boxed{\frac{dz}{dx} = -\frac{1+z^2}{2}}.$$

Proof. From the half-angle forms proved in S0.I, there are two branch types.

On the first branch,

$$z = \cot \frac{x}{2}.$$

Differentiate:

$$\frac{dz}{dx} = -\csc^2 \frac{x}{2} \cdot \frac{1}{2}.$$

Using

$$\csc^2 u = 1 + \cot^2 u,$$

we obtain

$$\frac{dz}{dx} = -\frac{1}{2} \left(1 + \cot^2 \frac{x}{2} \right).$$

Since

$$z = \cot \frac{x}{2},$$

this becomes

$$\frac{dz}{dx} = -\frac{1+z^2}{2}.$$

On the second branch,

$$z = -\tan \frac{x}{2}.$$

Differentiating,

$$\frac{dz}{dx} = -\sec^2 \frac{x}{2} \cdot \frac{1}{2}.$$

Since

$$\sec^2 u = 1 + \tan^2 u,$$

and

$$z^2 = \tan^2 \frac{x}{2},$$

we again obtain

$$\frac{dz}{dx} = -\frac{1+z^2}{2}.$$

Thus the same differential equation holds on every open quadrant:

$$\boxed{z' = -\frac{1+z^2}{2}}.$$

□

Corollary 8.3 — Strict monotonicity

Since

$$1 + z^2 > 0,$$

we have

$$z' < 0.$$

Therefore,

$$\boxed{\text{sr}\mathbf{x}(x) \text{ is strictly decreasing on every open quadrant.}}$$

9. Differential Laws of the Primitive Family**Theorem 9.1 — Cosinexpon differential equation**

Let

$$w = \exp .$$

Then

$$\boxed{\frac{dw}{dx} = \frac{1 + w^2}{2}}.$$

Proof. From Theorem 3.1,

$$w = \frac{z + \varepsilon}{\varepsilon z - 1}.$$

The quadrant sign ε is constant on the chosen quadrant. Differentiate with respect to z :

$$\frac{dw}{dz} = \frac{(\varepsilon z - 1) - \varepsilon(z + \varepsilon)}{(\varepsilon z - 1)^2}.$$

Simplify the numerator:

$$\varepsilon z - 1 - \varepsilon z - \varepsilon^2 = -1 - 1 = -2.$$

Therefore,

$$\frac{dw}{dz} = -\frac{2}{(\varepsilon z - 1)^2}.$$

By Theorem 8.2,

$$\frac{dz}{dx} = -\frac{1+z^2}{2}.$$

Using the chain rule,

$$\frac{dw}{dx} = \frac{dw}{dz} \frac{dz}{dx}.$$

Hence,

$$\frac{dw}{dx} = \left[-\frac{2}{(\varepsilon z - 1)^2} \right] \left[-\frac{1+z^2}{2} \right].$$

Thus,

$$\frac{dw}{dx} = \frac{1+z^2}{(\varepsilon z - 1)^2}.$$

Now compute $1 + w^2$:

$$1 + w^2 = 1 + \left(\frac{z + \varepsilon}{\varepsilon z - 1} \right)^2.$$

Using a common denominator,

$$1 + w^2 = \frac{(\varepsilon z - 1)^2 + (z + \varepsilon)^2}{(\varepsilon z - 1)^2}.$$

Expand the numerator:

$$(\varepsilon z - 1)^2 = z^2 - 2\varepsilon z + 1,$$

and

$$(z + \varepsilon)^2 = z^2 + 2\varepsilon z + 1.$$

Adding them gives

$$2z^2 + 2 = 2(1 + z^2).$$

Therefore,

$$1 + w^2 = \frac{2(1 + z^2)}{(\varepsilon z - 1)^2}.$$

Hence,

$$\frac{1 + w^2}{2} = \frac{1 + z^2}{(\varepsilon z - 1)^2}.$$

This is exactly dw/dx . Therefore,

$$w' = \frac{1 + w^2}{2}.$$

□

Corollary 9.2 — Primitive derivative table

The four primitive operators satisfy:

$$\text{sr}x' = -\frac{1 + \text{sr}x^2}{2},$$

$$\text{sxp}' = \frac{1 + \text{sxp}^2}{2},$$

$$\text{cxp}' = \frac{1 + \text{cxp}^2}{2},$$

and

$$\text{cr}x' = -\frac{1 + \text{cr}x^2}{2}.$$

Proof. The first and third identities are Theorems 8.2 and 9.1.

For

$$\text{sxp} = \frac{1}{z},$$

differentiate:

$$\text{sxp}' = -\frac{z'}{z^2}.$$

Using

$$z' = -\frac{1 + z^2}{2},$$

we obtain

$$\text{sxp}' = \frac{1 + z^2}{2z^2}.$$

Therefore,

$$\text{sxp}' = \frac{1}{2} \left(1 + \frac{1}{z^2} \right).$$

Since

$$\text{sxp} = \frac{1}{z},$$

it follows that

$$\text{sxp}' = \frac{1 + \text{sxp}^2}{2}.$$

The proof for

$$\text{crx} = \frac{1}{w}$$

is identical:

$$\text{crx}' = -\frac{w'}{w^2} = -\frac{1 + w^2}{2w^2} = -\frac{1 + \text{crx}^2}{2}.$$

□

Corollary 9.3 — Primitive monotonicity

On every open quadrant:

$$\text{srx}' < 0, \quad \text{crx}' < 0,$$

while

$$\text{sxp}' > 0, \quad \text{exp}' > 0.$$

Thus the two inverex operators decrease with phase, while the two expon operators increase with phase.

This is an exact local phase statement, not yet a temporal growth or decay law.

10. Quadrant-Local Differential Closure

Definition 10.1 — Rational expression field

Fix an open quadrant and its constant sign ε .

Let

$$\mathbb{R}(z)$$

denote the set of rational functions

$$F(z) = \frac{P(z)}{Q(z)},$$

where

$$P, Q \in \mathbb{R}[z]$$

and $Q(z) \neq 0$ on the declared evaluation interval.

Theorem 10.2 — Algebraic containment

Every named operator and named composite introduced in S0.I belongs to

$$\mathbb{R}(z)$$

on each open quadrant.

Proof. The canonical forms derived above are:

$$\text{stx} = z,$$

$$\text{sxp} = \frac{1}{z},$$

$$\text{cxp} = \frac{z + \varepsilon}{\varepsilon z - 1},$$

$$\text{ctx} = \frac{\varepsilon z - 1}{z + \varepsilon},$$

$$\text{urx} = \frac{z^2 + 1}{z + \varepsilon},$$

$$\text{uxp} = \frac{z^2 + 1}{z(\varepsilon z - 1)},$$

$$\text{saw}_r = \frac{z + \varepsilon}{z^2 + 1},$$

$$\text{saw}_x = \frac{z(\varepsilon z - 1)}{z^2 + 1},$$

and

$$\text{Flatwave} = \varepsilon.$$

Each is a rational function of z with real coefficients once ε is fixed.

Therefore, every named operator belongs to $\mathbb{R}(z)$.

□

Theorem 10.3 — Closure under differentiation

If

$$F(z) \in \mathbb{R}(z),$$

then

$$\frac{d}{dx}F(z(x)) \in \mathbb{R}(z)$$

on every open quadrant where F is defined.

Proof. Let

$$F(z) = \frac{P(z)}{Q(z)},$$

with $P, Q \in \mathbb{R}[z]$.

Differentiating with respect to z ,

$$\frac{dF}{dz} = \frac{P'(z)Q(z) - P(z)Q'(z)}{Q(z)^2}.$$

The numerator and denominator are polynomials in z . Therefore,

$$\frac{dF}{dz} \in \mathbb{R}(z).$$

By the chain rule,

$$\frac{dF}{dx} = \frac{dF}{dz} \frac{dz}{dx}.$$

From Theorem 8.2,

$$\frac{dz}{dx} = -\frac{1+z^2}{2},$$

which is a polynomial in z .

The product of two rational functions of z is again a rational function of z . Therefore,

$$\frac{dF}{dx} \in \mathbb{R}(z).$$

□

Corollary 10.4 — Closure under repeated differentiation

For every nonnegative integer n ,

$$\boxed{\frac{d^n}{dx^n} F(z(x)) \in \mathbb{R}(z)}.$$

Proof. The result follows by induction.

The case $n = 0$ is the assumption that $F \in \mathbb{R}(z)$.

If

$$\frac{d^n F}{dx^n} \in \mathbb{R}(z),$$

then Theorem 10.3 implies

$$\frac{d}{dx} \left(\frac{d^n F}{dx^n} \right) = \frac{d^{n+1} F}{dx^{n+1}} \in \mathbb{R}(z).$$

Therefore, the result holds for every $n \geq 0$.

□

11. Meaning of Closure

Theorem 11.1 — One-generator differential-field closure

On every open quadrant, the bounded phase operator calculus generated by the named R-operators is contained in the differential field

$$\boxed{\mathbb{R}(z)}$$

with derivation determined by

$$\boxed{z' = -\frac{1+z^2}{2}}.$$

Clarification 11.2 — What has been proved

The following statement is proved:

Every named operator, every rational combination of named operators, and every quadrant-local derivative of such a combination reduces to a rational function of one generator.

Clarification 11.3 — What has not been proved

The following stronger statements are not proved:

- that every rational function of z should be designated an R-operator;
- that the finite named list is closed under all rational operations;
- that no alternative generator exists;
- that every admissible phase function belongs to $\mathbb{R}(z)$;
- that the expression field has a physical interpretation.

The closure is a mathematical containment and reduction result.

12. Minimality of Generator Count

Theorem 12.1 — One nonconstant generator is sufficient

The named quadrant-local calculus requires no more than one nonconstant algebraic generator.

Proof. Theorem 10.2 explicitly expresses every named operator as a rational function of z .

Therefore, one nonconstant generator is sufficient.

□

Theorem 12.2 — Zero nonconstant generators are insufficient

No field containing only real constants can represent the named calculus.

Proof. A field generated by no nonconstant element over $\mathbb{R}$ contains only constant functions.
But

$$z(x) = \text{sr}x(x)$$

is strictly decreasing on every open quadrant and is therefore nonconstant.

Hence, a constant-only field cannot contain z , and therefore cannot contain the full named calculus.
At least one nonconstant generator is required.

□

Corollary 12.3 — Minimal generator count

Exactly one nonconstant generator is necessary and sufficient for the named calculus on each open quadrant.

This proves minimality of the generator count.

It does not prove that $\text{sr}x$ is the unique possible generator.

13. Terminal Statement

S0.I established the Pythagorean identities of the bounded phase operators.

S0.II establishes their algebraic organization.

On each open quadrant:

$$z = \text{sr}x$$

is sufficient to generate the entire named operator family.

The second primary envelope is not independent:

$$\text{c}xp = \frac{z + \varepsilon}{\varepsilon z - 1}.$$

The reciprocal, UNA, saw, and Flatwave operators are rational functions of z .

The double-angle phase readout is

$$\sin 2x = \frac{4z(z^2 - 1)}{(z^2 + 1)^2}.$$

The generator obeys

$$z' = -\frac{1 + z^2}{2}.$$

Therefore, the quadrant-local expression field is closed under differentiation:

$$F(z) \in \mathbb{R}(z) \quad \implies \quad \frac{dF}{dx} \in \mathbb{R}(z).$$

The result is a precise form of closure:

one generator, rational reduction, differential closure.

No additional primitive function is required to express or differentiate the named bounded phase operators within an open quadrant.

S0.III

Tangential Geometry and Bounded Transfer

Book 0 - Formal Core

Abstract

This essay develops the quadrant-local differential geometry of the bounded phase operators established in S0.I and reduced to one generator in S0.II.

The phase derivative is introduced as a derived operation rather than a new primitive. Each displayed operator satisfies an autonomous Riccati equation. Its normalized derivative reduces exactly to an absolute reciprocal trigonometric function:

$$\hat{T}[\text{sr}x] = -|\csc x|, \quad \hat{T}[\text{sx}p] = |\csc x|,$$

$$\hat{T}[\text{cx}p] = |\sec x|, \quad \hat{T}[\text{cr}x] = -|\sec x|.$$

Reciprocal pairing therefore reverses normalized tangential direction without introducing a new rate magnitude.

A pair of bounded auxiliary coordinates p and q is then defined. They satisfy

$$p' = -q, \quad q' = p, \quad p^2 + q^2 = 2.$$

These coordinates yield a canonical transfer variable

$$\lambda = \frac{1-p}{2},$$

which traverses strictly from 0 to 1 across every open quadrant. The two saw responses become

$$\text{saw}_r = \varepsilon\lambda, \quad \text{saw}_x = \varepsilon(1-\lambda),$$

where

$$\varepsilon = \text{sgn}(\sin 2x).$$

Their derivatives are equal and opposite:

$$\text{saw}'_r = \varepsilon \frac{q}{2}, \quad \text{saw}'_x = -\varepsilon \frac{q}{2}.$$

The total Flatwave derivative is therefore zero on every open quadrant. Transfer occurs entirely between the two saw components while their signed total remains fixed.

The transfer rate is bounded:

$$\frac{1}{2} < |\text{saw}'_r| = |\text{saw}'_x| \leq \frac{1}{\sqrt{2}}.$$

Each saw changes curvature only at the admissible midpoint of a quadrant, so curvature sign is fixed within every open octant.

No temporal, physical, or causal interpretation is assumed.

0. Scope and Method

0.1 Scope

This essay proves:

1. the autonomous phase-derivative equations of the four primitive operators;
2. their normalized tangential responses;
3. exact reversal of normalized response under reciprocal pairing;
4. the location of primary rate balance;
5. a bounded transfer coordinate for the two saw responses;
6. equal-and-opposite saw transfer;
7. the boundedness of the saw-transfer rate;
8. the curvature structure of the saw responses;
9. corresponding differential forms for the UNA operators;
10. closure of the tangential calculus.

0.2 Non-scope

This essay does not:

- identify phase x with time;
- specify a traversal law $x(\tau)$;
- introduce force, energy, mass, or information;

- interpret derivative sign as physical causation;
- assert a physical conservation law;
- integrate across excluded quadrant boundaries.

0.3 Inheritance

All definitions and proved results of S0.I and S0.II are inherited.

In particular, the admissible domain is

$$\mathcal{D} = \mathbb{R} \setminus \left\{ \frac{k\pi}{2} : k \in \mathbb{Z} \right\}.$$

The primitive operators are positive on $\mathcal{D}$, and the Flatwave satisfies

$$\text{Flatwave}(x) = \text{sgn}(\sin 2x).$$

On each open quadrant,

$$z(x) = \text{srx}(x)$$

satisfies

$$z' = -\frac{1 + z^2}{2}.$$

Every named operator is a rational function of z on that quadrant.

0.4 Analytic convention

Ordinary real differentiation is inherited on each connected component of $\mathcal{D}$.

No derivative is defined through an excluded boundary.

1. Tangential Notation

Definition 1.1 — Phase derivative

For a differentiable function f on an open quadrant, define its phase-tangential derivative by

$$\boxed{Tf = \frac{df}{dx}}.$$

The operator T is derived from ordinary differentiation. It is not a new primitive member of the R-operator family.

Definition 1.2 — Normalized tangential response

For a differentiable positive function f , define

$$\hat{T}[f] = \frac{f'(x)}{f(x)}.$$

Since the primitive operators are positive,

$$\hat{T}[f] = \frac{d}{dx} \ln f$$

for every primitive operator f .

Definition 1.3 — Tangential direction

Define

$$\text{dir}(f) = \text{sgn}(\hat{T}[f]).$$

Definition 1.4 — Tangential speed

Define

$$V_f = |\hat{T}[f]|.$$

These quantities describe change with respect to phase position x , not change with respect to physical time.

2. Primitive Differential Equations**Theorem 2.1 — Sininverex differential equation**

For every admissible x ,

$$\text{sr}x'(x) = -\frac{1 + \text{sr}x(x)^2}{2}.$$

Proof. By S0.II, on each open quadrant the generator

$$z = \text{sr}x$$

satisfies

$$z' = -\frac{1+z^2}{2}.$$

Substitution gives the stated identity.

□

Theorem 2.2 — Sinexpon differential equation

For every admissible x ,

$$\text{sxp}'(x) = \frac{1 + \text{sxp}(x)^2}{2}.$$

Proof. Since

$$\text{sxp} = \frac{1}{\text{sr}x},$$

differentiate:

$$\text{sxp}' = -\frac{\text{sr}x'}{\text{sr}x^2}.$$

Using Theorem 2.1,

$$\text{sxp}' = -\frac{-\frac{1}{2}(1 + \text{sr}x^2)}{\text{sr}x^2}.$$

Thus,

$$\text{sxp}' = \frac{1}{2} \left(1 + \frac{1}{\text{sr}x^2} \right).$$

Because

$$\text{sxp} = \frac{1}{\text{sr}x},$$

we obtain

$$\text{sxp}' = \frac{1 + \text{sxp}^2}{2}.$$

□

Theorem 2.3 — Cosinexpon differential equation

For every admissible x ,

$$\boxed{\text{cxp}'(x) = \frac{1 + \text{cxp}(x)^2}{2}}.$$

Proof. This was established from the canonical Möbius representation in S0.II. It may also be verified directly.

Let

$$M = |\sec x|.$$

On a fixed quadrant, let

$$\beta = \text{sgn}(\cos x).$$

Then

$$M = \beta \sec x,$$

because β is constant on that quadrant.

Differentiate:

$$M' = \beta \sec x \tan x = M \tan x.$$

Since

$$\text{cxp} = M + \tan x,$$

we have

$$\text{cxp}' = M \tan x + \sec^2 x.$$

Using

$$\sec^2 x = M^2,$$

this becomes

$$\text{cxp}' = M \tan x + M^2 = M(M + \tan x).$$

Therefore,

$$\text{exp}' = M \text{exp}.$$

From the reciprocal-conjugate identity,

$$\text{exp} + \text{crx} = 2M.$$

Since

$$\text{crx} = \frac{1}{\text{exp}},$$

we obtain

$$2M = \text{exp} + \frac{1}{\text{exp}}.$$

Multiplying by $\text{exp}/2$,

$$M \text{exp} = \frac{1 + \text{exp}^2}{2}.$$

Hence,

$$\boxed{\text{exp}' = \frac{1 + \text{exp}^2}{2}}.$$

□

Theorem 2.4 — Cosininverex differential equation

For every admissible x ,

$$\boxed{\text{crx}'(x) = -\frac{1 + \text{crx}(x)^2}{2}}.$$

Proof. Since

$$\text{crx} = \frac{1}{\text{exp}},$$

differentiate:

$$\text{crx}' = -\frac{\text{exp}'}{\text{exp}^2}.$$

Using Theorem 2.3,

$$\text{crx}' = -\frac{\frac{1}{2}(1 + \text{cxp}^2)}{\text{cxp}^2}.$$

Thus,

$$\text{crx}' = -\frac{1}{2} \left(1 + \frac{1}{\text{cxp}^2} \right).$$

Since

$$\text{crx} = \frac{1}{\text{cxp}},$$

we conclude that

$$\boxed{\text{crx}' = -\frac{1 + \text{crx}^2}{2}}.$$

□

Corollary 2.5 — Primitive direction table

On every open quadrant,

$$\text{srx}' < 0, \quad \text{crx}' < 0,$$

and

$$\text{sxp}' > 0, \quad \text{cxp}' > 0.$$

Thus the invrex operators decrease with increasing phase, while the expon operators increase with increasing phase.

This is an ordering statement in x , not yet a physical growth or decay law.

3. Normalized Primitive Responses

Lemma 3.1 — Sine-family sum

$$\boxed{\text{srx} + \text{sxp} = 2N}.$$

Proof. By the reciprocal-conjugate forms,

$$\text{srx} = N + \cot x,$$

and

$$\text{sxp} = N - \cot x.$$

Adding,

$$\text{srx} + \text{sxp} = 2N.$$

□

Lemma 3.2 — Cosine-family sum

$$\boxed{\text{exp} + \text{crx} = 2M}.$$

Proof. Using

$$\text{exp} = M + \tan x$$

and

$$\text{crx} = M - \tan x,$$

addition gives

$$\text{exp} + \text{crx} = 2M.$$

□

Theorem 3.3 — Sininverex normalized response

$$\boxed{\hat{T}[\text{srx}] = -N = -|\csc x|}.$$

Proof. From Theorem 2.1,

$$\hat{T}[\text{srx}] = \frac{\text{srx}'}{\text{srx}} = -\frac{1 + \text{srx}^2}{2 \text{srx}}.$$

Therefore,

$$\hat{T}[\text{srx}] = -\frac{1}{2} \left(\text{srx} + \frac{1}{\text{srx}} \right).$$

Since

$$\frac{1}{\text{srx}} = \text{sxp},$$

we have

$$\hat{T}[\text{srx}] = -\frac{1}{2}(\text{srx} + \text{sxp}).$$

By Lemma 3.1,

$$\hat{T}[\text{srx}] = -\frac{1}{2}(2N) = -N.$$

Thus,

$$\boxed{\hat{T}[\text{srx}] = -|\csc x|}.$$

□

Theorem 3.4 — Sinexpon normalized response

$$\boxed{\hat{T}[\text{sxp}] = N = |\csc x|}.$$

Proof. Since

$$\text{sxp} = \frac{1}{\text{srx}},$$

we have

$$\ln \text{sxp} = -\ln \text{srx}.$$

Differentiating,

$$\hat{T}[\text{sxp}] = -\hat{T}[\text{srx}].$$

Using Theorem 3.3,

$$\hat{T}[\text{sxp}] = N.$$

□

Theorem 3.5 — Cosinexpon normalized response

$$\boxed{\hat{T}[\text{cxp}] = M = |\sec x|}.$$

Proof. From Theorem 2.3,

$$\hat{T}[\text{cxp}] = \frac{1 + \text{cxp}^2}{2 \text{cxp}}.$$

Hence,

$$\hat{T}[\text{cxp}] = \frac{1}{2} \left(\text{cxp} + \frac{1}{\text{cxp}} \right).$$

Since

$$\frac{1}{\text{cxp}} = \text{crx},$$

Lemma 3.2 gives

$$\hat{T}[\text{cxp}] = \frac{1}{2}(2M) = M.$$

Therefore,

$$\boxed{\hat{T}[\text{cxp}] = |\sec x|}.$$

□

Theorem 3.6 — Cosininverex normalized response

$$\boxed{\hat{T}[\text{crx}] = -M = -|\sec x|}.$$

Proof. Because

$$\text{crx} = \frac{1}{\text{cxp}},$$

normalized differentiation reverses sign:

$$\hat{T}[\text{crx}] = -\hat{T}[\text{cxp}].$$

Theorem 3.5 then gives

$$\hat{T}[\text{crx}] = -M.$$

□

Corollary 3.7 — Complete normalized-response table

Operator	Normalized tangential response
srx	$-N$
sxp	$+N$
cxp	$+M$
crx	$-M$

4. Reciprocal Tangential Symmetry**Theorem 4.1 — Reciprocal reversal**

Let $f > 0$ be differentiable and define

$$f^* = \frac{1}{f}.$$

Then

$$\hat{T}[f^*] = -\hat{T}[f].$$

Proof. Differentiate:

$$(f^*)' = -\frac{f'}{f^2}.$$

Then

$$\hat{T}[f^*] = \frac{(f^*)'}{f^*} = \frac{-f'/f^2}{1/f}.$$

Therefore,

$$\hat{T}[f^*] = -\frac{f'}{f} = -\hat{T}[f].$$

□

Corollary 4.2 — Mirror rate magnitude

Reciprocal pairing preserves normalized speed:

$$\boxed{V_{f^*} = V_f}.$$

It reverses only normalized direction.

Corollary 4.3 — Family rates

The sine reciprocal pair shares rate magnitude

$$V_{\sin} = N,$$

while the cosine reciprocal pair shares rate magnitude

$$V_{\cos} = M.$$

Thus reciprocal access creates no independent rate scale.

5. Primary Rate Comparison

Definition 5.1 — Rate sum

Define the positive rate sum

$$\Sigma_T = \hat{T}[\text{exp}] - \hat{T}[\text{srx}].$$

Using Section 3,

$$\boxed{\Sigma_T = M + N}.$$

Definition 5.2 — Rate contrast

Define the signed rate contrast

$$\Delta_T = \hat{T}[\text{exp}] + \hat{T}[\text{srx}].$$

Therefore,

$$\boxed{\Delta_T = M - N}.$$

The sum appears here because the two primary normalized directions are opposite.

Theorem 5.3 — Positivity of total rate separation

For every $x \in \mathcal{D}$,

$$\boxed{\Sigma_T = M + N > 0}.$$

Thus the two primary normalized directions never coincide.

Theorem 5.4 — Unique rate balance within each quadrant

The primary rate magnitudes are equal exactly when

$$M = N.$$

This occurs exactly at

$$\boxed{x = \frac{(2k+1)\pi}{4}, \quad k \in \mathbb{Z}}.$$

Proof. The condition

$$M = N$$

is

$$\frac{1}{|\cos x|} = \frac{1}{|\sin x|}.$$

Since both denominators are positive,

$$|\sin x| = |\cos x|.$$

Squaring,

$$\sin^2 x = \cos^2 x.$$

Using

$$\sin^2 x + \cos^2 x = 1,$$

we obtain

$$\sin^2 x = \cos^2 x = \frac{1}{2}.$$

This occurs exactly at the odd multiples of $\pi/4$:

$$x = \frac{(2k+1)\pi}{4}.$$

Each open quadrant contains exactly one such point.

□

Corollary 5.5 — Rate balance and octant division

The unique primary rate-balance point in each quadrant is its midpoint and is also the boundary between its two octants.

At this point,

$$M = N = \sqrt{2}.$$

The neighboring dominant primitive operators tie there.

6. Bounded Transfer Coordinates

Definition 6.1 — Quadrant signs

On a fixed open quadrant, define

$$\alpha = \operatorname{sgn}(\sin x), \quad \beta = \operatorname{sgn}(\cos x).$$

Both are constant on that quadrant.

Define also

$$\varepsilon = \alpha\beta = \operatorname{sgn}(\sin x \cos x) = \operatorname{sgn}(\sin 2x).$$

Definition 6.2 — Transfer coordinates

Define

$$p = \alpha \cos x - \beta \sin x$$

and

$$q = \alpha \sin x + \beta \cos x.$$

Since

$$\alpha \sin x = |\sin x|$$

and

$$\beta \cos x = |\cos x|,$$

we have

$$q = |\sin x| + |\cos x|.$$

Therefore,

$$q > 0$$

throughout every open quadrant.

Theorem 6.3 — Transfer-circle identity

$$p^2 + q^2 = 2.$$

Proof. Expand:

$$p^2 = (\alpha c - \beta s)^2 = c^2 + s^2 - 2\alpha\beta sc.$$

Similarly,

$$q^2 = (\alpha s + \beta c)^2 = s^2 + c^2 + 2\alpha\beta sc.$$

Adding,

$$p^2 + q^2 = 2(s^2 + c^2).$$

By the Pythagorean identity,

$$s^2 + c^2 = 1.$$

Therefore,

$$p^2 + q^2 = 2.$$

□

Theorem 6.4 — Transfer-coordinate differential system

$$\boxed{p' = -q}$$

and

$$\boxed{q' = p}.$$

Proof. Because α and β are constant on the selected open quadrant,

$$p' = -\alpha \sin x - \beta \cos x.$$

Thus,

$$p' = -q.$$

Likewise,

$$q' = \alpha \cos x - \beta \sin x = p.$$

□

Corollary 6.5 — Harmonic closure

$$\boxed{p'' = -p}$$

and

$$\boxed{q'' = -q}.$$

The transfer coordinates therefore form a closed two-dimensional differential system.

Theorem 6.6 — Quadrant traversal of p

On every open quadrant,

$$-1 < p < 1,$$

and

$$p' < 0.$$

Moreover, under one-sided boundary limits,

$$p \longrightarrow 1$$

at the beginning of the quadrant, while

$$p \longrightarrow -1$$

at its end.

Proof. Since $q > 0$, Theorem 6.4 gives

$$p' = -q < 0.$$

Thus p is strictly decreasing.

At the left boundary of any quadrant, one of $|\sin x|$ or $|\cos x|$ approaches 0, while the other approaches 1. Direct substitution into the signed definition of p gives the left limit 1.

At the right boundary, the roles reverse and the limit is -1 .

Because the boundary points are excluded,

$$-1 < p < 1$$

inside the quadrant.

□

7. Canonical Transfer Variable

Definition 7.1 — Bounded transfer fraction

Define

$$\lambda = \frac{1-p}{2}.$$

Theorem 7.2 — Transfer range and monotonicity

On every open quadrant,

$$0 < \lambda < 1$$

and

$$\lambda' > 0.$$

Under one-sided boundary limits,

$$\lambda \longrightarrow 0$$

at the beginning of the quadrant and

$$\lambda \longrightarrow 1$$

at its end.

Proof. Since

$$-1 < p < 1,$$

we have

$$0 < 1 - p < 2.$$

Dividing by 2,

$$0 < \lambda < 1.$$

Differentiate:

$$\lambda' = -\frac{p'}{2}.$$

Using $p' = -q$,

$$\boxed{\lambda' = \frac{q}{2} > 0}.$$

The boundary limits follow directly from the limits $p \rightarrow 1$ and $p \rightarrow -1$.

□

Corollary 7.3 — Transfer acceleration

$$\boxed{\lambda'' = \frac{p}{2} = \frac{1}{2} - \lambda}.$$

Proof. Differentiate

$$\lambda' = \frac{q}{2}.$$

Then

$$\lambda'' = \frac{q'}{2} = \frac{p}{2}.$$

Since

$$p = 1 - 2\lambda,$$

we have

$$\lambda'' = \frac{1 - 2\lambda}{2} = \frac{1}{2} - \lambda.$$

□

Corollary 7.4 — Transfer differential equation

$$\lambda'' + \lambda = \frac{1}{2}.$$

This is a phase-domain identity. It is not yet an equation of physical motion.

8. Saw Responses as Complementary Transfer Channels

Theorem 8.1 — Saw representation by p

The two saw responses satisfy

$$\text{saw}_r = \varepsilon \frac{1 - p}{2}$$

and

$$\text{saw}_x = \varepsilon \frac{1 + p}{2}.$$

Proof. From S0.I, the UNA branch forms are

$$\text{urx} = \frac{1 + p}{sc},$$

and

$$\text{uxp} = \frac{1 - p}{sc}.$$

Also,

$$sc = \varepsilon |sc|.$$

From the proof of the UNA product identity,

$$1 - p^2 = 2|sc|.$$

Therefore,

$$sc = \varepsilon \frac{1 - p^2}{2}.$$

Now,

$$\text{saw}_r = \frac{1}{\text{urx}} = \frac{sc}{1 + p}.$$

Substitute the expression for sc :

$$\text{saw}_r = \varepsilon \frac{1 - p^2}{2(1 + p)}.$$

Since

$$1 - p^2 = (1 - p)(1 + p),$$

and $1 + p > 0$,

$$\text{saw}_r = \varepsilon \frac{1 - p}{2}.$$

Similarly,

$$\text{saw}_x = \frac{sc}{1 - p} = \varepsilon \frac{1 - p^2}{2(1 - p)} = \varepsilon \frac{1 + p}{2}.$$

□

Corollary 8.2 — Saw representation by λ

Since

$$\lambda = \frac{1 - p}{2},$$

we have

$$\text{saw}_r = \varepsilon \lambda$$

and

$$\text{saw}_x = \varepsilon(1 - \lambda).$$

Corollary 8.3 — Signed and unsigned partition

$$\text{saw}_r + \text{saw}_x = \varepsilon,$$

and

$$|\text{saw}_r| + |\text{saw}_x| = 1.$$

The transfer fraction λ gives the exact distribution of the signed Flatwave total between the two saw channels.

9. Tangential Saw Transfer

Theorem 9.1 — Equal-and-opposite transfer rates

$$\text{saw}'_r = \varepsilon \frac{q}{2}$$

and

$$\text{saw}'_x = -\varepsilon \frac{q}{2}.$$

Proof. From Corollary 8.2,

$$\text{saw}_r = \varepsilon \lambda.$$

Since ε is constant on the quadrant,

$$\text{saw}'_r = \varepsilon \lambda'.$$

Using

$$\lambda' = \frac{q}{2},$$

we get

$$\text{saw}'_r = \varepsilon \frac{q}{2}.$$

Likewise,

$$\text{saw}_x = \varepsilon(1 - \lambda),$$

so

$$\text{saw}'_x = -\varepsilon\lambda' = -\varepsilon \frac{q}{2}.$$

□

Corollary 9.2 — Flatwave tangential invariance

$$\boxed{\text{Flatwave}' = \text{saw}'_r + \text{saw}'_x = 0}.$$

The Flatwave is not itself a varying transfer channel within a quadrant. It is the fixed signed total under which the two saw components exchange magnitude.

Corollary 9.3 — Transfer-rate equality

$$\boxed{|\text{saw}'_r| = |\text{saw}'_x| = \frac{q}{2}}.$$

The two channels transfer at exactly equal magnitude and opposite direction.

10. Boundedness of Saw Transfer

Lemma 10.1 — Bounds on q

On every open quadrant,

$$\boxed{1 < q \leq \sqrt{2}}.$$

The upper bound is attained at the quadrant midpoint.

The lower value 1 is approached only at excluded quadrant boundaries.

Proof. Since

$$q = |\sin x| + |\cos x|,$$

we have

$$q^2 = \sin^2 x + \cos^2 x + 2|\sin x \cos x|.$$

Therefore,

$$q^2 = 1 + 2|sc|.$$

Inside an open quadrant,

$$|sc| > 0,$$

so

$$q^2 > 1.$$

Also,

$$2|sc| \leq s^2 + c^2 = 1.$$

Hence,

$$q^2 \leq 2.$$

Since $q > 0$,

$$1 < q \leq \sqrt{2}.$$

Equality $q = \sqrt{2}$ occurs exactly when

$$|\sin x| = |\cos x| = \frac{1}{\sqrt{2}},$$

which is the quadrant midpoint.

□

Theorem 10.2 — Bounded saw-transfer speed

$$\boxed{\frac{1}{2} < |\text{saw}'_r| = |\text{saw}'_x| \leq \frac{1}{\sqrt{2}}}.$$

Proof. By Corollary 9.3,

$$|\text{saw}'_r| = |\text{saw}'_x| = \frac{q}{2}.$$

Applying Lemma 10.1 gives

$$\frac{1}{2} < \frac{q}{2} \leq \frac{\sqrt{2}}{2}.$$

Since

$$\frac{\sqrt{2}}{2} = \frac{1}{\sqrt{2}},$$

the result follows. □

Corollary 10.3 — Maximum transfer point

The transfer-rate magnitude reaches its unique maximum at each quadrant midpoint:

$$x = \frac{(2k+1)\pi}{4}.$$

At that point,

$$\lambda = \frac{1}{2},$$

$$\text{saw}_r = \text{saw}_x = \frac{\varepsilon}{2},$$

and

$$|\text{saw}'_r| = |\text{saw}'_x| = \frac{1}{\sqrt{2}}.$$

Thus equality of the saw components and maximum transfer occur at the same admissible phase point.

11. Saw Curvature

Theorem 11.1 — Saw second derivatives

$$\boxed{\text{saw}''_r = \varepsilon \frac{p}{2}}$$

and

$$\boxed{\text{saw}''_x = -\varepsilon \frac{p}{2}}.$$

Proof. Differentiate the first-derivative forms:

$$\text{saw}'_r = \varepsilon \frac{q}{2}.$$

Therefore,

$$\text{saw}''_r = \varepsilon \frac{q'}{2}.$$

Since $q' = p$,

$$\text{saw}''_r = \varepsilon \frac{p}{2}.$$

The second identity follows similarly.

□

Corollary 11.2 — Saw differential equations

Both saw responses satisfy

$$y'' + y = \frac{\varepsilon}{2}.$$

Proof for saw_r. From

$$\text{saw}_r = \varepsilon \frac{1-p}{2}$$

and

$$\text{saw}''_r = \varepsilon \frac{p}{2},$$

we obtain

$$\text{saw}''_r + \text{saw}_r = \varepsilon \frac{p}{2} + \varepsilon \frac{1-p}{2} = \frac{\varepsilon}{2}.$$

The proof for saw_x is identical.

□

Theorem 11.3 — Unique inflection point

Each saw response has exactly one inflection point in every open quadrant, located at

$$x = \frac{(2k+1)\pi}{4}.$$

Proof. An inflection requires

$$\text{saw}_r'' = 0$$

or equivalently

$$p = 0.$$

Since p decreases strictly from 1 to -1 , it has exactly one zero in the quadrant.

The condition $p = 0$ is equivalent to

$$\alpha \cos x = \beta \sin x.$$

Taking magnitudes,

$$|\cos x| = |\sin x|.$$

This occurs at the quadrant midpoint

$$x = \frac{(2k+1)\pi}{4}.$$

□

Corollary 11.4 — Octant-local curvature

Within each open octant, p has fixed sign. Therefore, each saw response has fixed curvature sign on every open octant.

The phrase “quasi-linear” may be used only if it is explicitly defined as this octant-local monotonicity with fixed curvature sign. It does not mean that the saw functions are linear.

12. UNA Operators in Transfer Coordinates

Theorem 12.1 — Canonical UNA transfer forms

$$\text{urx} = \frac{\varepsilon}{\lambda}$$

and

$$\boxed{\text{uxp} = \frac{\varepsilon}{1 - \lambda}}.$$

Proof. Since

$$\text{saw}_r = \frac{1}{\text{urx}} = \varepsilon\lambda,$$

taking reciprocals gives

$$\text{urx} = \frac{1}{\varepsilon\lambda}.$$

Because

$$\varepsilon^2 = 1,$$

we have

$$\frac{1}{\varepsilon} = \varepsilon.$$

Thus,

$$\text{urx} = \frac{\varepsilon}{\lambda}.$$

The proof for uxp is identical.

□

Corollary 12.2 — UNA sum

$$\text{urx} + \text{uxp} = \varepsilon \left(\frac{1}{\lambda} + \frac{1}{1 - \lambda} \right).$$

Therefore,

$$\boxed{\text{urx} + \text{uxp} = \frac{\varepsilon}{\lambda(1 - \lambda)}}.$$

Lemma 12.3 — Transfer product relation

$$\boxed{\lambda(1 - \lambda) = \frac{|\sin 2x|}{4}}.$$

Proof. Using

$$\lambda = \frac{1-p}{2},$$

we obtain

$$\lambda(1-\lambda) = \frac{1-p}{2} \frac{1+p}{2}.$$

Therefore,

$$\lambda(1-\lambda) = \frac{1-p^2}{4}.$$

From S0.I,

$$1-p^2 = 2|sc|.$$

Thus,

$$\lambda(1-\lambda) = \frac{|sc|}{2}.$$

Since

$$|\sin 2x| = 2|sc|,$$

we conclude that

$$\lambda(1-\lambda) = \frac{|\sin 2x|}{4}.$$

□

Corollary 12.4 — Recovery of double-angle closure

Since

$$\varepsilon = \operatorname{sgn}(\sin 2x),$$

we have

$$\frac{\varepsilon}{|\sin 2x|} = \frac{1}{\sin 2x}.$$

Therefore,

$$\text{urx} + \text{uxp} = \frac{4}{\sin 2x}.$$

The double-angle theorem is thus equivalent to the bounded transfer-product relation.

13. UNA Tangential Responses

Theorem 13.1 — Unainverex phase derivative

$$\boxed{\text{urx}' = -\varepsilon \frac{q}{2\lambda^2}}.$$

Proof. From Theorem 12.1,

$$\text{urx} = \varepsilon \lambda^{-1}.$$

Differentiate:

$$\text{urx}' = -\varepsilon \lambda^{-2} \lambda'.$$

Since

$$\lambda' = \frac{q}{2},$$

we obtain

$$\text{urx}' = -\varepsilon \frac{q}{2\lambda^2}.$$

□

Theorem 13.2 — Unaexpon phase derivative

$$\boxed{\text{uxp}' = \varepsilon \frac{q}{2(1-\lambda)^2}}.$$

Proof. From

$$\text{uxp} = \varepsilon(1-\lambda)^{-1},$$

differentiate:

$$\text{uxp}' = \varepsilon(1-\lambda)^{-2} \lambda'.$$

Using $\lambda' = q/2$ gives the result.

□

Corollary 13.3 — UNA derivative signs

$$\text{sgn}(\text{urx}') = -\varepsilon$$

and

$$\text{sgn}(\text{uxp}') = \varepsilon.$$

The two UNA operators change in opposite phase directions within each quadrant.

Theorem 13.4 — Normalized UNA responses

$$\hat{T}[\text{urx}] = -\frac{q}{2\lambda}$$

and

$$\hat{T}[\text{uxp}] = \frac{q}{2(1-\lambda)}.$$

Proof. Divide Theorem 13.1 by

$$\text{urx} = \frac{\varepsilon}{\lambda}.$$

Then

$$\hat{T}[\text{urx}] = \frac{-\varepsilon q / (2\lambda^2)}{\varepsilon / \lambda} = -\frac{q}{2\lambda}.$$

The proof for uxp is identical.

□

Corollary 13.5 — UNA rate balance

The normalized UNA rate magnitudes are equal exactly when

$$\lambda = \frac{1}{2}.$$

This occurs at the quadrant midpoint, where both UNA magnitudes equal 2 and both saw components equal $\varepsilon/2$.

14. Differential Closure

Definition 14.1 — Generator derivation

On a fixed open quadrant, define the derivation

$$\mathcal{D}_z = -\frac{1+z^2}{2} \frac{d}{dz}.$$

For every rational function $F(z)$,

$$\frac{d}{dx}F(z(x)) = \mathcal{D}_z F.$$

Theorem 14.2 — Rational tangential closure

If

$$F(z) \in \mathbb{R}(z),$$

then

$$T[F] \in \mathbb{R}(z).$$

This result was established in S0.II and applies to all named primitive, UNA, saw, and Flatwave expressions.

Definition 14.3 — Transfer vector space

Define

$$\mathcal{V} = \text{span}_{\mathbb{R}}\{1, p, q\}.$$

Theorem 14.4 — Finite transfer closure

The vector space $\mathcal{V}$ is closed under phase differentiation.

Proof. The derivatives of its basis elements are

$$1' = 0,$$

$$p' = -q,$$

and

$$q' = p.$$

Each derivative belongs to

$$\text{span}_{\mathbb{R}}\{1, p, q\}.$$

Therefore,

$$T[\mathcal{V}] \subseteq \mathcal{V}.$$

□

Corollary 14.5 — Saw derivative closure

Because both saw responses are affine functions of p , every derivative of either saw response belongs to $\mathcal{V}$.

Their derivative sequence cycles through p and q , while their constant midpoint is fixed by $\varepsilon/2$.

15. Structural Interpretation

The exact mathematical results permit the following restricted statements.

15.1 Primitive envelopes

The four primitive operators have unbounded phase slopes near their inherited singular boundaries. Their normalized rates are exactly

$$\pm |\csc x| \quad \text{or} \quad \pm |\sec x|.$$

15.2 Reciprocal access

Reciprocal pairing reverses normalized tangential direction while preserving normalized rate magnitude.

15.3 Flatwave

Flatwave has no interior phase slope:

$$\text{Flatwave}' = 0.$$

It is a signed quadrant discriminator, not a continuously varying rate channel.

15.4 Saw transfer

The two saw components provide the bounded transfer geometry:

$$\text{saw}_r = \varepsilon\lambda, \quad \text{saw}_x = \varepsilon(1 - \lambda).$$

Their transfer is equal, opposite, monotone, and bounded.

15.5 Octant structure

The saw curvature changes sign only at the quadrant midpoint. Therefore, curvature sign is fixed within each octant.

These are structural statements about phase differentiation. Physical interpretation remains deferred.

16. Terminal Statement

S0.I derived the operator identities from the Pythagorean relation.

S0.II reduced the named calculus to one rational generator.

S0.III establishes the tangential geometry.

The primitive normalized responses are

$$\hat{T}[\text{sr}x] = -|\csc x|$$

$$\hat{T}[\text{sxp}] = |\csc x|$$

$$\hat{T}[\text{cxp}] = |\sec x|$$

$$\hat{T}[\text{cr}x] = -|\sec x|.$$

The bounded transfer coordinate satisfies

$$0 < \lambda < 1, \quad \lambda' > 0, \quad \lambda'' + \lambda = \frac{1}{2}.$$

The saw channels satisfy

$$\text{saw}_r = \varepsilon\lambda$$

and

$$\text{saw}_x = \varepsilon(1 - \lambda).$$

Their rates obey

$$\text{saw}'_r = -\text{saw}'_x$$

and

$$\frac{1}{2} < |\text{saw}'_r| = |\text{saw}'_x| \leq \frac{1}{\sqrt{2}}.$$

Their sum remains fixed:

$$\text{Flatwave}' = 0.$$

Thus the tangential core is:

unbounded primitive envelopes, reciprocal rate reversal, and bounded complementary transfer.

S0.IV

Phase Symmetry, Operator Orbits, and Octant Rotation

Book 0 - Formal Core

Abstract

This essay determines the discrete symmetry structure of the bounded phase operators.

Two transformations are introduced on the quotient phase circle

$$\mathbb{T} = \mathbb{R}/(2\pi\mathbb{Z}) :$$

$$R(x) = x + \frac{\pi}{2},$$

representing a quarter-turn, and

$$J(x) = -x,$$

representing reflection through the phase origin.

These transformations generate the dihedral group of the four-quadrant phase wheel. Throughout this corpus, D_4 denotes the order-eight group (called D_8 in some alternate conventions):

$$D_4 = \langle R, J \mid R^4 = J^2 = e, JRJ = R^{-1} \rangle.$$

Their action on the four primitive operators is derived exactly. A quarter-turn exchanges the sine and cosine families while preserving reciprocal position:

$$\text{srx} \leftrightarrow \text{crx}, \quad \text{sxp} \leftrightarrow \text{cxp}.$$

Phase reflection realizes reciprocal inversion:

$$\text{srx} \leftrightarrow \text{sxp}, \quad \text{cxp} \leftrightarrow \text{crx}.$$

The resulting permutation representation is not faithful. A half-turn acts trivially on every primitive operator because the complete operator family is π -periodic. The eight phase symmetries therefore reduce to a four-element operator symmetry group isomorphic to the Klein group:

$$D_4/\{e, R^2\} \cong V_4.$$

The actions on the UNA operators, saw responses, Flatwave, and bounded transfer coordinate are derived. Flatwave transforms as a sign representation: quarter-turn and phase reflection reverse its sign, while their composition preserves it.

Finally, the eighth-turn

$$H(x) = x + \frac{\pi}{4}$$

is shown to generate a cyclic action on the eight octants. It advances the dominant-operator label through

$$\text{srx} \rightarrow \text{cxp} \rightarrow \text{crx} \rightarrow \text{sxp} \rightarrow \text{srx}.$$

This octant action is combinatorial rather than an exact equality among shifted operator functions. The distinction between exact operator symmetry and dominance-label rotation is made explicit.

No physical, temporal, or observer interpretation is introduced.

0. Scope and Inheritance

0.1 Scope

This essay proves:

1. the exact quarter-turn transformation of the primitive operators;
2. the exact reflection transformation of the primitive operators;
3. the dihedral relations satisfied by the phase transformations;
4. the induced operator-permutation group;
5. the π -periodicity and half-turn degeneracy of the operator family;
6. transformation laws for UNA, saw, and Flatwave expressions;
7. transformation laws for the bounded transfer variables;
8. the cyclic octant action on dominant-operator labels;
9. the distinction between exact function symmetry and dominance rotation.

0.2 Non-scope

This essay does not:

- interpret reflection as a physical mirror universe;

- identify phase rotation with time evolution;
- introduce observers or reference frames;
- claim that an eighth-turn permutes the operator functions exactly;
- assign physical meaning to Flatwave sign;
- extend any function through an excluded boundary.

0.3 Inheritance

All definitions and theorems of S0.I–S0.III are inherited.

The admissible domain is

$$\mathcal{D} = \mathbb{R} \setminus \left\{ \frac{k\pi}{2} : k \in \mathbb{Z} \right\}.$$

The primitive operators are

$$\text{srx}(x) = |\csc x| + \cot x,$$

$$\text{sxp}(x) = |\csc x| - \cot x,$$

$$\text{cxp}(x) = |\sec x| + \tan x,$$

$$\text{crx}(x) = |\sec x| - \tan x.$$

The UNA operators are

$$\text{urx} = \text{srx} - \text{crx},$$

and

$$\text{uxp} = \text{cxp} - \text{sxp}.$$

The saw responses are

$$\text{saw}_r = \frac{1}{\text{urx}}, \quad \text{saw}_x = \frac{1}{\text{uxp}}.$$

Flatwave is

$$\text{Flatwave} = \text{saw}_r + \text{saw}_x = \text{sgn}(\sin 2x).$$

1. Elementary Rotation Formulas

Inherited theorem 1.1 — Angle addition

For real x and y ,

$$\sin(x + y) = \sin x \cos y + \cos x \sin y,$$

and

$$\cos(x + y) = \cos x \cos y - \sin x \sin y.$$

Lemma 1.2 — Quarter-turn formulas

For every admissible x ,

$$\sin\left(x + \frac{\pi}{2}\right) = \cos x$$

and

$$\cos\left(x + \frac{\pi}{2}\right) = -\sin x.$$

Proof. Using

$$\cos \frac{\pi}{2} = 0, \quad \sin \frac{\pi}{2} = 1,$$

the addition formulas give

$$\sin\left(x + \frac{\pi}{2}\right) = \sin x \cdot 0 + \cos x \cdot 1 = \cos x,$$

and

$$\cos\left(x + \frac{\pi}{2}\right) = \cos x \cdot 0 - \sin x \cdot 1 = -\sin x.$$

□

Corollary 1.3 — Quarter-turn quotient formulas

$$\tan\left(x + \frac{\pi}{2}\right) = -\cot x$$

and

$$\cot\left(x + \frac{\pi}{2}\right) = -\tan x.$$

Proof. Since $x \in \mathcal{D}$,

$$\tan\left(x + \frac{\pi}{2}\right) = \frac{\cos x}{-\sin x} = -\cot x.$$

The second identity follows by reciprocal exchange.

□

Corollary 1.4 — Quarter-turn absolute reciprocal formulas

$$\left|\csc\left(x + \frac{\pi}{2}\right)\right| = |\sec x|$$

and

$$\left|\sec\left(x + \frac{\pi}{2}\right)\right| = |\csc x|.$$

2. Quarter-Turn Operator Symmetry

Definition 2.1 — Quarter-turn transformation

Define the quotient action

$$R([x]) = \left[x + \frac{\pi}{2}\right]$$

on $\mathbb{T}$. In formulas involving the periodic operators, the shorter notation $Rx = x + \pi/2 \pmod{2\pi}$ is retained.

Theorem 2.2 — Quarter-turn action on Sininverex

$$\text{srx}(Rx) = \text{crx}(x).$$

Proof. By definition,

$$\text{srx}(Rx) = \left|\csc\left(x + \frac{\pi}{2}\right)\right| + \cot\left(x + \frac{\pi}{2}\right).$$

Using Corollaries 1.3–1.4,

$$\text{sr}\mathbf{x}(Rx) = |\sec x| - \tan x.$$

But

$$|\sec x| - \tan x = \text{cr}\mathbf{x}(x).$$

Therefore,

$$\text{sr}\mathbf{x}(Rx) = \text{cr}\mathbf{x}(x).$$

□

Theorem 2.3 — Complete quarter-turn action

$\begin{aligned}\text{sr}\mathbf{x}(Rx) &= \text{cr}\mathbf{x}(x), \\ \text{sxp}(Rx) &= \text{exp}(x), \\ \text{exp}(Rx) &= \text{sxp}(x), \\ \text{cr}\mathbf{x}(Rx) &= \text{sr}\mathbf{x}(x).\end{aligned}$
--

Proof. The first identity is Theorem 2.2.

For Sinexpon,

$$\text{sxp}(Rx) = \left| \csc \left(x + \frac{\pi}{2} \right) \right| - \cot \left(x + \frac{\pi}{2} \right).$$

Thus,

$$\text{sxp}(Rx) = |\sec x| + \tan x = \text{exp}(x).$$

Similarly,

$$\text{exp}(Rx) = \left| \sec \left(x + \frac{\pi}{2} \right) \right| + \tan \left(x + \frac{\pi}{2} \right)$$

becomes

$$\text{exp}(Rx) = |\csc x| - \cot x = \text{sxp}(x).$$

Finally,

$$\text{cr}\mathbf{x}(Rx) = |\csc x| + \cot x = \text{sr}\mathbf{x}(x).$$

□

Corollary 2.4 — Quarter-turn permutation

On the ordered primitive set

$$\mathcal{O} = \{\text{srx}, \text{sxp}, \text{cxp}, \text{crx}\}.$$

the quarter-turn induces the permutation

$$\boxed{\rho(R) = (\text{srx} \ \text{crx})(\text{sxp} \ \text{cxp})}.$$

This induced permutation has order 2, even though the phase transformation R has order 4.

3. Half-Turn Invariance

Lemma 3.1 — Half-turn trigonometric formulas

$$\sin(x + \pi) = -\sin x,$$

$$\cos(x + \pi) = -\cos x,$$

$$\tan(x + \pi) = \tan x,$$

and

$$\cot(x + \pi) = \cot x.$$

Furthermore,

$$|\csc(x + \pi)| = |\csc x|,$$

and

$$|\sec(x + \pi)| = |\sec x|.$$

Theorem 3.2 — Primitive half-turn invariance

For every primitive operator $f \in \mathcal{O}$,

$$\boxed{f(x + \pi) = f(x)}.$$

Proof. Each primitive consists of:

- an absolute reciprocal function, which is π -periodic; and
- either tangent or cotangent, which is also π -periodic.

Therefore each complete primitive is π -periodic.

□

Corollary 3.3 — Operator period

The displayed operator family has fundamental phase repetition no greater than π :

$$\mathcal{O}(x + \pi) = \mathcal{O}(x).$$

Corollary 3.4 — Half-turn kernel

Although

$$R^2(x) = x + \pi$$

is not the identity transformation on the 2π -phase circle, it acts trivially on every primitive operator:

$$\rho(R^2) = e.$$

The operator representation therefore does not distinguish phase positions separated by π .

4. Reflection and Reciprocal Inversion

Definition 4.1 — Phase reflection

Define the quotient action

$$J([x]) = [-x]$$

on $\mathbb{T}$. The shorter notation $Jx = -x \pmod{2\pi}$ is retained in operator formulas.

Lemma 4.2 — Reflection formulas

$$\sin(-x) = -\sin x, \quad \cos(-x) = \cos x,$$

$$\tan(-x) = -\tan x, \quad \cot(-x) = -\cot x.$$

Absolute reciprocal magnitudes satisfy

$$|\csc(-x)| = |\csc x|,$$

and

$$|\sec(-x)| = |\sec x|.$$

Theorem 4.3 — Reflection action on the primitive operators

$$\begin{aligned} \text{srx}(Jx) &= \text{sxp}(x), \\ \text{sxp}(Jx) &= \text{srx}(x), \\ \text{cxp}(Jx) &= \text{crx}(x), \\ \text{crx}(Jx) &= \text{cxp}(x). \end{aligned}$$

Proof. For Sininverex,

$$\text{srx}(-x) = |\csc(-x)| + \cot(-x).$$

Therefore,

$$\text{srx}(-x) = |\csc x| - \cot x = \text{sxp}(x).$$

The remaining identities follow by the same substitution.

□

Corollary 4.4 — Reflection permutation

Reflection induces

$$\rho(J) = (\text{srx} \ \text{sxp})(\text{cxp} \ \text{crx}).$$

Corollary 4.5 — Reciprocal inversion is geometrically realized

Since

$$\text{sxp} = \frac{1}{\text{srx}}$$

and

$$\text{crx} = \frac{1}{\text{cxp}},$$

phase reflection realizes reciprocal exchange:

$$f(-x) = \frac{1}{f(x)}$$

for each primary envelope $f \in \{\text{srx}, \text{exp}\}$.

This statement applies to the paired primary envelopes. It is not a general rule for every composite expression.

5. The Phase Symmetry Group

Theorem 5.1 — Dihedral relations

The phase transformations R and J satisfy

$$R^4 = e, \quad J^2 = e, \quad J R J = R^{-1}.$$

Proof. Four quarter-turns give

$$R^4(x) = x + 4\frac{\pi}{2} = x + 2\pi \equiv x.$$

Thus,

$$R^4 = e.$$

Two reflections give

$$J^2(x) = -(-x) = x,$$

so

$$J^2 = e.$$

Finally,

$$J R J(x) = J(R(-x)).$$

Now,

$$R(-x) = -x + \frac{\pi}{2}.$$

Applying J ,

$$J\left(-x + \frac{\pi}{2}\right) = x - \frac{\pi}{2}.$$

But

$$x - \frac{\pi}{2} = R^{-1}(x).$$

Therefore,

$$J R J = R^{-1}.$$

□

Corollary 5.2 — Quadrant phase group

The transformations generated by R and J form the dihedral group

$$D_4 = \langle R, J \mid R^4 = J^2 = e, J R J = R^{-1} \rangle.$$

Under the convention fixed above, it contains eight phase transformations:

$$e, R, R^2, R^3, J, R J, R^2 J, R^3 J.$$

6. The Induced Operator Group

Definition 6.1 — Operator representation

Let

$$\rho : D_4 \longrightarrow \text{Sym}(\mathcal{O})$$

assign to each phase symmetry its induced permutation of the primitive operator set.

Lemma 6.2 — Generator images

$$\rho(R) = (\text{srx crx})(\text{sxp exp}),$$

and

$$\rho(J) = (\text{srx sxp})(\text{exp crx}).$$

Both induced permutations have order 2.

Theorem 6.3 — Induced operator group

The image of ρ is

$$\boxed{\operatorname{im} \rho = \{e, Q, I, QI\}}.$$

where

$$Q = (\operatorname{srx} \ \operatorname{crx})(\operatorname{sxp} \ \operatorname{cxp}),$$

$$I = (\operatorname{srx} \ \operatorname{sxp})(\operatorname{cxp} \ \operatorname{crx}),$$

and

$$QI = (\operatorname{srx} \ \operatorname{cxp})(\operatorname{sxp} \ \operatorname{crx}).$$

These three nonidentity permutations satisfy

$$Q^2 = I^2 = (QI)^2 = e$$

and commute.

Therefore,

$$\boxed{\operatorname{im} \rho \cong V_4},$$

the Klein four-group.

Proof. The two generator images Q and I are distinct commuting involutions.

Their product is

$$QI = (\operatorname{srx} \ \operatorname{cxp})(\operatorname{sxp} \ \operatorname{crx}).$$

Thus the generated group contains exactly

$$e, \ Q, \ I, \ QI.$$

Every nonidentity element has order 2. This is the defining structure of V_4 .

□

Theorem 6.4 — Representation kernel

$$\ker \rho = \{e, R^2\}.$$

Proof. By half-turn invariance,

$$\rho(R^2) = e.$$

Thus,

$$\{e, R^2\} \subseteq \ker \rho.$$

The phase group has eight elements, while its image has four. By the homomorphism counting relation,

$$|\ker \rho| = \frac{|D_4|}{|\text{im } \rho|} = \frac{8}{4} = 2.$$

Therefore the kernel contains exactly the two already identified elements:

$$\ker \rho = \{e, R^2\}.$$

□

Corollary 6.5 — Quotient symmetry

$$D_4/\{e, R^2\} \cong V_4.$$

The primitive operators retain only half of the full phase-wheel symmetry information.

7. The Third Primitive Symmetry**Definition 7.1 — Diagonal reflection**

Define the composition

$$K = RJ,$$

with J applied first.

Then

$$K(x) = \frac{\pi}{2} - x.$$

Theorem 7.2 — Diagonal-reflection action

$$\begin{aligned} \text{srx}(Kx) &= \text{cxp}(x), \\ \text{sxp}(Kx) &= \text{crx}(x), \\ \text{cxp}(Kx) &= \text{srx}(x), \\ \text{crx}(Kx) &= \text{sxp}(x). \end{aligned}$$

Proof. The induced permutation is

$$\rho(K) = \rho(R)\rho(J) = QI.$$

The explicit permutation QI exchanges

$$\text{srx} \leftrightarrow \text{cxp}$$

and

$$\text{sxp} \leftrightarrow \text{crx}.$$

The same result follows by direct substitution using

$$\sin\left(\frac{\pi}{2} - x\right) = \cos x$$

and

$$\cos\left(\frac{\pi}{2} - x\right) = \sin x.$$

□

Corollary 7.3 — Operator orbit

The orbit of any primitive operator under the phase symmetry group contains all four primitive operators.

For example,

$$\text{srx} \xrightarrow{J} \text{sxp},$$

$$\text{srx} \xrightarrow{R} \text{crx},$$

and

$$\text{srx} \xrightarrow{K} \text{cxp}.$$

Thus the primitive family is one symmetry orbit.

8. Transformation of the UNA Operators

Theorem 8.1 — Quarter-turn UNA reversal

$$\boxed{\text{urx}(Rx) = -\text{urx}(x)}$$

and

$$\boxed{\text{uxp}(Rx) = -\text{uxp}(x)}.$$

Proof. Using the primitive quarter-turn transformations,

$$\text{urx}(Rx) = \text{srx}(Rx) - \text{crx}(Rx).$$

Therefore,

$$\text{urx}(Rx) = \text{crx}(x) - \text{srx}(x) = -\text{urx}(x).$$

Likewise,

$$\text{uxp}(Rx) = \text{sxp}(x) - \text{cxp}(x) = -\text{uxp}(x).$$

□

Theorem 8.2 — Reflection exchanges the UNA operators with sign reversal

$$\boxed{\text{urx}(Jx) = -\text{uxp}(x)}$$

and

$$\boxed{\text{uxp}(Jx) = -\text{urx}(x)}.$$

Proof. Under reflection,

$$\text{srx} \mapsto \text{sxp}, \quad \text{crx} \mapsto \text{cxp}.$$

Therefore,

$$\text{urx}(Jx) = \text{sxp} - \text{cxp} = -\text{uxp}.$$

The second identity follows similarly.

□

Theorem 8.3 — Diagonal reflection exchanges the UNA operators

$$\text{urx}(Kx) = \text{uxp}(x)$$

and

$$\text{uxp}(Kx) = \text{urx}(x).$$

There is no sign reversal under K .

9. Transformation of the Saw Responses**Theorem 9.1 — Quarter-turn saw reversal**

$$\text{saw}_r(Rx) = -\text{saw}_r(x)$$

and

$$\text{saw}_x(Rx) = -\text{saw}_x(x).$$

Proof. Quarter-turn negates both UNA denominators. Taking reciprocals preserves that negative factor.

□

Theorem 9.2 — Reflection exchanges the saw channels with sign reversal

$$\text{saw}_r(Jx) = -\text{saw}_x(x)$$

and

$$\text{saw}_x(Jx) = -\text{saw}_r(x).$$

Theorem 9.3 — Diagonal reflection exchanges the saw channels

$$\text{saw}_r(Kx) = \text{saw}_x(x)$$

and

$$\text{saw}_x(Kx) = \text{saw}_r(x).$$

10. Flatwave as a Sign Representation

Theorem 10.1 — Quarter-turn Flatwave reversal

$$\text{Flatwave}(Rx) = -\text{Flatwave}(x).$$

Proof. Since

$$\sin(2x + \pi) = -\sin 2x,$$

we have

$$\text{sgn}(\sin 2Rx) = -\text{sgn}(\sin 2x).$$

The result follows from the exact Flatwave theorem. □

Theorem 10.2 — Reflection Flatwave reversal

$$\text{Flatwave}(Jx) = -\text{Flatwave}(x).$$

Proof.

$$\sin(-2x) = -\sin 2x.$$

Therefore its sign reverses. □

Theorem 10.3 — Diagonal-reflection invariance

$$\text{Flatwave}(Kx) = \text{Flatwave}(x).$$

Proof. Since

$$Kx = \frac{\pi}{2} - x,$$

we have

$$\sin(2Kx) = \sin(\pi - 2x) = \sin 2x.$$

Therefore Flatwave is unchanged. □

Definition 10.4 — Flatwave character

Define

$$\chi : D_4 \rightarrow \{-1, +1\}$$

by

$$\text{Flatwave}(gx) = \chi(g) \text{Flatwave}(x).$$

The generators satisfy

$$\chi(R) = -1, \quad \chi(J) = -1.$$

Therefore,

$$\boxed{\chi(R^a J^b) = (-1)^{a+b}.$$

Corollary 10.5 — Flatwave-preserving subgroup

The phase symmetries that preserve Flatwave form the four-element subgroup

$$\boxed{\ker \chi = \{e, R^2, RJ, R^3 J\}.$$

Quarter-turn and origin reflection reverse quadrant orientation; their composition preserves it.

11. Transformation of the Transfer Coordinates

Recall from S0.III:

$$\alpha = \text{sgn}(\sin x), \quad \beta = \text{sgn}(\cos x),$$

$$\varepsilon = \alpha\beta,$$

$$p = \alpha \cos x - \beta \sin x,$$

$$q = \alpha \sin x + \beta \cos x,$$

and

$$\lambda = \frac{1-p}{2}.$$

Theorem 11.1 — Quarter-turn transfer invariance

Under R ,

$$\boxed{p(Rx) = p(x),}$$

$$\boxed{q(Rx) = q(x),}$$

$$\boxed{\lambda(Rx) = \lambda(x),}$$

while

$$\boxed{\varepsilon(Rx) = -\varepsilon(x).}$$

Proof. Under the quarter-turn,

$$\sin Rx = \cos x, \quad \cos Rx = -\sin x.$$

Therefore,

$$\alpha_R = \beta, \quad \beta_R = -\alpha.$$

Then

$$p(Rx) = \alpha_R \cos Rx - \beta_R \sin Rx.$$

Substituting,

$$p(Rx) = \beta(-\sin x) - (-\alpha)(\cos x).$$

Thus,

$$p(Rx) = \alpha \cos x - \beta \sin x = p(x).$$

Similarly,

$$q(Rx) = \beta \cos x + (-\alpha)(-\sin x) = q(x).$$

Because λ depends only on p , it is unchanged.

Finally,

$$\varepsilon_R = \alpha_R \beta_R = \beta(-\alpha) = -\varepsilon.$$

□

Theorem 11.2 — Reflection transfer complement

Under J ,

$$\boxed{p(Jx) = -p(x),}$$

$$\boxed{q(Jx) = q(x),}$$

$$\boxed{\lambda(Jx) = 1 - \lambda(x),}$$

and

$$\boxed{\varepsilon(Jx) = -\varepsilon(x).}$$

Proof. Under reflection,

$$\sin(-x) = -\sin x, \quad \cos(-x) = \cos x,$$

so

$$\alpha_J = -\alpha, \quad \beta_J = \beta.$$

Therefore,

$$p(Jx) = (-\alpha) \cos x - \beta(-\sin x) = -p(x).$$

Likewise,

$$q(Jx) = (-\alpha)(-\sin x) + \beta \cos x = q(x).$$

Hence,

$$\lambda(Jx) = \frac{1+p}{2} = 1 - \lambda(x).$$

Also,

$$\varepsilon_J = (-\alpha)\beta = -\varepsilon.$$

□

Corollary 11.3 — Diagonal-reflection transfer complement

Under $K = RJ$,

$$p(Kx) = -p(x),$$

$$q(Kx) = q(x),$$

$$\lambda(Kx) = 1 - \lambda(x),$$

while

$$\varepsilon(Kx) = \varepsilon(x).$$

Thus K exchanges the two transfer channels without changing their signed total.

12. Exact Symmetry and Symmetric Invariants**Theorem 12.1 — Primitive-set invariance**

Every phase symmetry in D_4 permutes the primitive operator set without introducing a new primitive function.

Therefore, the unordered set

$$\{\text{srx}, \text{sxp}, \text{cxp}, \text{crx}\}$$

is invariant under D_4 .

Corollary 12.2 — Symmetric-expression invariance

Every symmetric rational expression in the four primitive operators is invariant under the induced operator group.

Examples include

$$\text{srx} + \text{sxp} + \text{cxp} + \text{crx},$$

and

$$\text{srx} \text{ sxp} \text{ cxp} \text{ crx} .$$

Corollary 12.3 — Total primitive product

$$\boxed{\text{srx sxp exp crx} = 1}.$$

This follows from the two reciprocal pairs and is invariant under every phase symmetry.

13. Octant Rotation**Definition 13.1 — Open octants**

For $j \in \mathbb{Z}/8\mathbb{Z}$, define

$$O_j = \left(\frac{j\pi}{4}, \frac{(j+1)\pi}{4} \right).$$

Definition 13.2 — Eighth-turn transformation

Define

$$\boxed{H(x) = x + \frac{\pi}{4} \pmod{2\pi}.}$$

Then

$$H(O_j) = O_{j+1}.$$

Theorem 13.3 — Octant rotation group

The transformation H satisfies

$$H^8 = e.$$

It generates a cyclic action on the eight octants:

$$\boxed{\langle H \rangle \cong C_8.}$$

Proof. Eight eighth-turns give

$$H^8(x) = x + 8\frac{\pi}{4} = x + 2\pi \equiv x.$$

No smaller positive power fixes every phase position modulo 2π . Therefore H has order 8.

□

14. Dominance-Label Rotation

Definition 14.1 — Dominant-operator map

For each open octant, let $W(O_j)$ denote its unique magnitude-dominant primitive operator.

From S0.I,

$$W(O_0) = \text{srx},$$

$$W(O_1) = \text{cxp},$$

$$W(O_2) = \text{crx},$$

$$W(O_3) = \text{sxp},$$

with repetition after four octants.

Definition 14.2 — Winner-cycle permutation

Define

$$C = (\text{srx} \ \text{cxp} \ \text{crx} \ \text{sxp}).$$

Thus,

$$C(\text{srx}) = \text{cxp},$$

$$C(\text{cxp}) = \text{crx},$$

$$C(\text{crx}) = \text{sxp},$$

and

$$C(\text{sxp}) = \text{srx}.$$

Theorem 14.3 — Dominance advancement

For every open octant,

$$W(HO_j) = C(W(O_j)).$$

Proof. Since

$$H(O_j) = O_{j+1},$$

the result follows directly from the proved octant sequence:

$$\text{srx} \rightarrow \text{cxp} \rightarrow \text{crx} \rightarrow \text{sxp} \rightarrow \text{srx}.$$

□

Corollary 14.4 — Dominance quotient

The eighth-turn phase action has order 8, while the winner-cycle permutation has order 4:

$$C^4 = e.$$

The half-turn

$$H^4(x) = x + \pi$$

acts trivially on winner labels.

Therefore, the dominance-label action factors through

$$\boxed{C_8 / \{e, H^4\} \cong C_4.}$$

15. Exact Operator Symmetry Versus Dominance Rotation

Theorem 15.1 — Quarter-turn consistency

A quarter-turn equals two eighth-turns:

$$R = H^2.$$

The dominance permutation generated by two octant steps is

$$C^2 = (\text{srx} \text{ crx})(\text{cxp} \text{ sxp}).$$

This is exactly the primitive quarter-turn permutation $\rho(R)$.

Thus exact operator symmetry and dominance-label rotation agree at the quarter-turn scale.

Theorem 15.2 — Eighth-turn is not generally an exact primitive permutation

In general,

$$\boxed{\text{srx} \left(x + \frac{\pi}{4} \right) \neq \text{cxp}(x).}$$

Therefore, the winner-cycle permutation C is not an exact function-substitution symmetry.

Proof. In the first quadrant,

$$\text{srx}\left(x + \frac{\pi}{4}\right) = \cot\left(\frac{x}{2} + \frac{\pi}{8}\right),$$

while

$$\exp(x) = \tan\left(\frac{\pi}{4} + \frac{x}{2}\right).$$

Equality would require

$$\cot\left(\frac{x}{2} + \frac{\pi}{8}\right) = \tan\left(\frac{\pi}{4} + \frac{x}{2}\right).$$

Using

$$\cot u = \tan\left(\frac{\pi}{2} - u\right),$$

this becomes

$$\tan\left(\frac{3\pi}{8} - \frac{x}{2}\right) = \tan\left(\frac{\pi}{4} + \frac{x}{2}\right).$$

Within the relevant interval, equality requires

$$\frac{3\pi}{8} - \frac{x}{2} = \frac{\pi}{4} + \frac{x}{2}.$$

Therefore,

$$x = \frac{\pi}{8}.$$

Thus the equality holds only at the midpoint of the first octant, not throughout the octant.

Hence an eighth-turn advances the dominant label but does not generally permute the primitive functions exactly.

□

Corollary 15.3 — Two distinct symmetry layers

The bounded phase calculus contains two related but distinct discrete structures:

1. Exact operator symmetry: generated by quarter-turn and reflection, with induced group V_4 .
2. Dominance-label rotation: generated by the eighth-turn, with induced cycle C_4 .

They agree when an eighth-turn is applied twice, but not after a single eighth-turn.

16. Terminal Statement

S0.IV establishes the discrete symmetry architecture of the bounded phase operators.

The phase wheel carries the dihedral symmetry

$$D_4 = \langle R, J \mid R^4 = J^2 = e, J R J = R^{-1} \rangle.$$

The primitive operators provide a nonfaithful representation of this group. The half-turn acts trivially, so the induced operator group is

$$D_4 / \{e, R^2\} \cong V_4.$$

Quarter-turn acts by

$$\text{srx} \leftrightarrow \text{crx}, \quad \text{sxp} \leftrightarrow \text{cxp}.$$

Reflection acts by reciprocal exchange:

$$\text{srx} \leftrightarrow \text{sxp}, \quad \text{cxp} \leftrightarrow \text{crx}.$$

Diagonal reflection exchanges the sine and cosine families:

$$\text{srx} \leftrightarrow \text{cxp}, \quad \text{sxp} \leftrightarrow \text{crx}.$$

Flatwave transforms according to a sign character:

$$\text{Flatwave}(Rx) = -\text{Flatwave}(x),$$

$$\text{Flatwave}(Jx) = -\text{Flatwave}(x),$$

and

$$\text{Flatwave}(Kx) = \text{Flatwave}(x).$$

The transfer fraction obeys

$$\lambda(Rx) = \lambda(x),$$

$$\lambda(Jx) = 1 - \lambda(x),$$

and

$$\lambda(Kx) = 1 - \lambda(x).$$

Finally, the eighth-turn advances the dominant operator through

$$\text{srx} \rightarrow \text{cxp} \rightarrow \text{crx} \rightarrow \text{sxp} \rightarrow \text{srx}.$$

This is an exact rotation of dominance labels, but not generally an exact permutation of the shifted primitive functions.

The resulting structure is:

dihedral phase symmetry, Klein operator symmetry, and cyclic octant dominance.

S0.V

Boundary Structure, Singular Limits, and Admissible Extension

Book 0 - Formal Core

Abstract

The bounded phase operators are defined on the punctured phase domain

$$\mathcal{D} = \mathbb{R} \setminus \left\{ \frac{k\pi}{2} : k \in \mathbb{Z} \right\}.$$

The excluded points divide the phase line into open quadrants. The preceding essays established exact algebraic, differential, transfer, and symmetry structures inside those quadrants. This essay determines what the core calculus establishes at their boundaries.

The primitive operators exhibit alternating zero–pole exchanges. At a sine-zero boundary, the Sininverex–Sinexpon pair exchanges 0 and $+\infty$, while the cosine-family pair tends to 1. At a cosine-zero boundary, the Cosinexpon–Cosininverex pair exchanges $+\infty$ and 0, while the sine-family pair tends to 1.

Within every open quadrant, the transfer coordinate

$$\lambda = \frac{1-p}{2}$$

increases continuously from 0 to 1. It therefore extends continuously to the corresponding closed quadrant. The saw responses extend with it:

$$\text{saw}_r = \varepsilon\lambda, \quad \text{saw}_x = \varepsilon(1-\lambda),$$

so the signed unit total is transferred from the expon saw channel to the inverex saw channel.

At a common boundary between adjacent quadrants, however, the Flatwave sign reverses. The one-sided limits satisfy

$$\lim_{x \rightarrow b^-} \text{Flatwave}(x) = - \lim_{x \rightarrow b^+} \text{Flatwave}(x).$$

Consequently, Flatwave has no continuous extension to the ordinary phase circle. The individual saw responses also possess finite but unequal one-sided limits at each boundary.

A split-boundary extension domain is introduced in which every boundary has a distinct entering and exiting copy. All bounded transfer quantities extend continuously to this enlarged domain. Identifying those copies requires an additional seam rule. The resulting countable disjoint union is not asserted to be compact as a whole.

The core calculus therefore determines one-sided limits but does not determine boundary-crossing dynamics.

0. Scope and Inheritance

0.1 Scope

This essay proves:

1. the complete one-sided boundary limits of the primitive operators;
2. the first-order asymptotic behavior of the zero–pole exchanges;
3. the one-sided limits of the UNA operators;
4. the finite boundary limits of the saw responses;
5. continuous compactification of the transfer coordinate on each quadrant;
6. the Flatwave sign-jump theorem;
7. the impossibility of a global continuous Flatwave extension on the ordinary phase circle;
8. continuous extension on a split-boundary phase domain, without claiming global compactness;
9. the underdetermination of any boundary-crossing law.

0.2 Non-scope

This essay does not:

- define a physical phase transition;
- prescribe what happens when a trajectory reaches an excluded point;
- identify divergence with infinite physical energy or mass;
- identify a sign reversal with annihilation, creation, or temporal reversal;
- introduce a new evolution law;
- remove the original exclusions from the analytic definitions.

0.3 Inheritance

All definitions and theorems of S0.I–S0.IV are inherited.

In particular, the primitive operators are

$$\text{srx}(x) = |\csc x| + \cot x,$$

$$\text{sxp}(x) = |\csc x| - \cot x = \frac{1}{\text{srx}(x)},$$

$$\text{cxp}(x) = |\sec x| + \tan x,$$

and

$$\text{crx}(x) = |\sec x| - \tan x = \frac{1}{\text{cxp}(x)}.$$

The UNA operators are

$$\text{urx} = \text{srx} - \text{crx},$$

and

$$\text{uxp} = \text{cxp} - \text{sxp}.$$

The saw responses are

$$\text{saw}_r = \frac{1}{\text{urx}}, \quad \text{saw}_x = \frac{1}{\text{uxp}}.$$

Flatwave is

$$\text{Flatwave} = \text{saw}_r + \text{saw}_x = \text{sgn}(\sin 2x).$$

On every open quadrant,

$$\text{saw}_r = \varepsilon \lambda,$$

and

$$\text{saw}_x = \varepsilon(1 - \lambda),$$

where

$$\varepsilon = \text{sgn}(\sin 2x)$$

is constant on that quadrant and

$$0 < \lambda < 1.$$

1. Boundary Indexing

Definition 1.1 — Boundary sequence

For each integer k , define

$$b_k = \frac{k\pi}{2}.$$

The forbidden set is

$$\mathbb{B} = \{b_k : k \in \mathbb{Z}\}.$$

Definition 1.2 — Open quadrants

Define

$$Q_k = (b_k, b_{k+1}).$$

Then

$$\mathcal{D} = \bigcup_{k \in \mathbb{Z}} Q_k.$$

The quadrants are pairwise disjoint connected components of $\mathcal{D}$.

Definition 1.3 — Quadrant sign

For $x \in Q_k$, define

$$\varepsilon_k = \operatorname{sgn}(\sin 2x).$$

Since

$$2x \in (k\pi, (k+1)\pi),$$

the sine function has constant sign on that interval. Therefore,

$$\varepsilon_k = (-1)^k.$$

Adjacent quadrants have opposite signs:

$$\varepsilon_{k+1} = -\varepsilon_k.$$

Definition 1.4 — Boundary types

A boundary b_k is called a sine boundary when k is even. Then

$$b_k = m\pi$$

for some integer m , and

$$\sin b_k = 0.$$

A boundary b_k is called a cosine boundary when k is odd. Then

$$b_k = \frac{\pi}{2} + m\pi,$$

and

$$\cos b_k = 0.$$

2. Standard Small-Angle Limits**Inherited limit 2.1**

As $h \rightarrow 0$,

$$\frac{\sin h}{h} \rightarrow 1.$$

Equivalently,

$$\sin h \sim h.$$

Inherited limit 2.2

As $h \rightarrow 0$,

$$1 - \cos h \sim \frac{h^2}{2},$$

while

$$1 + \cos h \rightarrow 2.$$

Definition 2.3 — Asymptotic equivalence

For functions f and g , the notation

$$f(h) \sim g(h) \quad \text{as } h \rightarrow 0$$

means

$$\lim_{h \rightarrow 0} \frac{f(h)}{g(h)} = 1.$$

These standard limits will be used to determine the order of each boundary zero or pole.

3. Primitive Limits at Sine Boundaries**Theorem 3.1 — Sine-family one-sided limits**

Let

$$b = m\pi$$

be a sine boundary. Then

$$\lim_{x \rightarrow b^-} \text{srx}(x) = 0,$$

$$\lim_{x \rightarrow b^+} \text{srx}(x) = +\infty,$$

$$\lim_{x \rightarrow b^-} \text{sxp}(x) = +\infty,$$

and

$$\lim_{x \rightarrow b^+} \text{sxp}(x) = 0.$$

Proof. Because all primitive operators are π -periodic, it is sufficient to examine $b = 0$.

First approach from the right. Let

$$x = h, \quad h \rightarrow 0^+.$$

Then

$$\text{srx}(h) = \csc h + \cot h = \frac{1 + \cos h}{\sin h}.$$

Therefore,

$$\text{srx}(h) \sim \frac{2}{h},$$

so

$$\text{srx}(h) \rightarrow +\infty.$$

Since

$$\text{sxp} = \frac{1}{\text{srx}},$$

it follows that

$$\text{sxp}(h) \rightarrow 0.$$

Now approach from the left. Let

$$x = -h, \quad h \rightarrow 0^+.$$

Then

$$|\csc(-h)| = \csc h$$

and

$$\cot(-h) = -\cot h.$$

Thus,

$$\text{srx}(-h) = \csc h - \cot h = \frac{1 - \cos h}{\sin h}.$$

Using the standard limits,

$$\text{srx}(-h) \sim \frac{h^2/2}{h} = \frac{h}{2}.$$

Therefore,

$$\text{srx}(-h) \rightarrow 0.$$

Taking reciprocals gives

$$\text{sxp}(-h) \rightarrow +\infty.$$

The same limits hold at every $m\pi$ by π -periodicity.

□

Theorem 3.2 — Cosine-family limits at sine boundaries

At every sine boundary $b = m\pi$,

$$\lim_{x \rightarrow b^\pm} \text{cxp}(x) = 1$$

and

$$\lim_{x \rightarrow b^\pm} \text{crx}(x) = 1.$$

Proof. As $x \rightarrow m\pi$,

$$|\sec x| \rightarrow 1, \quad \tan x \rightarrow 0.$$

Therefore,

$$\text{cxp}(x) = |\sec x| + \tan x \rightarrow 1.$$

Likewise,

$$\text{crx}(x) = |\sec x| - \tan x \rightarrow 1.$$

□

Corollary 3.3 — Sine-boundary primitive table

At every sine boundary,

	$x \rightarrow b^-$	$x \rightarrow b^+$
srx	0	$+\infty$
sxp	$+\infty$	0
cxp	1	1
crx	1	1

4. Primitive Limits at Cosine Boundaries

Theorem 4.1 — Cosine-family one-sided limits

Let

$$b = \frac{\pi}{2} + m\pi$$

be a cosine boundary. Then

$$\lim_{x \rightarrow b^-} \text{csp}(x) = +\infty,$$

$$\lim_{x \rightarrow b^+} \text{csp}(x) = 0,$$

$$\lim_{x \rightarrow b^-} \text{crs}(x) = 0,$$

and

$$\lim_{x \rightarrow b^+} \text{crs}(x) = +\infty.$$

Proof. By π -periodicity, it is sufficient to examine $b = \pi/2$.

Approach from the left. Let

$$x = \frac{\pi}{2} - h, \quad h \rightarrow 0^+.$$

Then

$$\cos x = \sin h, \quad \sin x = \cos h.$$

Therefore,

$$\text{csp}(x) = \frac{1 + \sin x}{\cos x} = \frac{1 + \cos h}{\sin h}.$$

Hence,

$$\text{csp}(x) \sim \frac{2}{h},$$

so

$$\text{csp}(x) \rightarrow +\infty.$$

Its reciprocal satisfies

$$\text{crs}(x) \rightarrow 0.$$

Approach from the right. Let

$$x = \frac{\pi}{2} + h.$$

Then

$$\cos x = -\sin h, \quad \sin x = \cos h.$$

Since

$$|\sec x| = \frac{1}{\sin h}$$

and

$$\tan x = -\frac{\cos h}{\sin h},$$

we have

$$\exp(x) = \frac{1 - \cos h}{\sin h}.$$

Thus,

$$\exp(x) \sim \frac{h}{2},$$

so

$$\exp(x) \rightarrow 0.$$

Its reciprocal therefore diverges:

$$\exp(x) \rightarrow +\infty.$$

□

Theorem 4.2 — Sine-family limits at cosine boundaries

At every cosine boundary,

$$\lim_{x \rightarrow b^\pm} \operatorname{srx}(x) = 1$$

and

$$\lim_{x \rightarrow b^\pm} \operatorname{sxp}(x) = 1.$$

Proof. As $x \rightarrow \pi/2 + m\pi$,

$$|\csc x| \rightarrow 1, \quad \cot x \rightarrow 0.$$

Therefore,

$$\text{sr}x \rightarrow 1$$

and

$$\text{sx}p \rightarrow 1.$$

□

Corollary 4.3 — Cosine-boundary primitive table

At every cosine boundary,

	$x \rightarrow b^-$	$x \rightarrow b^+$
sr x	1	1
sx p	1	1
cx p	$+\infty$	0
cr x	0	$+\infty$

5. First-Order Primitive Asymptotics

Theorem 5.1 — Sine-boundary zero–pole scale

Let $b = m\pi$ and $h \rightarrow 0^+$. Then

$$\text{sr}x(b - h) \sim \frac{h}{2},$$

$$\text{sx}p(b - h) \sim \frac{2}{h},$$

$$\text{sr}x(b + h) \sim \frac{2}{h},$$

and

$$\text{sx}p(b + h) \sim \frac{h}{2}.$$

Proof. These asymptotic relations were obtained in the proof of Theorem 3.1. The reciprocal relations follow because

$$\text{srx sxp} = 1.$$

□

Theorem 5.2 — Cosine-boundary zero–pole scale

Let

$$b = \frac{\pi}{2} + m\pi$$

and $h \rightarrow 0^+$. Then

$$\boxed{\text{cxp}(b - h) \sim \frac{2}{h},}$$

$$\boxed{\text{crx}(b - h) \sim \frac{h}{2},}$$

$$\boxed{\text{cxp}(b + h) \sim \frac{h}{2},}$$

and

$$\boxed{\text{crx}(b + h) \sim \frac{2}{h}.}$$

Corollary 5.3 — Reciprocal pole order

Every primitive divergence at an excluded boundary is a first-order pole in the distance to the boundary.

Every reciprocal zero vanishes linearly in that same distance.

The products

$$\text{srx sxp} = 1$$

and

$$\text{cxp crx} = 1$$

remain exact throughout the approach, even though neither factor is defined at the boundary itself.

6. Transfer Coordinate at Quadrant Boundaries

Fix an open quadrant

$$Q_k = (b_k, b_{k+1})$$

and write

$$\varepsilon = \varepsilon_k.$$

Theorem 6.1 — Transfer-coordinate endpoint limits

Within Q_k ,

$$\lim_{x \rightarrow b_k^+} \lambda(x) = 0$$

and

$$\lim_{x \rightarrow b_{k+1}^-} \lambda(x) = 1.$$

Proof. From S0.III,

$$\lambda = \frac{1-p}{2},$$

where p decreases continuously from 1 to -1 across every open quadrant.

Therefore,

$$p \rightarrow 1 \quad \text{as } x \rightarrow b_k^+,$$

which gives

$$\lambda \rightarrow 0.$$

Likewise,

$$p \rightarrow -1 \quad \text{as } x \rightarrow b_{k+1}^-,$$

which gives

$$\lambda \rightarrow 1.$$

□

Theorem 6.2 — First-order transfer asymptotics

Let $h \rightarrow 0^+$. Then

$$\lambda(b_k + h) \sim \frac{h}{2}$$

and

$$1 - \lambda(b_{k+1} - h) \sim \frac{h}{2}.$$

Proof. From S0.III,

$$\lambda' = \frac{q}{2},$$

where

$$q = |\sin x| + |\cos x|.$$

At either boundary of a quadrant,

$$q \rightarrow 1.$$

Therefore,

$$\lambda' \rightarrow \frac{1}{2}$$

at the entering boundary.

Since $\lambda \rightarrow 0$ there,

$$\lambda(b_k + h) \sim \frac{h}{2}.$$

At the exiting boundary,

$$(1 - \lambda)' = -\lambda' \rightarrow -\frac{1}{2}.$$

Since $1 - \lambda \rightarrow 0$,

$$1 - \lambda(b_{k+1} - h) \sim \frac{h}{2}.$$

□

7. UNA Boundary Structure

Theorem 7.1 — Quadrant-relative UNA limits

Within the quadrant Q_k , with sign $\varepsilon = \varepsilon_k$,

$$\lim_{x \rightarrow b_k^+} \text{urx}(x) = \varepsilon \infty,$$

$$\lim_{x \rightarrow b_k^+} \text{uxp}(x) = \varepsilon,$$

$$\lim_{x \rightarrow b_{k+1}^-} \text{urx}(x) = \varepsilon,$$

and

$$\lim_{x \rightarrow b_{k+1}^-} \text{uxp}(x) = \varepsilon \infty.$$

Here $\varepsilon \infty$ denotes divergence with sign ε .

Proof. From S0.III,

$$\text{urx} = \frac{\varepsilon}{\lambda},$$

and

$$\text{uxp} = \frac{\varepsilon}{1 - \lambda}.$$

At the entering boundary,

$$\lambda \rightarrow 0^+, \quad 1 - \lambda \rightarrow 1.$$

Thus,

$$\text{urx} \rightarrow \varepsilon \infty$$

and

$$\text{uxp} \rightarrow \varepsilon.$$

At the exiting boundary,

$$\lambda \rightarrow 1, \quad 1 - \lambda \rightarrow 0^+.$$

Therefore,

$$\text{urx} \rightarrow \varepsilon$$

and

$$\text{uxp} \rightarrow \varepsilon\infty.$$

□

Corollary 7.2 — UNA divergence transfer

Across every open quadrant, the first-order pole transfers from Unainverex to Unaexpon:

$$(\varepsilon\infty, \varepsilon) \longrightarrow (\varepsilon, \varepsilon\infty).$$

The two operators retain the same sign throughout the quadrant.

Theorem 7.3 — UNA first-order asymptotics

Let $h \rightarrow 0^+$. Within Q_k ,

$$\text{urx}(b_k + h) \sim \frac{2\varepsilon}{h},$$

and

$$\text{uxp}(b_{k+1} - h) \sim \frac{2\varepsilon}{h}.$$

Proof. Use

$$\text{urx} = \frac{\varepsilon}{\lambda}$$

with

$$\lambda(b_k + h) \sim \frac{h}{2}.$$

Then

$$\text{urx}(b_k + h) \sim \frac{\varepsilon}{h/2} = \frac{2\varepsilon}{h}.$$

The proof for uxp is identical.

□

8. Saw Boundary Structure

Theorem 8.1 — Quadrant-relative saw limits

Within Q_k ,

$$\lim_{x \rightarrow b_k^+} \text{saw}_r(x) = 0,$$

$$\lim_{x \rightarrow b_k^+} \text{saw}_x(x) = \varepsilon_k,$$

$$\lim_{x \rightarrow b_{k+1}^-} \text{saw}_r(x) = \varepsilon_k,$$

and

$$\lim_{x \rightarrow b_{k+1}^-} \text{saw}_x(x) = 0.$$

Proof. Use

$$\text{saw}_r = \varepsilon_k \lambda$$

and

$$\text{saw}_x = \varepsilon_k(1 - \lambda).$$

The result follows from the endpoint limits of λ .

□

Corollary 8.2 — Signed unit transfer

Across every open quadrant,

$$(\text{saw}_r, \text{saw}_x) : (0, \varepsilon_k) \longrightarrow (\varepsilon_k, 0).$$

Thus the complete signed unit weight passes continuously from the expon saw channel to the inverex saw channel.

Theorem 8.3 — Saw first-order asymptotics

Let $h \rightarrow 0^+$. Then

$$\text{saw}_r(b_k + h) \sim \varepsilon_k \frac{h}{2},$$

$$\text{saw}_x(b_k + h) = \varepsilon_k \left(1 - \frac{h}{2} + o(h) \right),$$

$$\text{saw}_r(b_{k+1} - h) = \varepsilon_k \left(1 - \frac{h}{2} + o(h) \right),$$

and

$$\text{saw}_x(b_{k+1} - h) \sim \varepsilon_k \frac{h}{2}.$$

Corollary 8.4 — No saw divergence

Neither saw response diverges as an excluded boundary is approached from within a quadrant.

The saw expressions remain undefined at the boundary under their original reciprocal definitions, but their quadrant-relative limits are finite.

9. Two-Sided Saw Limits at a Common Boundary

Let b_k be a common boundary of Q_{k-1} and Q_k .

Since

$$\varepsilon_{k-1} = -\varepsilon_k,$$

the two neighboring quadrants carry opposite Flatwave signs.

Theorem 9.1 — Two-sided saw limit table

At b_k ,

$$\boxed{\lim_{x \rightarrow b_k^-} \text{saw}_r(x) = \varepsilon_{k-1} = -\varepsilon_k}.$$

while

$$\boxed{\lim_{x \rightarrow b_k^+} \text{saw}_r(x) = 0}.$$

Likewise,

$$\lim_{x \rightarrow b_k^-} \text{saw}_x(x) = 0,$$

while

$$\lim_{x \rightarrow b_k^+} \text{saw}_x(x) = \varepsilon_k.$$

Proof. The left side of b_k is the exiting boundary of Q_{k-1} . Apply Theorem 8.1 with sign ε_{k-1} . The right side is the entering boundary of Q_k . Apply Theorem 8.1 with sign ε_k . □

Corollary 9.2 — No global continuous saw extension

Neither individual saw response has equal left and right limits at a boundary.

Therefore, neither

$$\text{saw}_r$$

nor

$$\text{saw}_x$$

admits a continuous extension to the ordinary phase circle at any excluded boundary.

Their exclusions are removable only relative to a chosen closed quadrant, not globally after neighboring quadrants are identified.

10. Flatwave Boundary Discontinuity

Theorem 10.1 — Flatwave one-sided limits

At every boundary b_k ,

$$\lim_{x \rightarrow b_k^-} \text{Flatwave}(x) = \varepsilon_{k-1},$$

and

$$\lim_{x \rightarrow b_k^+} \text{Flatwave}(x) = \varepsilon_k.$$

Since

$$\varepsilon_k = -\varepsilon_{k-1},$$

we have

$$\lim_{x \rightarrow b_k^-} \text{Flatwave}(x) = - \lim_{x \rightarrow b_k^+} \text{Flatwave}(x).$$

Proof. Flatwave is constant on every open quadrant:

$$\text{Flatwave}(x) = \varepsilon_j \quad \text{for } x \in Q_j.$$

The left-hand limit at b_k is therefore ε_{k-1} , while the right-hand limit is ε_k .

Adjacent signs are opposite.

□

Corollary 10.2 — Nonexistence of the two-sided Flatwave limit

For every $b_k \in \mathbb{B}$,

$$\lim_{x \rightarrow b_k} \text{Flatwave}(x) \text{ does not exist.}$$

Corollary 10.3 — No continuous real-valued extension

There is no function

$$\overline{\text{Flatwave}} : \mathbb{R} \rightarrow \mathbb{R}$$

that is continuous at every b_k and agrees with Flatwave on $\mathcal{D}$.

Proof. A continuous extension at b_k would require equal left and right limits. Theorem 10.1 proves that these limits are opposite.

□

Definition 10.4 — Flatwave phase reversal

The exact mathematical event at a boundary is the sign reversal

$$\varepsilon_{k-1} \longrightarrow \varepsilon_k = -\varepsilon_{k-1}.$$

This may be called a Flatwave phase reversal.

The term denotes a discontinuity of the signed quadrant invariant. It does not, by itself, identify a physical transition.

11. Quadrant-Wise Compactification

Definition 11.1 — Closed quadrant

For each integer k , define

$$\overline{Q}_k = [b_k, b_{k+1}].$$

The ordinary operator formulas remain defined only on the interior Q_k .

Definition 11.2 — Extended transfer coordinate

Define

$$\overline{\lambda}_k : \overline{Q}_k \rightarrow [0, 1]$$

by

$$\overline{\lambda}_k(x) = \lambda(x) \quad \text{for } x \in Q_k,$$

together with

$$\overline{\lambda}_k(b_k) = 0$$

and

$$\overline{\lambda}_k(b_{k+1}) = 1.$$

Theorem 11.3 — Continuous transfer compactification

The function

$$\overline{\lambda}_k$$

is continuous on the closed quadrant $\overline{Q}_k$.

Proof. The original λ is continuous on Q_k .

By Theorem 6.1,

$$\lim_{x \rightarrow b_k^+} \lambda(x) = 0 = \overline{\lambda}_k(b_k),$$

and

$$\lim_{x \rightarrow b_{k+1}^-} \lambda(x) = 1 = \bar{\lambda}_k(b_{k+1}).$$

Therefore, the endpoint assignments produce a continuous extension.

□

Definition 11.4 — Extended saw responses

Define

$$\overline{\text{saw}}_{r,k} = \varepsilon_k \bar{\lambda}_k,$$

and

$$\overline{\text{saw}}_{x,k} = \varepsilon_k (1 - \bar{\lambda}_k).$$

Corollary 11.5 — Continuous saw compactification

Both extended saw responses are continuous on $\overline{Q}_k$.

Their endpoint values are

$$\overline{\text{saw}}_{r,k}(b_k) = 0,$$

$$\overline{\text{saw}}_{x,k}(b_k) = \varepsilon_k,$$

$$\overline{\text{saw}}_{r,k}(b_{k+1}) = \varepsilon_k,$$

and

$$\overline{\text{saw}}_{x,k}(b_{k+1}) = 0.$$

Definition 11.6 — Quadrant-relative Flatwave extension

Define

$$\overline{\text{Flatwave}}_k(x) = \varepsilon_k$$

for every

$$x \in \overline{Q}_k.$$

Then

$$\overline{\text{Flatwave}}_k = \overline{\text{saw}}_{r,k} + \overline{\text{saw}}_{x,k}$$

on the entire closed quadrant.

Corollary 11.7 — Local but incompatible extensions

Every bounded transfer expression extends continuously to each closed quadrant separately.

However, adjacent closed quadrants assign incompatible values to their shared boundary.

For example,

$$\overline{\text{Flatwave}}_{k-1}(b_k) = \varepsilon_{k-1},$$

while

$$\overline{\text{Flatwave}}_k(b_k) = \varepsilon_k = -\varepsilon_{k-1}.$$

Thus the obstruction is not failure of one-sided compactification. It is incompatibility under boundary identification.

12. Split-Boundary Phase Domain

Definition 12.1 — Split-boundary extension domain

Define

$$\widehat{\mathcal{D}} = \bigsqcup_{k \in \mathbb{Z}} \overline{Q}_k,$$

the disjoint union of all closed quadrants.

Each component $\overline{Q}_k$ is compact, but the countable disjoint union $\widehat{\mathcal{D}}$ is not thereby a compact space. Accordingly, *compactification* in this essay applies only componentwise to $Q_k \subset \overline{Q}_k$; the global object is the split-boundary extension domain.

In this space, the exiting copy of a boundary and the entering copy of the same numerical phase are distinct points.

For example, the numerical boundary b_k appears as:

$$b_k^{\text{exit}} \in \overline{Q}_{k-1},$$

and

$$b_k^{\text{entry}} \in \overline{Q}_k.$$

Theorem 12.2 — Continuous bounded extension on the split domain

The functions

$$\lambda, \quad \text{saw}_r, \quad \text{saw}_x, \quad \text{Flatwave}$$

extend continuously to $\widehat{\mathcal{D}}$.

Proof. Each component of $\widehat{\mathcal{D}}$ is one closed quadrant $\overline{Q}_k$.

Section 11 constructs a continuous extension of every listed function to each $\overline{Q}_k$.

Because the components are disjoint, no equality between entering and exiting boundary values is required.

Therefore, these component-wise extensions define continuous functions on the disjoint union. □

Corollary 12.3 — Boundary state distinction

The core bounded variables distinguish an exiting boundary state from an entering boundary state.

At b_k ,

$$(\lambda, \text{saw}_r, \text{saw}_x, \text{Flatwave})_{\text{exit}} = (1, \varepsilon_{k-1}, 0, \varepsilon_{k-1}),$$

while

$$(\lambda, \text{saw}_r, \text{saw}_x, \text{Flatwave})_{\text{entry}} = (0, 0, \varepsilon_k, \varepsilon_k).$$

Since

$$\varepsilon_k = -\varepsilon_{k-1},$$

these are distinct boundary states.

13. Classification of Boundary Behavior

Definition 13.1 — Divergent boundary

A function has a divergent boundary at b from one side if its magnitude tends to $+\infty$ from that side.

Definition 13.2 — Finite one-sided boundary

A function has a finite one-sided boundary value if the corresponding one-sided limit exists in $\mathbb{R}$.

Definition 13.3 — Globally removable exclusion

An excluded point is globally removable for a real-valued function when the left and right limits exist, are finite, and are equal.

Definition 13.4 — Quadrant-removable exclusion

An excluded endpoint is quadrant-removable when the interior function has a finite limit from within that quadrant, allowing a continuous extension to that closed quadrant.

Theorem 13.5 — Boundary classification

The operator families satisfy the following classification.

Primitive operators

At every excluded boundary:

- one reciprocal pair undergoes a zero–pole exchange;
- the other reciprocal pair approaches 1;
- at least one primitive diverges from one side.

Therefore, the primitive family does not admit a finite global extension.

UNA operators

Within each quadrant:

- Unainverex diverges at entry and is finite at exit;
- Unaexpon is finite at entry and diverges at exit.

Therefore, neither UNA operator extends finitely to both endpoints of a closed quadrant.

Saw operators

Each saw has finite one-sided limits at every boundary and extends continuously to every closed quadrant.

However, its two-sided limits are unequal.

Therefore, each saw exclusion is quadrant-removable but not globally removable.

Flatwave

Flatwave extends continuously to every closed quadrant as a constant.

Its neighboring one-sided values are opposite.

Therefore, its boundary exclusion is quadrant-removable but globally nonremovable.

Transfer coordinate

The transfer coordinate extends continuously to every closed quadrant from 0 to 1.

Its exit value and the next quadrant's entry value differ:

$$1 \neq 0.$$

Therefore, global identification requires an explicit reset or gluing rule.

14. Seam Maps**Definition 14.1 — Exit and entry sets**

Let

$$E_k = b_{k+1}^{\text{exit}} \in \overline{Q}_k$$

denote the exit endpoint of Q_k .

Let

$$I_{k+1} = b_{k+1}^{\text{entry}} \in \overline{Q}_{k+1}$$

denote the entering endpoint of the next quadrant.

Definition 14.2 — Seam map

A seam map is an additional rule assigning an entering state to an exiting state:

$$\Sigma : E \rightarrow I,$$

where E is the set of exit endpoints and I is the set of entry endpoints.

A seam map may specify, for example:

- which quadrant follows an exit;
- whether phase direction is preserved or reversed;

- whether λ resets from 1 to 0;
- whether saw labels are exchanged;
- whether Flatwave sign reverses;
- whether additional state variables are retained.

Proposition 14.3 — Interior independence of seam choice

The identities proved in S0.I–S0.IV and the one-sided limits proved in this essay hold independently of the choice of seam map.

Proof. Every earlier theorem is stated either:

- at an interior point $x \in Q_k$; or
- as a one-sided limit taken from within a specified quadrant.

A seam map acts only after an exit endpoint has been reached in the split-boundary description.

Changing the seam map does not alter any value or derivative at an interior point, nor does it alter a one-sided limit already determined from within the quadrant.

Therefore, all core theorems remain valid under more than one possible seam map.

□

Theorem 14.4 — No unique boundary-crossing law

The core bounded phase calculus does not uniquely determine a seam map.

Proof. The core calculus determines:

$$\lambda \rightarrow 1, \quad \text{saw}_r \rightarrow \varepsilon_k, \quad \text{saw}_x \rightarrow 0$$

at the exit of Q_k .

It also determines the possible entry data of every quadrant:

$$\lambda = 0, \quad \text{saw}_r = 0, \quad \text{saw}_x = \varepsilon_j$$

at the entry of Q_j .

Nothing in the interior identities selects which entry copy must be paired with a given exit copy.

At minimum, one may define:

1. a forward phase seam pairing the exit of Q_k with the entry of Q_{k+1} ;
2. a reflected seam pairing that exit with an entry selected by a phase reflection;

3. a periodic or lifted seam retaining additional winding information;
4. a terminating rule under which no continuation is defined.

Each choice leaves all interior identities and one-sided limits unchanged.

Therefore, no unique seam map follows from the core calculus.

□

Corollary 14.5 — Boundary dynamics require a new assumption

Any statement specifying what happens after a boundary is reached is additional to Book 0's interior operator calculus.

Such a statement must be labeled as one of:

- a boundary axiom;
- an evolution rule;
- a projection convention;
- or a model-specific hypothesis.

It may not be presented as a consequence of the Flatwave identity alone.

15. Boundary Ledger

15.1 Sine boundary

For

$$b = m\pi,$$

the primitive limits are

	b^-	b^+
sr x	0	$+\infty$
sr p	$+\infty$	0
cr p	1	1
cr x	1	1

15.2 Cosine boundary

For

$$b = \frac{\pi}{2} + m\pi,$$

the primitive limits are

	b^-	b^+
srx	1	1
sxp	1	1
exp	$+\infty$	0
crx	0	$+\infty$

15.3 General seam b_k

Let

$$\varepsilon_k = (-1)^k.$$

Then

	b_k^-	b_k^+
urx	$-\varepsilon_k$	$\varepsilon_k \infty$
uxp	$-\varepsilon_k \infty$	ε_k
saw _r	$-\varepsilon_k$	0
saw _x	0	ε_k
Flatwave	$-\varepsilon_k$	ε_k

This table records one-sided limits only. No row assigns an actual value at b_k on the original domain.

16. Terminal Statement

S0.I established the Pythagorean operator identities.

S0.II established one-generator rational reduction.

S0.III established tangential transfer.

S0.IV established discrete phase and operator symmetry.

S0.V establishes the boundary structure.

At a sine boundary, the sine reciprocal pair exchanges

$$0 \longleftrightarrow +\infty,$$

while the cosine pair approaches

$$1.$$

At a cosine boundary, the cosine reciprocal pair exchanges

$$+\infty \longleftrightarrow 0,$$

while the sine pair approaches

$$1.$$

Within each quadrant,

$$\lambda : 0 \longrightarrow 1$$

and

$$(\text{saw}_r, \text{saw}_x) : (0, \varepsilon) \longrightarrow (\varepsilon, 0).$$

The saw responses have finite one-sided limits, but their neighboring limits do not agree.

Flatwave obeys

$$\lim_{x \rightarrow b^-} \text{Flatwave}(x) = - \lim_{x \rightarrow b^+} \text{Flatwave}(x).$$

Therefore,

$$\text{Flatwave has no continuous extension across an identified boundary.}$$

A continuous bounded extension exists only when entering and exiting boundary copies remain distinct:

$$\widehat{\mathcal{D}} = \bigsqcup_k \overline{Q}_k.$$

The core calculus determines the approach to every boundary but does not determine passage through it.

Thus the boundary result is:

$$\text{exact one-sided limits and componentwise endpoint compactification,} \\ \text{with no canonical seam dynamics.}$$

S0.VI

Axiomatic Status, Minimality, and the Book 0 Theorem Ledger

Book 0 - Formal Core

Abstract

This essay closes Book 0 by auditing the logical status of the bounded phase operator calculus developed in S0.I–S0.V.

The audit distinguishes inherited mathematical background, definitions, conventions, proved theorems, corollaries, conditional statements, conjectures, and interpretations. It shows that, relative to ordinary real trigonometry and elementary real analysis, Book 0 requires no independent R-specific mathematical axiom. The R-operators are definitions; their reciprocal, differential, transfer, symmetry, and boundary properties are derived consequences.

The four displayed primitive operators are not algebraically independent. They form two reciprocal pairs, and on every open quadrant the entire named calculus reduces to rational functions of one nonconstant generator,

$$z = \text{sr}x.$$

One nonconstant generator is therefore necessary and sufficient for quadrant-local generation. This is a theorem about generator count, not a uniqueness theorem for the particular choice of generator.

The finite named list is not closed under unrestricted rational operations. The correct closure object is the quadrant-local rational differential field

$$\mathbb{R}(z),$$

with derivation

$$\mathcal{D}_z = -\frac{1+z^2}{2} \frac{d}{dz}.$$

Globally, the calculus is piecewise rational and requires branch information identifying the open quadrant or the sign

$$\varepsilon = \text{sgn}(\sin 2x).$$

The exact Flatwave theorem,

$$\text{Flatwave}(x) = \text{sgn}(\sin 2x),$$

proves a signed partition invariant. It does not by itself establish conservation of physical mass, energy, or information. Likewise, the operator calculus determines one-sided boundary limits but does not determine a seam-crossing or evolution law.

The final result of Book 0 is therefore a bounded phase operator calculus with exact algebraic identities, one-generator local reduction, differential closure, complementary transfer, discrete symmetry, and rigorously classified boundary behavior.

0. Purpose of the Closing Essay

Book 0 has introduced and analyzed the mathematical core of the S Essay series.

Its purpose has not been to produce a physical theory. Its purpose has been to determine exactly what follows from the bounded phase operators before any external interpretation, projection, calibration, or evolution law is introduced.

This closing essay answers six questions.

1. What mathematical background was inherited?
2. What objects were merely defined?
3. Which statements were proved?
4. In what sense is the calculus minimal?
5. In what sense is the calculus closed?
6. What claims remain unavailable at the end of Book 0?

No new operator is introduced in this essay.

No new dynamical rule is introduced.

No physical interpretation is promoted to theorem status.

1. Status Vocabulary

Definition 1.1 — Background result

A background result is a standard theorem or rule inherited from ordinary mathematics rather than proved within Book 0.

Examples include:

$$\sin^2 x + \cos^2 x = 1,$$

the angle-addition formulas, the ordinary derivative rules for sine and cosine, and standard one-sided limits.

A background result is not an R-specific axiom.

Definition 1.2 — Convention

A convention selects notation, coordinates, orientation, or presentation without asserting a new mathematical fact.

Examples include:

- measuring phase in radians;
- writing phase modulo 2π ;
- naming an operator Sininverex ;
- choosing the sign convention

$$\text{uxp} = \text{cxp} - \text{sxp}.$$

Definition 1.3 — Definition

A definition declares a mathematical object.

A definition is neither true nor false in the manner of a theorem. It is judged by consistency, clarity, and usefulness.

Definition 1.4 — Axiom

An axiom is an assumed proposition used without derivation inside the formal system.

A genuine axiom must assert more than a choice of notation or the definition of an object.

Definition 1.5 — Lemma

A lemma is a proved supporting proposition used primarily to establish a later result.

Definition 1.6 — Theorem

A theorem is a principal proposition derived from earlier definitions, background results, axioms, lemmas, or theorems.

Definition 1.7 — Corollary

A corollary is an immediate or nearly immediate consequence of a theorem.

Definition 1.8 — Conditional theorem

A conditional theorem is proved only after an additional condition, candidate class, branch selection, or model rule has been declared.

Its conclusion may not be generalized beyond those conditions.

Definition 1.9 — Conjecture

A conjecture is a mathematically precise claim for which no proof has yet been supplied.

Definition 1.10 — Interpretation

An interpretation assigns conceptual, physical, informational, biological, economic, or metaphysical meaning to a mathematical object.

An interpretation is not a theorem unless a separate projection and validation argument establishes it.

2. Inherited Mathematical Background

Book 0 depends on ordinary real mathematics.

The background may be stated compactly as follows.

Background B.1 — Real-number structure

The real numbers are assumed with their ordinary field operations, order relation, absolute value, and limits.

Background B.2 — Real trigonometric functions

The functions

$$\sin x \quad \text{and} \quad \cos x$$

are inherited with their ordinary real definitions.

The quotient and reciprocal functions are defined where their denominators are nonzero:

$$\tan x = \frac{\sin x}{\cos x},$$

$$\cot x = \frac{\cos x}{\sin x},$$

$$\sec x = \frac{1}{\cos x},$$

and

$$\csc x = \frac{1}{\sin x}.$$

Background B.3 — Pythagorean identity

$$\sin^2 x + \cos^2 x = 1.$$

This is the primary algebraic background theorem of S0.I.

Background B.4 — Angle-addition identities

$$\sin(x + y) = \sin x \cos y + \cos x \sin y,$$

and

$$\cos(x + y) = \cos x \cos y - \sin x \sin y.$$

These support the rotation and reflection analysis in S0.IV.

Background B.5 — Elementary differentiation

On open intervals where the functions are defined,

$$\frac{d}{dx} \sin x = \cos x,$$

$$\frac{d}{dx} \cos x = -\sin x,$$

together with the product, quotient, reciprocal, and chain rules.

These support S0.II and S0.III.

Background B.6 — Standard local limits

As $h \rightarrow 0$,

$$\frac{\sin h}{h} \rightarrow 1,$$

and

$$\frac{1 - \cos h}{h^2/2} \rightarrow 1.$$

These support the boundary asymptotics of S0.V.

Theorem 2.1 — No independent R-specific axiom is required

Relative to Background B.1–B.6, all Book 0 results follow from definitions and derivations.

Proof. The operators are introduced by explicit definitions using ordinary trigonometric functions, absolute value, arithmetic operations, and reciprocation.

Their algebraic identities are derived from the Pythagorean identity.

Their symmetry properties are derived from angle-addition identities.

Their differential properties are derived from ordinary differentiation.

Their boundary properties are derived from standard limits.

No additional unproved proposition concerning the R-operators is needed to obtain the Book 0 theorem set.

Therefore, relative to the inherited mathematical background, Book 0 requires no independent R-specific axiom.

□

Clarification 2.2 — Earlier uses of the word “axiom”

Earlier essays occasionally used “axiom” for declarations such as periodic phase, boundary exclusion, or absolute-value placement.

In the final logical classification:

- periodic phase is a coordinate convention inherited from trigonometric periodicity;
- boundary exclusion is part of the domain definition;
- absolute-value placement is part of the operator definitions;
- the UNA sign convention is a naming convention.

None is an independent mathematical proposition comparable to the Pythagorean identity.

3. Canonical Definitions of Book 0

Definition 3.1 — Phase domain

The analytic carrier is the lifted real line

$$x \in \mathbb{R}.$$

The operator family is 2π -periodic. The identified phase circle is

$$\mathbb{T} = \mathbb{R}/(2\pi\mathbb{Z}),$$

with quotient relation

$$x \equiv x + 2\pi.$$

Analytic expressions, domains, derivatives, and one-sided limits use the lifted coordinate in radians; finite rotation groups act on $\mathbb{T}$.

Definition 3.2 — Forbidden set

$$\mathbb{B} = \left\{ \frac{k\pi}{2} : k \in \mathbb{Z} \right\}.$$

Definition 3.3 — Admissible domain

$$\boxed{\mathcal{D} = \mathbb{R} \setminus \mathbb{B}.}$$

Thus, for $x \in \mathcal{D}$,

$$\sin x \neq 0 \quad \text{and} \quad \cos x \neq 0.$$

Definition 3.4 — Absolute reciprocal magnitudes

$$M(x) = |\sec x| = \frac{1}{|\cos x|},$$

and

$$N(x) = |\csc x| = \frac{1}{|\sin x|}.$$

Definition 3.5 — Primitive displayed operators

$$\boxed{\text{srx}(x) = N(x) + \cot x}$$

$$\boxed{\text{sxp}(x) = \frac{1}{\text{srx}(x)}}$$

$$\text{exp}(x) = M(x) + \tan x$$

$$\text{crx}(x) = \frac{1}{\text{exp}(x)}.$$

Definition 3.6 — UNA operators

$$\text{urx} = \text{srx} - \text{crx}$$

and

$$\text{uxp} = \text{cxp} - \text{sxp}.$$

Definition 3.7 — Saw responses

$$\text{saw}_r = \frac{1}{\text{urx}}$$

and

$$\text{saw}_x = \frac{1}{\text{uxp}}.$$

Definition 3.8 — Flatwave

$$\text{Flatwave} = \text{saw}_r + \text{saw}_x.$$

Definition 3.9 — Open quadrants

$$Q_k = \left(\frac{k\pi}{2}, \frac{(k+1)\pi}{2} \right).$$

Definition 3.10 — Quadrant sign

For $x \in Q_k$,

$$\varepsilon_k = \text{sgn}(\sin 2x) = (-1)^k.$$

Definition 3.11 — Canonical generator

On an open quadrant,

$$z = \text{sr}x.$$

Definition 3.12 — Transfer coordinates

Let

$$\alpha = \text{sgn}(\sin x), \quad \beta = \text{sgn}(\cos x).$$

Define

$$p = \alpha \cos x - \beta \sin x,$$

$$q = \alpha \sin x + \beta \cos x,$$

and

$$\lambda = \frac{1-p}{2}.$$

Definition 3.13 — Phase-tangential derivative

For a differentiable function f on an open quadrant,

$$T[f] = \frac{df}{dx}.$$

For $f > 0$, define

$$\hat{T}[f] = \frac{f'}{f}.$$

These definitions exhaust the canonical mathematical vocabulary of Book 0.

4. Primary Dependency Chain

The central proof structure of Book 0 is:

$$\text{Pythagorean identity}$$

$\Downarrow$

secant–tangent and cosecant–cotangent identities

 $\Downarrow$

reciprocal-conjugate forms

 $\Downarrow$

positivity and reciprocal pairing

 $\Downarrow$

UNA sum, product, and common sign

 $\Downarrow$

Flatwave and saw partition

 $\Downarrow$

one-generator reduction

 $\Downarrow$

differential and transfer structure

 $\Downarrow$

symmetry and boundary classification.

No physical premise occurs in this chain.

5. Theorem Ledger for S0.I

Theorem I.1 — Reciprocal Pythagorean identities

$$M^2 - \tan^2 x = 1$$

and

$$N^2 - \cot^2 x = 1.$$

Status: Proved theorem.

Dependencies: Pythagorean identity and ordinary reciprocal definitions.

Theorem I.2 — Reciprocal-conjugate forms

$$\text{crx} = M - \tan x$$

and

$$\text{sxp} = N - \cot x.$$

Status: Proved theorem.

Corollary I.3 — Exact reciprocal pairs

$$\text{exp crx} = 1$$

and

$$\text{srx sxp} = 1.$$

Status: Proved corollary.

Corollary I.4 — Primitive positivity

$$\text{srx}, \text{sxp}, \text{exp}, \text{crx} > 0$$

on $\mathcal{D}$.

Status: Proved corollary.

Theorem I.5 — Common UNA sign

$$\boxed{\operatorname{sgn}(\operatorname{urx}) = \operatorname{sgn}(\operatorname{uxp}) = \operatorname{sgn}(\sin 2x)}.$$

Status: Proved theorem.

Theorem I.6 — UNA sum

$$\boxed{\operatorname{urx} + \operatorname{uxp} = \frac{4}{\sin 2x}}.$$

Status: Proved theorem.

Corollary I.7 — Inverse UNA sum

$$\boxed{\frac{1}{\operatorname{urx} + \operatorname{uxp}} = \frac{\sin 2x}{4}}.$$

Status: Proved corollary.

Theorem I.8 — UNA product

$$\boxed{\operatorname{urx} \operatorname{uxp} = \frac{4}{|\sin 2x|}}.$$

Status: Proved theorem.

Theorem I.9 — Exact Flatwave

$$\boxed{\operatorname{Flatwave} = \operatorname{sgn}(\sin 2x)}.$$

Status: Proved theorem.

Corollary I.10 — Flatwave magnitude

$$\boxed{|\operatorname{Flatwave}| = 1}.$$

Status: Proved corollary.

Corollary I.11 — Signed saw partition

$$\boxed{\operatorname{saw}_r + \operatorname{saw}_x = \operatorname{sgn}(\sin 2x)}.$$

Status: Proved corollary.

Corollary I.12 — Unsigned saw partition

$$\boxed{|\text{saw}_r| + |\text{saw}_x| = 1}.$$

Status: Proved corollary.

Theorem I.13 — Octant dominance cycle

The unique dominant primitive on successive open octants follows

$$\boxed{\text{srx} \rightarrow \text{cxp} \rightarrow \text{crx} \rightarrow \text{sxp} \rightarrow \text{srx} .}$$

Status: Proved theorem.

Qualification: At odd multiples of $\pi/4$, adjacent winners tie. Those points are admissible but are not open-octant interiors.

6. Theorem Ledger for S0.II**Theorem II.1 — One-generator primary reduction**

On a fixed open quadrant,

$$\boxed{\text{cxp} = \frac{z + \varepsilon}{\varepsilon z - 1}},$$

where

$$z = \text{srx}$$

and

$$\varepsilon = \text{sgn}(\sin 2x).$$

Status: Proved quadrant-local theorem.

Corollary II.2 — Canonical primitive forms

$$\text{srx} = z,$$

$$\text{sxp} = \frac{1}{z},$$

$$\text{cxp} = \frac{z + \varepsilon}{\varepsilon z - 1},$$

and

$$\text{crx} = \frac{\varepsilon z - 1}{z + \varepsilon}.$$

Status: Proved quadrant-local corollary.

Theorem II.3 — Generator range

$$\varepsilon = +1 \implies z > 1,$$

while

$$\varepsilon = -1 \implies 0 < z < 1.$$

Status: Proved theorem.

Corollary II.4 — Branch sign from the generator

$$\varepsilon = \text{sgn}(z - 1) = \text{sgn}(z^2 - 1).$$

Status: Proved corollary.

Theorem II.5 — Canonical UNA forms

$$\text{urx} = \frac{z^2 + 1}{z + \varepsilon}$$

and

$$\text{uxp} = \frac{z^2 + 1}{z(\varepsilon z - 1)}.$$

Status: Proved theorem.

Theorem II.6 — Canonical saw forms

$$\text{saw}_r = \frac{z + \varepsilon}{z^2 + 1}$$

and

$$\text{saw}_x = \frac{z(\varepsilon z - 1)}{z^2 + 1}.$$

Status: Proved theorem.

Theorem II.7 — Rational double-angle readout

$$\sin 2x = \frac{4z(z^2 - 1)}{(z^2 + 1)^2}.$$

Status: Proved theorem.

Theorem II.8 — Generator differential equation

$$z' = -\frac{1 + z^2}{2}.$$

Status: Proved quadrant-local theorem.

Theorem II.9 — Quadrant-local rational closure

Every named Book 0 operator belongs to

$$\mathbb{R}(z)$$

on a fixed open quadrant.

Status: Proved theorem.

Theorem II.10 — Differential closure

If

$$F(z) \in \mathbb{R}(z),$$

then

$$\frac{d}{dx} F(z(x)) \in \mathbb{R}(z).$$

Status: Proved theorem.

Corollary II.11 — Repeated differential closure

For every nonnegative integer n ,

$$\frac{d^n}{dx^n} F(z(x)) \in \mathbb{R}(z).$$

Status: Proved corollary.

7. Theorem Ledger for S0.III

Theorem III.1 — Primitive Riccati equations

$$\text{sr}x' = -\frac{1 + \text{sr}x^2}{2}$$

$$\text{sxp}' = \frac{1 + \text{sxp}^2}{2}$$

$$\text{cxp}' = \frac{1 + \text{cxp}^2}{2}$$

$$\text{cr}x' = -\frac{1 + \text{cr}x^2}{2}.$$

Status: Proved quadrant-local theorems.

Theorem III.2 — Normalized primitive rates

$$\hat{T}[\text{sr}x] = -|\csc x|$$

$$\hat{T}[\text{sxp}] = |\csc x|$$

$$\hat{T}[\text{cxp}] = |\sec x|$$

$$\hat{T}[\text{cr}x] = -|\sec x|.$$

Status: Proved theorems.

Theorem III.3 — Reciprocal rate reversal

For $f > 0$,

$$\hat{T}[1/f] = -\hat{T}[f].$$

Status: Proved general theorem.

Theorem III.4 — Transfer-circle identity

$$p^2 + q^2 = 2.$$

Status: Proved theorem.

Theorem III.5 — Transfer differential system

$$p' = -q$$

and

$$q' = p.$$

Status: Proved theorem.

Corollary III.6 — Harmonic transfer closure

$$p'' = -p$$

and

$$q'' = -q.$$

Status: Proved corollary.

Theorem III.7 — Transfer interval

$$0 < \lambda < 1$$

and

$$\lambda' > 0$$

on every open quadrant.

Status: Proved theorem.

Corollary III.8 — Transfer equation

$$\lambda'' + \lambda = \frac{1}{2}.$$

Status: Proved corollary.

Theorem III.9 — Saw transfer representation

$$\boxed{\text{saw}_r = \varepsilon \lambda}$$

and

$$\boxed{\text{saw}_x = \varepsilon(1 - \lambda)}.$$

Status: Proved theorem.

Theorem III.10 — Equal-and-opposite saw rates

$$\boxed{\text{saw}'_r = -\text{saw}'_x}.$$

More exactly,

$$\text{saw}'_r = \varepsilon \frac{q}{2},$$

and

$$\text{saw}'_x = -\varepsilon \frac{q}{2}.$$

Status: Proved theorem.

Corollary III.11 — Flatwave tangential invariance

$$\boxed{\text{Flatwave}' = 0}$$

inside every open quadrant.

Status: Proved corollary.

Theorem III.12 — Bounded saw-transfer rate

$$\boxed{\frac{1}{2} < |\text{saw}'_r| = |\text{saw}'_x| \leq \frac{1}{\sqrt{2}}}.$$

Status: Proved theorem.

Theorem III.13 — Saw curvature and midpoint inflection

Each saw has exactly one inflection point within each open quadrant, located at its midpoint,

$$x = \frac{(2k+1)\pi}{4}.$$

Status: Proved theorem.

8. Theorem Ledger for S0.IV**Theorem IV.1 — Quarter-turn primitive action**

For

$$R(x) = x + \frac{\pi}{2},$$

$$\boxed{\text{sr}\mathbf{x}(Rx) = \text{cr}\mathbf{x}(x)}$$

$$\boxed{\text{sx}\mathbf{p}(Rx) = \text{cx}\mathbf{p}(x)}$$

$$\boxed{\text{cx}\mathbf{p}(Rx) = \text{sx}\mathbf{p}(x)}$$

$$\boxed{\text{cr}\mathbf{x}(Rx) = \text{sr}\mathbf{x}(x)}.$$

Status: Proved theorem.

Theorem IV.2 — Half-turn invariance

For every primitive f ,

$$\boxed{f(x + \pi) = f(x)}.$$

Status: Proved theorem.

Theorem IV.3 — Reflection as reciprocal exchange

For

$$J(x) = -x,$$

$$\boxed{\text{sr}\mathbf{x}(Jx) = \text{sx}\mathbf{p}(x)}$$

and

$$\boxed{\text{cx}\mathbf{p}(Jx) = \text{cr}\mathbf{x}(x),}$$

with the reciprocal reverse relations also holding.

Status: Proved theorem.

Theorem IV.4 — Phase symmetry group

On $\mathbb{T}$, and using the convention that D_4 has eight elements, the quarter-turn and reflection generate

$$\boxed{D_4 = \langle R, J \mid R^4 = J^2 = e, JRJ = R^{-1} \rangle.}$$

Under the convention used in Book 0, D_4 has eight elements.

Status: Proved theorem.

Theorem IV.5 — Induced operator group

The primitive operator action has kernel

$$\{e, R^2\}$$

and image isomorphic to the Klein four-group:

$$\boxed{D_4/\{e, R^2\} \cong V_4.}$$

Status: Proved theorem.

Theorem IV.6 — Flatwave sign character

$$\boxed{\text{Flatwave}(Rx) = -\text{Flatwave}(x)}$$

$$\boxed{\text{Flatwave}(Jx) = -\text{Flatwave}(x).}$$

while diagonal reflection preserves Flatwave.

Status: Proved theorem.

Theorem IV.7 — Transfer transformation

Under quarter-turn,

$$\lambda(Rx) = \lambda(x).$$

Under reflection,

$$\lambda(Jx) = 1 - \lambda(x).$$

Status: Proved theorem.

Theorem IV.8 — Octant-label rotation

The eighth-turn

$$H(x) = x + \frac{\pi}{4}$$

advances the dominant label by

$$\text{srx} \rightarrow \text{cxp} \rightarrow \text{crx} \rightarrow \text{sxp}.$$

Status: Proved combinatorial theorem.

Negative result IV.9 — Eighth-turn nonidentity

In general,

$$\text{srx}\left(x + \frac{\pi}{4}\right) \neq \text{cxp}(x).$$

Status: Proved negative theorem.

Meaning: Dominance-label rotation is not the same as exact function permutation.

9. Theorem Ledger for S0.V**Theorem V.1 — Primitive zero–pole exchange**

At sine boundaries, the sine pair exchanges

$$0 \longleftrightarrow +\infty,$$

while the cosine pair approaches 1.

At cosine boundaries, the cosine pair exchanges

$$+\infty \leftrightarrow 0,$$

while the sine pair approaches 1.

Status: Proved one-sided limit theorem.

Theorem V.2 — First-order asymptotics

Every primitive pole is first order in boundary distance, with leading behavior

$$\frac{2}{h}.$$

Every paired zero is first order, with leading behavior

$$\frac{h}{2}.$$

Status: Proved asymptotic theorem.

Theorem V.3 — Transfer endpoint limits

Within every open quadrant,

$$\boxed{\lambda : 0 \longrightarrow 1.}$$

Status: Proved one-sided limit theorem.

Theorem V.4 — UNA boundary transfer

Within a quadrant of sign ε ,

$$(\text{urx}, \text{uxp}) : (\varepsilon\infty, \varepsilon) \longrightarrow (\varepsilon, \varepsilon\infty).$$

Status: Proved theorem.

Theorem V.5 — Saw boundary transfer

Within a quadrant,

$$\boxed{(\text{saw}_r, \text{saw}_x) : (0, \varepsilon) \longrightarrow (\varepsilon, 0).}$$

Status: Proved theorem.

Theorem V.6 — Flatwave sign jump

At every excluded boundary b ,

$$\boxed{\lim_{x \rightarrow b^-} \text{Flatwave}(x) = - \lim_{x \rightarrow b^+} \text{Flatwave}(x)}.$$

Status: Proved theorem.

Corollary V.7 — No continuous global Flatwave extension

Flatwave has no continuous real-valued extension across an identified quadrant boundary.

Status: Proved corollary.

Theorem V.8 — Componentwise quadrant compactification

The bounded transfer coordinate and saw responses extend continuously to each closed quadrant. This is a compactification of each individual component, not of the full lifted domain.

Status: Proved theorem.

Theorem V.9 — Split-boundary extension domain

The bounded variables extend continuously to

$$\widehat{\mathcal{D}} = \bigsqcup_{k \in \mathbb{Z}} \overline{Q}_k.$$

Status: Proved theorem.

Negative result V.10 — No canonical seam dynamics

The interior identities and one-sided boundary limits do not uniquely determine a boundary-crossing rule.

Status: Proved underdetermination theorem.

10. Minimality of the Primitive Presentation

The word minimality can refer to several different questions. They must not be conflated.

Definition 10.1 — Display minimality

A displayed family is minimal when no displayed member may be removed without losing a desired symmetry, naming pattern, or visual role.

Definition 10.2 — Algebraic generator minimality

A set of functions is algebraically minimal when no smaller set generates the same expression field under the declared operations.

Definition 10.3 — Axiomatic minimality

An axiom set is minimal when no axiom can be removed without losing at least one theorem.

Definition 10.4 — Interpretive minimality

An interpretive vocabulary is minimal when each label contributes meaning not supplied by another label.

Interpretive minimality is outside the scope of Book 0 mathematics.

Theorem 10.5 — The four displayed primitives are not algebraically independent

The four displayed operators contain two exact reciprocal dependencies:

$$\text{srx} = \frac{1}{\text{exp}},$$

and

$$\text{crx} = \frac{1}{\text{exp}}.$$

Therefore, the four displayed primitives are not algebraically independent.

Corollary 10.6 — Two primary envelopes suffice before quadrant reduction

If reciprocation is allowed, the displayed primitive family is generated by

$$\text{srx} \quad \text{and} \quad \text{exp}.$$

Thus two generators suffice without using the Flatwave branch relation.

Theorem 10.7 — One generator suffices quadrant-locally

On a fixed open quadrant, every named Book 0 operator is rational in

$$z = \text{srx}.$$

Therefore, one nonconstant generator is sufficient quadrant-locally.

Proof. S0.II supplies explicit rational expressions for every named operator in terms of z and the constant branch sign ε .

Once a quadrant is fixed, ε is a real constant equal to $+1$ or -1 .

Therefore, every named expression belongs to $\mathbb{R}(z)$.

□

Theorem 10.8 — Zero nonconstant generators are insufficient

A field generated over $\mathbb{R}$ by no nonconstant function contains only constant functions.

Since

$$z' = -\frac{1+z^2}{2} < 0,$$

the function z is nonconstant on every open quadrant.

Therefore, zero nonconstant generators cannot produce the Book 0 calculus.

Corollary 10.9 — Quadrant-local generator-count minimality

Exactly one nonconstant generator is necessary and sufficient on each open quadrant.

Minimal quadrant-local generator count = 1.

Clarification 10.10 — The generator is not uniquely determined

The theorem proves the number of required nonconstant generators.

It does not prove that

$$z = \text{sr}x$$

is the only possible generator.

For example, any invertible rational transform of z may generate the same field where the inverse transform is defined.

Possible alternative generators include other primitive operators on appropriate quadrants.

Therefore:

generator-count minimality does not imply generator uniqueness.

Clarification 10.11 — Global generation requires branch data

The formula

$$\text{cxp} = \frac{z + \varepsilon}{\varepsilon z - 1}$$

contains the quadrant sign ε .

Although

$$\varepsilon = \text{sgn}(z - 1),$$

the sign operation is not rational.

Thus the global calculus is not a single ordinary rational field in z without branch qualification.

A global presentation requires at least one of:

- a quadrant index;
- the branch sign ε ;
- a piecewise expression rule;
- or an enlarged function class containing sign functions.

11. Correct Meaning of Closure

Definition 11.1 — Finite named catalogue

Let

$$\mathcal{N}$$

denote the finite list consisting of the four primitive operators, the two UNA operators, the two saw responses, and Flatwave.

Proposition 11.2 — The finite named catalogue is not closed under unrestricted rational operations

For example, the expression

$$\text{srx} + \text{cxp}$$

is generally not identical to any single member of $\mathcal{N}$.

Likewise,

$$\text{sr}x^2, \quad \frac{\text{sr}x+2}{\exp-3},$$

and indefinitely many other rational expressions need not reduce to one of the nine named functions. Therefore, $\mathcal{N}$ is not a finite algebra closed under unrestricted rational operations.

Definition 11.3 — Quadrant-local rational field

On a fixed open quadrant, define

$$\mathcal{F}_k = \mathbb{R}(z).$$

Elements of $\mathcal{F}_k$ are rational functions

$$F(z) = \frac{P(z)}{Q(z)}$$

with real polynomial numerator and denominator, evaluated where $Q(z) \neq 0$.

Definition 11.4 — Phase derivation

Define

$$\mathcal{D}_z = -\frac{1+z^2}{2} \frac{d}{dz}.$$

Then

$$\frac{d}{dx} F(z(x)) = \mathcal{D}_z F.$$

Theorem 11.5 — Differential-field closure

For every fixed open quadrant,

$$\mathcal{D}_z(\mathcal{F}_k) \subseteq \mathcal{F}_k.$$

Therefore,

$$(\mathbb{R}(z), \mathcal{D}_z)$$

is the correct quadrant-local differential closure object.

Definition 11.6 — Global piecewise-rational class

Define

$$\mathcal{F} = \left\{ F : \mathcal{D} \rightarrow \mathbb{R} : F|_{Q_k} \in \mathbb{R}(z) \text{ for every } k \right\},$$

subject to the requirement that every denominator remain nonzero on its declared evaluation set.

Theorem 11.7 — Component-wise global closure

The class $\mathcal{F}$ is closed under:

- addition;
- subtraction;
- multiplication;
- division where the divisor is nonzero;
- and differentiation on each open quadrant.

Proof. Each operation is performed separately on every connected component Q_k .

The rational functions are closed under the four field operations where denominators are nonzero.

S0.II proves closure under differentiation.

No equality across an excluded boundary is required.

Therefore, $\mathcal{F}$ is component-wise closed.

□

Final closure statement

The defensible closure theorem is:

The Book 0 calculus is a piecewise rational differential calculus, not a finite closed list of named functions.

12. Uniqueness Audit**12.1 Reciprocal-conjugate uniqueness**

Given the definitions

$$\text{cxp} = M + \tan x$$

and

$$M^2 - \tan^2 x = 1,$$

the reciprocal is uniquely determined:

$$\frac{1}{\text{cxp}} = M - \tan x.$$

Likewise,

$$\frac{1}{\text{srx}} = N - \cot x.$$

This is ordinary functional uniqueness of a reciprocal.

12.2 Flatwave value

Once Flatwave is defined as

$$\frac{1}{\text{urx}} + \frac{1}{\text{uxp}},$$

its value is uniquely determined:

$$\text{Flatwave} = \text{sgn}(\sin 2x).$$

This is a theorem about the defined expression.

12.3 Flatwave-form uniqueness is not proved

Book 0 does not prove that Flatwave is the only possible composite satisfying some broad list of desired properties.

There may exist many functions that:

- equal +1 on Quadrants I and III;
- equal −1 on Quadrants II and IV;
- are invariant under half-turn;
- reverse under quarter-turn.

Indeed,

$$\text{sgn}(\sin 2x)$$

itself may be represented in multiple algebraically equivalent forms.

A nontrivial uniqueness theorem would require a restricted candidate class, such as:

- symmetric rational expressions of bounded degree;
- expressions built from specified generators;
- fixed pole orders;
- fixed normalization;
- fixed behavior under reciprocal exchange;
- fixed boundary asymptotics.

No such complete classification is proved in Book 0.

12.4 Primitive-family uniqueness is not proved

Book 0 does not prove that the four named operators are the only positive reciprocal trigonometric family with related properties.

Such a theorem would require a formally declared search space and equivalence relation.

Final uniqueness statement

Book 0 proves exact evaluation, not universal classification uniqueness.

13. The Status of “Non-Additivity”

Addition occurs throughout the core definitions and proofs:

$$\text{sr}x = N + \cot x,$$

$$\text{cx}p = M + \tan x,$$

$$\text{Flatwave} = \text{saw}_r + \text{saw}_x,$$

and

$$\text{ur}x + \text{ux}p = \frac{4}{\sin 2x}.$$

Therefore, a general axiom forbidding addition would contradict the formal construction.

A narrower modeling rule may later prohibit arbitrary superposition in a particular projection or physical interpretation. Such a rule would belong to a later book.

The mathematical conclusion is:

Book 0 contains no general non-additivity axiom.

14. The Status of Growth and Decay Language

The following derivative signs are exact:

$$\text{sr}x' < 0, \quad \text{cr}x' < 0,$$

$$\text{sx}p' > 0, \quad \text{cx}p' > 0.$$

Therefore, with respect to increasing phase x :

- the invrex pair decreases;
- the expon pair increases.

This justifies the restricted mathematical vocabulary:

phase-increasing and phase-decreasing.

It does not yet justify:

- temporal growth;
- temporal decay;
- causal production;
- dissipation;
- biological growth;
- economic expansion;
- cosmological development.

Those require a map

$$x = x(\tau)$$

or another declared projection relating phase to the external variable of interest.

The Book 0 status is:

growth and decay are phase-direction labels only.

15. The Status of Conservation Language

Book 0 proves:

$$\text{saw}_r + \text{saw}_x = \varepsilon,$$

and

$$|\text{saw}_r| + |\text{saw}_x| = 1.$$

It also proves:

$$\text{saw}'_r + \text{saw}'_x = 0$$

inside every open quadrant.

These are exact normalized partition identities.

They justify the phrase:

signed partition invariant.

They do not by themselves prove conservation of:

- mass;
- energy;
- momentum;
- electric charge;
- probability;
- thermodynamic information;
- algorithmic information.

To establish a physical conservation law, a later theory must provide:

1. a state space;
2. a physical quantity C ;
3. a projection identifying C with a function of the saw variables;
4. an evolution parameter;
5. a proof that $dC/d\tau = 0$;
6. dimensional consistency;
7. empirical validation.

Therefore:

Flatwave is mathematically invariant within a quadrant, but no physical conservation law is proved in Book 0.

16. The Status of Boundaries and Phase Change

Book 0 proves exact one-sided limits.

It does not assign values to the original operators at

$$x = \frac{k\pi}{2}.$$

It proves that:

- primitive zero–pole exchanges occur;
- the saw responses have finite one-sided limits;
- Flatwave reverses sign;
- the transfer coordinate runs from 0 to 1;
- adjacent boundary states are incompatible under ordinary identification.

The term phase reversal may therefore denote the exact sign change

$$\varepsilon \longrightarrow -\varepsilon.$$

It does not establish:

- a singularity in physical spacetime;
- a Big Bang;
- a heat death;
- a bounce;
- annihilation;
- creation;
- a mirror universe;
- reversal of physical time.

Any rule mapping one exit state to another entry state is a seam rule introduced after Book 0.

Therefore:

Book 0 proves boundary approach, not boundary passage.

17. Domain Ledger

Every theorem in Book 0 must be read with its domain.

17.1 Global admissible-domain theorems

The following hold for every $x \in \mathcal{D}$:

- reciprocal-conjugate identities;
- primitive positivity;
- reciprocal products;
- UNA sum and product;
- common UNA sign;
- exact Flatwave;
- signed and unsigned saw partitions.

17.2 Quadrant-local theorems

The following require a fixed open quadrant:

- constant ε ;
- one-generator rational reduction;
- rational differential-field closure;
- ordinary differentiation without crossing a boundary;
- transfer-coordinate monotonicity;
- saw-transfer rates;
- curvature classification.

17.3 Octant-local theorems

The following are naturally stated on open octants:

- unique primitive dominance;
- fixed saw-curvature sign.

17.4 Boundary theorems

The following are one-sided statements:

- primitive zero–pole limits;
- UNA asymptotics;
- saw endpoint limits;
- Flatwave sign jumps;
- componentwise quadrant compactification.

17.5 Split-domain theorems

Continuous boundary extensions of the bounded variables hold on

$$\widehat{\mathcal{D}} = \bigsqcup_k \overline{Q}_k,$$

not on the ordinary identified phase circle.

18. Book 0 Negative Theorems

Negative theorems are important because they prevent later interpretation from being mistaken for deduction.

Negative Theorem 18.1 — No finite named algebra

The nine named expressions do not form a finite algebra under unrestricted rational operations.

Negative Theorem 18.2 — No global rational single branch

The complete global calculus is not represented by one branch-independent rational formula in z alone without a sign, quadrant, or piecewise rule.

Negative Theorem 18.3 — No universal uniqueness theorem

Book 0 does not classify all possible reciprocal operator families or all possible Flatwave-like expressions.

Negative Theorem 18.4 — No continuously varying Flatwave interior

Inside every open quadrant,

$$\text{Flatwave}' = 0.$$

Therefore, Flatwave itself does not encode a continuously varying interior magnitude.

Negative Theorem 18.5 — No continuous Flatwave seam

Flatwave cannot be continuously extended across an identified quadrant wall.

Negative Theorem 18.6 — No canonical seam map

The core calculus does not determine how a state passes from one quadrant to another.

Negative Theorem 18.7 — No physical ontology

Book 0 does not establish physical meanings for its mathematical operators.

Negative Theorem 18.8 — No temporal dynamics

Differentiation with respect to x is not, without further assumptions, differentiation with respect to physical time.

19. Conjecture Ledger

Book 0 leaves the following possible research questions open.

Conjecture C.1 — Restricted Flatwave uniqueness

Within a precisely declared class of symmetric rational expressions of bounded degree, Flatwave may be unique up to algebraic equivalence and normalization.

No proof is currently supplied.

Conjecture C.2 — Classification of positive reciprocal families

The R-family may belong to a broader class of positive reciprocal half-angle constructions admitting comparable partition identities.

A complete classification is not supplied.

Conjecture C.3 — Canonical generator classification

The generators of the quadrant-local field $\mathbb{R}(z)$ may be classifiable under Möbius equivalence.

Book 0 proves only the sufficiency of $z = \text{srx}$.

Conjecture C.4 — Natural seam selection

A later dynamical or topological extension may select a seam map satisfying additional symmetry, continuity, reversibility, or optimization conditions.

No seam is selected by Book 0 alone.

Conjecture C.5 — External projection utility

The transfer coordinate, saw partition, or primitive rates may provide useful projections into external systems.

This is a research hypothesis requiring the projection discipline of Book 1.

20. Book 0 Formal Summary

20.1 Mathematical carrier

$$\mathcal{D} = \mathbb{R} \setminus \left\{ \frac{k\pi}{2} : k \in \mathbb{Z} \right\}.$$

20.2 Primitive reciprocal family

$$\text{srx} = |\csc x| + \cot x$$

$$\text{sxp} = |\csc x| - \cot x = \frac{1}{\text{srx}}$$

$$\text{cxp} = |\sec x| + \tan x$$

$$\text{crx} = |\sec x| - \tan x = \frac{1}{\text{cxp}}$$

20.3 UNA closure

$$\text{urx} + \text{uxp} = \frac{4}{\sin 2x}$$

and

$$\text{urx} \text{ uxp} = \frac{4}{|\sin 2x|}.$$

20.4 Flatwave

$$\text{Flatwave} = \text{sgn}(\sin 2x).$$

20.5 Saw partition

$$\text{saw}_r = \varepsilon \lambda$$

$$\text{saw}_x = \varepsilon(1 - \lambda)$$

and

$$|\text{saw}_r| + |\text{saw}_x| = 1$$

20.6 Generator reduction

On a fixed open quadrant,

$$\mathcal{N} \subset \mathbb{R}(z), \quad z = \text{sr}x.$$

20.7 Differential law

$$z' = -\frac{1 + z^2}{2}.$$

Therefore,

$$\mathcal{D}_z = -\frac{1 + z^2}{2} \frac{d}{dz}$$

preserves $\mathbb{R}(z)$.

20.8 Transfer law

$$p^2 + q^2 = 2, \quad p' = -q, \quad q' = p.$$

20.9 Symmetry

The phase transformations generate D_4 , while the induced primitive operator action is

$$V_4.$$

20.10 Boundary structure

Within each quadrant,

$$\lambda : 0 \rightarrow 1$$

and

$$(\text{saw}_r, \text{saw}_x) : (0, \varepsilon) \rightarrow (\varepsilon, 0).$$

Across each identified wall,

$$\text{Flatwave reverses sign.}$$

21. Book 0 Certification

Book 0 establishes a rigorous mathematical calculus with the following certified properties.

Certified C0.1 — Exactness

The principal identities are algebraic equalities, not numerical approximations.

Certified C0.2 — Domain control

Every theorem has an explicit admissible domain or one-sided boundary domain.

Certified C0.3 — Minimal generator count

One nonconstant generator is necessary and sufficient quadrant-locally.

Certified C0.4 — Differential closure

The quadrant-local rational field is closed under phase differentiation.

Certified C0.5 — Complementary transfer

The saw responses form an exact signed and unsigned partition.

Certified C0.6 — Discrete symmetry

The primitive family is a symmetry orbit under the induced Klein action, while octant dominance follows a four-label cycle.

Certified C0.7 — Boundary classification

Primitive poles, reciprocal zeros, saw limits, Flatwave jumps, and split-boundary extensions are explicitly classified.

Certified C0.8 — Interpretive restraint

No physical law is included among the certified Book 0 results.

22. Terminal Theorem of Book 0**Theorem 22.1 — Foundational closure of Book 0**

Relative to ordinary real trigonometry and elementary real analysis, the bounded phase operator calculus of Book 0 is completely specified by its definitions.

On every open quadrant:

1. all named operators reduce to rational functions of one nonconstant generator;
2. the resulting rational field is closed under phase differentiation;
3. the saw responses form a complementary signed partition;
4. the primitive and transfer structures obey exact discrete symmetries;
5. every approach to an excluded boundary has a classified one-sided limit.

The calculus does not determine a physical interpretation, dimensional calibration, temporal evolution, or seam-crossing law.

Proof. Statements 1–5 are the principal results of S0.I–S0.V and are recorded in the theorem ledger above.

The absence of physical interpretation, calibration, temporal evolution, and seam dynamics follows from the fact that no such structures are included among the Book 0 definitions or inherited mathematical background.

Therefore, Book 0 is mathematically closed at the level of its declared structural purpose, while remaining intentionally open to later projection and model layers.

□

23. Closing Declaration

Book 0 began with the Pythagorean identity

$$\sin^2 x + \cos^2 x = 1.$$

From that starting point, it derived:

positive reciprocal operators

signed UNA differences

a double-angle closure

a unit Flatwave

complementary saw transfer

one-generator rational reduction

differential-field closure

phase and operator symmetry

and

exact boundary classification.

Its strongest justified conclusion is:

The bounded phase operators form a piecewise rational differential calculus generated quadrant-locally by one nonconstant function, with exact reciprocal, transfer, symmetry, and boundary structures.

Its strongest necessary restraint is:

No physical conservation law, evolution law, measurement rule, or ontology follows from the structural calculus alone.

This completes Book 0.

END OF BOOK 0

Book 1: Projection and Measurement

S1.I

Structural Mathematics and Projection Contracts

Book 1 - Projection and Measurement

Abstract

Book 0 established a dimensionless bounded phase operator calculus on the admissible domain

$$\mathcal{D} = \mathbb{R} \setminus \left\{ \frac{k\pi}{2} : k \in \mathbb{Z} \right\}.$$

Its theorems concern exact relations among phase functions. They do not, by themselves, identify external objects, measurable quantities, physical units, or empirical observations.

This essay introduces the projection layer.

A projection is defined as an explicit composition

$$\Omega_{\mathcal{P}} \xrightarrow{\Phi} \mathcal{D} \xrightarrow{\mathbf{F}} \mathbb{R}^m \xrightarrow{\Gamma_{\theta}} \mathcal{Y},$$

where:

- $\Omega_{\mathcal{P}}$ is the declared valid subset of an external domain;
- Φ assigns an admissible phase to each valid external instance;
- $\mathbf{F}$ evaluates one or more Book 0 structural expressions;
- Γ_{θ} converts the resulting dimensionless structural profile into a declared output space;
- θ contains every calibration parameter, scale, unit, or convention required by the readout.

The complete projection contract also declares uncertainty and failure behavior.

It is proved that exact structural identities remain exact after lawful composition. The converse is weaker: agreement of projected outputs implies structural equality only when the readout is injective and the inferred phases cover the structural domain under examination.

It is further proved that all dimensional content enters through the external domain, the calibration parameters, or the readout map. Book 0 alone supplies no dimensional scale.

A boundary guard is established: an interior Book 0 expression may not be evaluated when the inferred phase lies in the forbidden set. Boundary-aware projections must instead use the split-boundary domain and the bounded extensions proved in S0.V.

Finally, an admissible reparameterization theorem shows that different phase assignments and structural profiles can generate identical external predictions. Projection data therefore do not uniquely determine the internal phase description without additional identification conditions.

The projection contract is the formal boundary between structural mathematics and external interpretation.

IDENTIFICATION DIVISION

PROGRAM-ID

S1-I-PROJECTION-CONTRACTS

PURPOSE

To define the minimal formal machinery required to apply the dimensionless structural results of Book 0 to an external domain.

PRINCIPAL QUESTION

Given an external object or observation ω , under what declared conditions may a Book 0 expression be used to produce an external output?

NON-PURPOSE

This essay does not:

- identify a particular physical system;
- define time, distance, mass, energy, charge, or information;
- select a phase-inference rule for any empirical domain;
- calibrate any physical constant;
- prove that an external system is governed by the R-operators;
- introduce a boundary-crossing dynamics;
- establish that a projection is unique;
- replace empirical validation with structural algebra.

ENVIRONMENT DIVISION

1. Inherited Structural Environment

All definitions and theorems of Book 0 are inherited.

Definition 1.1 — Interior structural domain

$$\mathcal{D} = \mathbb{R} \setminus \left\{ \frac{k\pi}{2} : k \in \mathbb{Z} \right\}.$$

Definition 1.2 — Forbidden boundary set

$$\mathbb{B} = \left\{ \frac{k\pi}{2} : k \in \mathbb{Z} \right\}.$$

Definition 1.3 — Split-boundary domain

From S0.V, define

$$\widehat{\mathcal{D}} = \bigsqcup_{k \in \mathbb{Z}} \overline{Q}_k,$$

where each entering and exiting copy of a numerical boundary remains distinct.

Definition 1.4 — Structural expression class

Let

$$\mathcal{F}$$

denote the global piecewise-rational differential class of Book 0.

Thus, for every open quadrant Q_k ,

$$F|_{Q_k} \in \mathbb{R}(z)$$

for each

$$F \in \mathcal{F},$$

where

$$z = \text{sr}x.$$

Definition 1.5 — Structural profile

For a positive integer m , a structural profile is an ordered tuple

$$\mathbf{F} = (F_1, \dots, F_m)$$

with

$$F_i \in \mathcal{F}.$$

The profile defines a map

$$\mathbf{F} : \mathcal{D}_{\mathbf{F}} \rightarrow \mathbb{R}^m,$$

where

$$\mathcal{D}_{\mathbf{F}} \subseteq \mathcal{D}$$

is the common domain on which every component F_i is defined.

The common domain may be smaller than $\mathcal{D}$ if a composite expression introduces additional denominator restrictions.

AXIOM DIVISION

Book 1 introduces one methodological axiom.

Axiom P.1 — Explicit Projection Mediation

Every claim assigning Book 0 structure to an external object or observation must be mediated by an explicitly declared projection contract.

Equivalently, no external interpretation is admitted merely because a Book 0 expression resembles, correlates with, or can be verbally associated with an external phenomenon.

The external claim must declare:

1. what is being projected;
2. how phase is inferred;
3. which structural expressions are evaluated;
4. how structural values become external outputs;
5. where the projection is valid;
6. what uncertainty applies;

7. what happens when the contract fails.

This is a methodological axiom governing the S Essay series. It is not a theorem of trigonometry.

DATA DIVISION

2. External Objects and Outputs

Definition 2.1 — External instance space

Let

$$\Omega$$

be a nonempty set whose elements represent external instances.

An element

$$\omega \in \Omega$$

may represent an observation, state, object, event, record, configuration, or other externally specified item.

No topology, metric, probability distribution, temporal order, or physical interpretation is assumed merely by declaring Ω .

Definition 2.2 — Output space

Let

$$\mathcal{Y}$$

be the declared output space of a projection.

Depending on the contract, $\mathcal{Y}$ may be:

- a subset of $\mathbb{R}$;
- a vector space;
- a set of labels;
- an interval;
- a probability space;
- or another explicitly defined set.

The structure of $\mathcal{Y}$ must be declared before it is used.

Definition 2.3 — Validity domain

A projection contract declares a subset

$$\Omega_{\mathcal{P}} \subseteq \Omega$$

called its declared validity domain.

This is a scope declaration, not a claim that the domain is maximal. S1.II distinguishes the declared domain from the maximal operational admissibility envelope determined by the non-domain guards.

The contract makes no prediction outside $\Omega_{\mathcal{P}}$, except through an explicitly declared failure rule.

3. Phase Inference**Definition 3.1 — Interior phase-inference map**

An interior phase-inference map is a function

$$\Phi : \Omega_{\mathcal{P}} \rightarrow \mathcal{D}.$$

For each valid external instance ω ,

$$\Phi(\omega)$$

is the phase at which the selected Book 0 structural profile is evaluated.

Definition 3.2 — Boundary-aware phase-inference map

A boundary-aware phase-inference map is a function

$$\widehat{\Phi} : \Omega_{\mathcal{P}} \rightarrow \widehat{\mathcal{D}}.$$

Such a map distinguishes entering and exiting copies of a boundary.

A boundary-aware contract may use only those Book 0 quantities for which the required extensions to $\widehat{\mathcal{D}}$ have been proved.

Definition 3.3 — Phase-inference rule

The phase-inference rule consists of:

1. the map Φ or $\widehat{\Phi}$;
2. the data from ω used to determine phase;

3. the method of determination;
4. any branch-selection rule;
5. any parameters used;
6. the conditions under which phase inference fails.

A numerical phase assignment without this declaration is not a complete projection rule.

4. Structural Evaluation

Definition 4.1 — Structural evaluation map

Given a structural profile

$$\mathbf{F} = (F_1, \dots, F_m),$$

define

$$E_{\mathbf{F}}(x) = (F_1(x), \dots, F_m(x)).$$

Thus,

$$E_{\mathbf{F}} : \mathcal{D}_{\mathbf{F}} \rightarrow \mathbb{R}^m.$$

Definition 4.2 — Evaluated structural state

For

$$\omega \in \Omega_{\mathcal{P}},$$

the evaluated structural state is

$$\mathbf{s}_{\mathcal{P}}(\omega) = E_{\mathbf{F}}(\Phi(\omega)).$$

This value is defined only when

$$\Phi(\omega) \in \mathcal{D}_{\mathbf{F}}.$$

5. Readout and Calibration

Definition 5.1 — Calibration parameter

Let

$$\theta \in \Theta$$

denote the complete collection of parameters used by the readout map.

The parameter object θ may contain:

- scale factors;
- offsets;
- unit definitions;
- reference values;
- empirical coefficients;
- thresholds;
- orientation conventions;
- fitted parameters;
- conversion constants.

Every such element must be declared.

Definition 5.2 — Readout map

A readout map is a function

$$\Gamma_{\theta} : \mathcal{S}_{\mathcal{P}} \rightarrow \mathcal{Y},$$

where

$$\mathcal{S}_{\mathcal{P}} \subseteq \mathbb{R}^m$$

contains the structural states produced by the contract.

The readout map converts dimensionless structural values into the declared external output.

Definition 5.3 — Calibration rule

A calibration rule specifies:

1. the parameter space Θ ;
2. the selected parameter θ ;
3. the origin or method of obtaining θ ;
4. the units carried by θ , if any;
5. whether θ is fixed, measured, estimated, or fitted;
6. the uncertainty attached to θ .

Book 0 supplies no calibration parameter.

6. Uncertainty and Failure

Definition 6.1 — Uncertainty space

Let

$$\mathfrak{U}(\mathcal{Y})$$

be a declared class of uncertainty descriptions over $\mathcal{Y}$.

Examples may include:

- nonnegative error bounds;
- intervals;
- covariance matrices;
- probability distributions;
- confidence sets;
- qualitative uncertainty labels.

No particular uncertainty formalism is assumed in S1.I.

Definition 6.2 — Uncertainty assignment

An uncertainty assignment is a map

$$U_{\mathcal{P}} : \Omega_{\mathcal{P}} \rightarrow \mathfrak{U}(\mathcal{Y}).$$

Definition 6.3 — Failure-witness space

Let

$\mathfrak{W}$

be a declared set of failure witnesses.

A failure witness records why a projection could not be lawfully evaluated.

Definition 6.4 — Failure map

A failure map is

$$W_{\mathcal{P}} : \Omega \setminus \Omega_{\mathcal{P}} \rightarrow \mathfrak{W}.$$

Possible failure witnesses include:

- PHASE-NOT-INFERABLE;
- PHASE-AT-FORBIDDEN-BOUNDARY;
- BRANCH-AMBIGUOUS;
- STRUCTURAL-DENOMINATOR-ZERO;
- OUTSIDE-CALIBRATION-RANGE;
- READOUT-NOT-DEFINED;
- UNCERTAINTY-NOT-AVAILABLE;
- MODEL-DOMAIN-EXCEEDED.

The presence of a failure witness is not a defect in the theory. It is part of a complete contract.

7. Projection Contract

Definition 7.1 — Projection contract

A projection contract is the tuple

$$\mathcal{P} = (\Omega, \Omega_{\mathcal{P}}, \Phi, \mathbf{F}, \Gamma_{\theta}, U_{\mathcal{P}}, W_{\mathcal{P}}).$$

For a boundary-aware contract, Φ is replaced by $\widehat{\Phi}$.

Definition 7.2 — Prediction map

For

$$\omega \in \Omega_{\mathcal{P}},$$

define

$$\hat{y}_{\mathcal{P}}(\omega) = \Gamma_{\theta}(E_{\mathbf{F}}(\Phi(\omega))).$$

Equivalently,

$$\hat{y}_{\mathcal{P}} = \Gamma_{\theta} \circ E_{\mathbf{F}} \circ \Phi.$$

Definition 7.3 — Complete contract

A projection contract is complete when all of the following are declared:

1. the external instance space Ω ;
2. the valid subset $\Omega_{\mathcal{P}}$;
3. the phase-inference map;
4. the structural profile;
5. the common structural domain;
6. the readout map;
7. every calibration parameter;
8. the output space;
9. the uncertainty assignment;
10. the failure map.

Definition 7.4 — Lawful contract

A complete projection contract is lawful when:

1. $\Phi(\Omega_{\mathcal{P}}) \subseteq \mathcal{D}_{\mathbf{F}}$;
2. $E_{\mathbf{F}}$ is defined on every inferred phase;
3. Γ_{θ} is defined on every produced structural state;
4. all dimensional scales and parameters are declared;
5. boundary states are rejected or treated by a declared boundary-aware contract;
6. no theorem is asserted outside its structural or projection domain.

PROCEDURE DIVISION

8. Well-Defined Composition

Theorem 8.1 — Projection composition theorem

Let

$$\mathcal{P} = (\Omega, \Omega_{\mathcal{P}}, \Phi, \mathbf{F}, \Gamma_{\theta}, U_{\mathcal{P}}, W_{\mathcal{P}})$$

be a lawful interior projection contract.

Then the prediction map

$$\hat{y}_{\mathcal{P}} = \Gamma_{\theta} \circ E_{\mathbf{F}} \circ \Phi$$

is well-defined on $\Omega_{\mathcal{P}}$.

Proof. Take any

$$\omega \in \Omega_{\mathcal{P}}.$$

Because the contract is lawful,

$$\Phi(\omega) \in \mathcal{D}_{\mathbf{F}}.$$

Therefore,

$$E_{\mathbf{F}}(\Phi(\omega))$$

is defined and belongs to the declared structural-state domain

$$\mathcal{S}_{\mathcal{P}}.$$

Because

$$\Gamma_{\theta} : \mathcal{S}_{\mathcal{P}} \rightarrow \mathcal{Y},$$

the value

$$\Gamma_{\theta}(E_{\mathbf{F}}(\Phi(\omega)))$$

is defined and belongs to $\mathcal{Y}$.

This holds for every

$$\omega \in \Omega_{\mathcal{P}}.$$

Therefore, the composed prediction map is well-defined on the validity domain.

□

Corollary 8.2 — No lawful prediction outside the contract

For

$$\omega \notin \Omega_{\mathcal{P}},$$

the contract produces a failure witness rather than a prediction:

$$W_{\mathcal{P}}(\omega) \in \mathfrak{W}.$$

Extending the prediction beyond $\Omega_{\mathcal{P}}$ requires a new or enlarged contract.

9. Preservation of Structural Identities

Theorem 9.1 — Identity preservation under phase inference

Let

$$F, G \in \mathcal{F}$$

satisfy

$$F(x) = G(x)$$

for every

$$x \in S \subseteq \mathcal{D}.$$

Let

$$\Phi : \Omega_{\mathcal{P}} \rightarrow S.$$

Then

$$\boxed{F(\Phi(\omega)) = G(\Phi(\omega))}$$

for every

$$\omega \in \Omega_p.$$

Proof. For each valid ω ,

$$\Phi(\omega) \in S.$$

The structural identity holds at every point of S . Therefore, it holds at the particular point $\Phi(\omega)$:

$$F(\Phi(\omega)) = G(\Phi(\omega)).$$

□

Theorem 9.2 — Identity preservation under readout

Under the assumptions of Theorem 9.1, let

$$\Gamma_\theta$$

be any function defined on the common values of F and G .

Then

$$\boxed{\Gamma_\theta(F(\Phi(\omega))) = \Gamma_\theta(G(\Phi(\omega)))}$$

for every valid ω .

Proof. Theorem 9.1 gives equal inputs to Γ_θ .

A function assigns the same output to equal inputs.

Therefore, the readout values are equal.

□

Corollary 9.3 — Projected UNA identity

For every lawful phase inference into $\mathcal{D}$,

$$\text{urx}(\Phi(\omega)) + \text{uxp}(\Phi(\omega)) = \frac{4}{\sin(2\Phi(\omega))}.$$

Any common readout applied to these equal quantities preserves their equality.

Corollary 9.4 — Projected Flatwave identity

For every lawful phase inference,

$$\text{Flatwave}(\Phi(\omega)) = \text{sgn}(\sin(2\Phi(\omega))).$$

The projection does not weaken the structural theorem.

10. Limits of the Converse

Projected agreement does not generally prove structural equality.

Theorem 10.1 — Injective converse on the inferred image

Let

$$F, G \in \mathcal{F},$$

and suppose

$$\Gamma_\theta$$

is injective on the union of the values

$$F(\Phi(\Omega_{\mathcal{P}})) \cup G(\Phi(\Omega_{\mathcal{P}})).$$

If

$$\Gamma_\theta(F(\Phi(\omega))) = \Gamma_\theta(G(\Phi(\omega)))$$

for every valid ω , then

$$\boxed{F(x) = G(x)}$$

for every

$$x \in \Phi(\Omega_{\mathcal{P}}).$$

Proof. For every valid ω ,

$$\Gamma_\theta(F(\Phi(\omega))) = \Gamma_\theta(G(\Phi(\omega))).$$

Because Γ_θ is injective on the relevant values,

$$F(\Phi(\omega)) = G(\Phi(\omega)).$$

Therefore, F and G agree at every point in the image

$$\Phi(\Omega_{\mathcal{P}}).$$

□

Corollary 10.2 — Structural equality requires phase coverage

If, in addition,

$$\Phi(\Omega_{\mathcal{P}}) = S,$$

then

$$F = G$$

throughout S .

If the inferred image is only a proper subset of S , equality outside the sampled image does not follow.

Negative Theorem 10.3 — Noninjective readout cannot establish structural equality

If Γ_{θ} is noninjective, projected equality may occur even when the structural values differ.

Proof. Since Γ_{θ} is noninjective, there exist distinct values

$$a \neq b$$

such that

$$\Gamma_{\theta}(a) = \Gamma_{\theta}(b).$$

Choose structural expressions and an inferred phase for which

$$F(\Phi(\omega)) = a$$

and

$$G(\Phi(\omega)) = b.$$

Then the projected outputs agree even though the structural values do not.
Therefore, projected agreement alone does not prove structural equality.

□

Example 10.4 — Sign-only readout

Suppose

$$\Gamma(u) = \text{sgn}(u).$$

Then every positive structural value has the same projected output $+1$, regardless of its magnitude.
A sign-only readout cannot distinguish

$$1, \quad 2, \quad 100.$$

11. Dimensional Isolation

Definition 11.1 — Structural dimensionlessness

The phase variable x , the Book 0 primitive operators, UNA operators, saw responses, Flatwave, p , q , and λ are dimensionless.

Definition 11.2 — Dimension-bearing element

A dimension-bearing element is any external datum, parameter, or output assigned a physical or conventional unit.

Examples include units of:

- seconds;
- metres;
- kilograms;
- kelvin;
- volts;
- dollars.

Theorem 11.3 — Dimensional isolation theorem

In a lawful projection contract, every dimension carried by the output must enter through at least one of:

1. the external instance data;
2. the calibration parameter θ ;
3. the readout map Γ_θ ;
4. the declared structure of the output space $\mathcal{Y}$.

No dimensional unit is generated by the Book 0 structural profile alone.

Proof. The evaluated structural state

$$E_{\mathbf{F}}(\Phi(\omega))$$

is a tuple of dimensionless real values.

A composition of dimensionless structural expressions remains dimensionless until an operation involving a dimension-bearing element is applied.

By definition, the only post-structural operation in the contract is the readout

$$\Gamma_\theta.$$

Any external data or calibration parameters used by this map must be explicitly declared.

Therefore, every output dimension must enter through the external or readout layer rather than arising from the Book 0 expressions alone.

□

Corollary 11.4 — Identity readout remains dimensionless

If

$$\Gamma(u) = u,$$

then the projected output remains dimensionless.

Corollary 11.5 — A unit cannot be derived from a pure number alone

A dimensionless structural value cannot, without a declared scale or unit assignment, determine a dimensional quantity.

For example, the equation

$$\lambda = \frac{1}{2}$$

does not determine:

- one-half second;
- one-half metre;
- one-half joule;
- one-half dollar.

The unit and scale must enter through the contract.

12. Boundary Guard

Theorem 12.1 — Interior boundary guard

Let $\mathcal{P}$ be an interior projection contract using

$$\Phi : \Omega_{\mathcal{P}} \rightarrow \mathcal{D}.$$

If an external instance ω would require

$$\Phi(\omega) \in \mathbb{B},$$

then ω is not admissible under that contract.

Proof. By definition,

$$\mathcal{D} \cap \mathbb{B} = \emptyset.$$

An interior phase-inference map has codomain $\mathcal{D}$. It therefore cannot lawfully assign a forbidden boundary phase.

Moreover, the primitive and UNA expressions are not defined at those points.

Hence the contract must reject the instance or replace the interior map with a boundary-aware contract.

□

Corollary 12.2 — Required boundary failure witness

An interior contract encountering a boundary state must return a witness such as

$$\text{PHASE-AT-FORBIDDEN-BOUNDARY.}$$

It may not silently evaluate the interior formulas by substituting a boundary value.

Theorem 12.3 — Boundary-aware admissibility

A boundary-aware contract may use

$$\widehat{\Phi} : \Omega_{\mathcal{P}} \rightarrow \widehat{\mathcal{D}}$$

provided that:

1. the entering or exiting boundary copy is declared;
2. every selected structural quantity has a proved extension to that copy;
3. no divergent primitive or UNA quantity is assigned a finite value without an additional rule;
4. no seam transition is inferred merely from the boundary state.

Proof. S0.V proves continuous split-boundary extensions for the bounded transfer coordinate, saw responses, and Flatwave.

It does not provide finite extensions for all primitive and UNA expressions, nor does it provide a seam map.

Therefore, a boundary-aware contract is lawful only for the quantities and side labels actually supplied by S0.V or by an explicitly added extension rule.

□

13. Reparameterization and Non-Identification

Different internal descriptions may produce the same external prediction.

Definition 13.1 — Admissible phase reparameterization

Let

$$S, T \subseteq \mathcal{D}.$$

An admissible phase reparameterization is a bijection

$$\boxed{h : S \rightarrow T}$$

such that, for the selected structural profile,

$$\widetilde{\mathbf{F}} = \mathbf{F} \circ h^{-1}$$

is defined on T .

When a contract requires $\widetilde{\mathbf{F}}$ to remain inside a restricted structural class, that requirement must also be satisfied.

Theorem 13.2 — Reparameterization invariance

Let

$$\Phi : \Omega_{\mathcal{P}} \rightarrow S$$

and define

$$\tilde{\Phi} = h \circ \Phi.$$

Define

$$\tilde{\mathbf{F}} = \mathbf{F} \circ h^{-1}.$$

Then

$$\boxed{\tilde{\mathbf{F}} \circ \tilde{\Phi} = \mathbf{F} \circ \Phi.}$$

Consequently, for the same readout map,

$$\boxed{\Gamma_{\theta} \circ \tilde{\mathbf{F}} \circ \tilde{\Phi} = \Gamma_{\theta} \circ \mathbf{F} \circ \Phi.}$$

Proof. For any valid ω ,

$$\tilde{\mathbf{F}}(\tilde{\Phi}(\omega)) = \mathbf{F}(h^{-1}(h(\Phi(\omega)))) .$$

Since h is bijective,

$$h^{-1}(h(\Phi(\omega))) = \Phi(\omega).$$

Therefore,

$$\tilde{\mathbf{F}}(\tilde{\Phi}(\omega)) = \mathbf{F}(\Phi(\omega)).$$

Applying the same readout map preserves equality.

□

Corollary 13.3 — Prediction does not uniquely identify phase

External prediction data alone do not necessarily determine a unique phase-inference map.

A different phase map may be compensated by a corresponding transformed structural profile.

Corollary 13.4 — Identification conditions are additional

To identify a unique phase description, a contract must impose additional conditions such as:

- fixing the structural profile in advance;
- fixing a phase origin;
- fixing orientation;
- fixing a branch convention;
- requiring a particular symmetry;
- requiring a particular calibration form;
- requiring identifiability from data.

These conditions do not follow from Book 0.

14. Structural Truth and Empirical Adequacy

Definition 14.1 — Structural truth

A statement is structurally true when it is proved for the Book 0 functions on its declared mathematical domain.

Example:

$$\text{Flatwave}(x) = \text{sgn}(\sin 2x)$$

for every

$$x \in \mathcal{D}.$$

Definition 14.2 — Contract validity

A projection contract is valid for an external domain when its assumptions, inference rule, readout, and calibration are appropriate on its declared validity set.

Definition 14.3 — Empirical adequacy

A projection is empirically adequate when its outputs agree with relevant observations to the standard declared by the contract.

Theorem 14.4 — Structural truth does not imply empirical adequacy

The correctness of a Book 0 identity does not, by itself, establish that a proposed external projection is empirically adequate.

Proof. A projected prediction depends on three distinct components:

$$\Phi, \quad \mathbf{F}, \quad \Gamma_{\theta}.$$

A structural theorem constrains $\mathbf{F}$.

It does not prove that:

- Φ assigns the correct phase to the external instance;
- the selected structural profile is relevant to the external phenomenon;
- Γ_{θ} is the correct readout;
- θ is correctly calibrated;
- the uncertainty model is adequate.

Therefore, a true structural identity may participate in an inaccurate projection.

□

Theorem 14.5 — Empirical agreement does not prove structural necessity

A projection may agree with observations without proving that its internal structural representation is uniquely necessary.

Proof. The reparameterization theorem shows that distinct internal descriptions may generate identical predictions.

Noninjective readouts may also erase structural distinctions.

Finite observations may fail to cover the full structural domain.

Therefore, empirical agreement establishes compatibility with the tested observations, not unique structural necessity.

□

Corollary 14.6 — Three separate burdens

Every external application carries three separate burdens:

1. mathematical correctness of the structural expressions;
2. contract validity of the inference and readout;

3. empirical adequacy of the resulting predictions.

Passing one burden does not automatically pass the others.

15. Minimal Projection Contract

Theorem 15.1 — Minimal field count

A projection cannot be fully specified without at least the following four functional elements:

1. a validity domain;
2. a phase-inference map;
3. a structural profile;
4. a readout map.

Proof. Without a validity domain, the contract does not state where it applies.

Without phase inference, no external instance is assigned a structural phase.

Without a structural profile, no Book 0 quantity is selected.

Without a readout map, no structural value becomes an external output.

Therefore, removing any one of the four elements leaves the external prediction undefined or its domain unspecified.

□

Corollary 15.2 — Operational completeness requires more than mathematical composition

For responsible empirical use, a contract must additionally declare:

- calibration parameters;
- uncertainty;
- failure behavior.

These do not alter the formal composition but are required to make its use auditable.

16. Abstract Worked Projection

This example demonstrates the contract form without assigning a physical interpretation.

Definition 16.1 — External phase-labelled instances

Let

$$\Omega$$

be a set of instances for which a phase-inference rule

$$\Phi : \Omega_{\mathcal{P}} \rightarrow Q_k$$

has been declared.

Definition 16.2 — Structural profile

Choose the single bounded structural function

$$\mathbf{F} = (\lambda).$$

Thus,

$$E_{\mathbf{F}}(x) = \lambda(x).$$

From Book 0,

$$0 < \lambda(x) < 1$$

on the open quadrant.

Definition 16.3 — Affine readout

Let

$$a, b \in \mathbb{R}$$

with

$$a < b.$$

Define

$$\Gamma_{a,b}(u) = a + (b - a)u.$$

The parameters a and b may carry units if the output space is dimensional.

Definition 16.4 — Projected output

$$\hat{y}(\omega) = a + (b - a)\lambda(\Phi(\omega)).$$

Theorem 16.5 — Bounded affine projection

For every valid ω ,

$$a < \hat{y}(\omega) < b.$$

Proof. Since

$$0 < \lambda(\Phi(\omega)) < 1$$

and

$$b - a > 0,$$

we have

$$0 < (b - a)\lambda(\Phi(\omega)) < b - a.$$

Adding a gives

$$a < a + (b - a)\lambda(\Phi(\omega)) < b.$$

Therefore,

$$a < \hat{y}(\omega) < b.$$

□

Corollary 16.6 — Dimensional source of the bounds

If a and b carry units, then $\hat{y}$ carries those units.

The units originate in the calibration parameters, not in λ .

Clarification 16.7 — What the example does not prove

The example does not prove that any actual phenomenon should be modeled by λ .

It proves only that, once such a projection contract is declared, its output is bounded between its declared anchors.

17. Projection Contract Ledger

Every later projection essay or applied model should begin with the following ledger.

17.1 External declaration

Ω declared external instance space.

17.2 Validity declaration

$\Omega_{\mathcal{P}}$ declared valid subset.

17.3 Phase declaration

Φ declared phase-inference map.

17.4 Structural declaration

$\mathbf{F}$ declared Book 0 profile.

17.5 Readout declaration

Γ_{θ} declared conversion to external output.

17.6 Calibration declaration

θ declared external scales and parameters.

17.7 Uncertainty declaration

$U_{\mathcal{P}}$ declared uncertainty assignment.

17.8 Failure declaration

$W_{\mathcal{P}}$ declared failure behavior.

17.9 Prediction declaration

$$\hat{y}_{\mathcal{P}} = \Gamma_{\theta} \circ E_{\mathbf{F}} \circ \Phi.$$

No applied claim is complete until these fields are supplied.

18. Negative Theorems of S1.I

Negative Theorem 18.1 — No external meaning by resemblance

Similarity between a structural graph and an observed graph does not, by itself, define a projection contract.

Negative Theorem 18.2 — No hidden units

A dimensional unit may not be attributed to a Book 0 quantity unless the unit enters through a declared external or readout element.

Negative Theorem 18.3 — No silent boundary substitution

An interior structural expression may not be evaluated at an excluded boundary by inserting an arbitrary finite value.

Negative Theorem 18.4 — No inference from output equality alone

Equal projected outputs do not prove equal structural states unless the readout is injective on the relevant image.

Negative Theorem 18.5 — No unique phase from fit alone

A successful external fit does not, by itself, establish a unique phase-inference map.

Negative Theorem 18.6 — No physical law from structural identity alone

A Book 0 theorem becomes part of a physical or empirical model only through a declared and tested projection contract.

19. Terminal Theorem

Theorem 19.1 — Projection Separation Theorem

Let Book 0 provide a structural expression class $\mathcal{F}$ on $\mathcal{D}$.

Every lawful external prediction using that structure factors through three logically distinct stages:

external instance $\longrightarrow$ inferred phase $\longrightarrow$ structural state $\longrightarrow$ external output.

More formally,

$$\hat{y}_{\mathcal{P}} = \Gamma_{\theta} \circ E_{\mathbf{F}} \circ \Phi.$$

The structural theorems constrain $E_{\mathbf{F}}$. They do not independently determine Φ , Γ_{θ} , θ , the validity domain, uncertainty, or failure behavior.

Proof. The factorization is the defining form required by Axiom P.1 and Definition 7.1.

Book 0 defines and constrains the structural expressions comprising $E_{\mathbf{F}}$.

The external instance space, phase inference, readout, calibration, uncertainty, and failure behavior are introduced only in the projection contract.

They are therefore logically distinct from the Book 0 structural calculus.

□

20. Closing Declaration

Book 0 answered:

What does the bounded phase calculus prove?

Book 1 begins by asking:

How may that structure be lawfully applied?

The answer established in S1.I is the projection contract:

$$\mathcal{P} = (\Omega, \Omega_{\mathcal{P}}, \Phi, \mathbf{F}, \Gamma_{\theta}, U_{\mathcal{P}}, W_{\mathcal{P}}).$$

Its prediction map is

$$\hat{y}_{\mathcal{P}} = \Gamma_{\theta} \circ E_{\mathbf{F}} \circ \Phi.$$

Exact structural identities survive lawful composition.

Dimensional quantities enter only through declared external or readout elements.

Boundary states require explicit side-aware treatment.

Projected agreement does not guarantee structural uniqueness.

Structural truth, contract validity, and empirical adequacy remain separate burdens.

Thus the opening law of Book 1 is:

No external interpretation without explicit inference;
no measurement without declared readout;
no unit without declared scale;
no application outside its validity contract.

END OF S1.I

S1.II

Admissibility, Failure Witnesses, and Falsification

Book 1 - Projection and Measurement

Abstract

S1.I defined a projection contract as the explicit composition

$$\Omega_{\mathcal{P}} \xrightarrow{\Phi} \mathcal{D} \xrightarrow{E_F} \mathbb{R}^m \xrightarrow{\Gamma_\theta} \mathcal{Y}.$$

This essay determines how such a contract is audited.

A projection is not admitted merely because its formulas can be evaluated at selected examples. Every instance must pass a declared sequence of guards concerning domain membership, phase inference, branch selection, structural evaluation, boundary status, calibration, readout, and uncertainty.

For each external instance ω , a failure-witness set

$$\mathcal{W}_{\mathcal{P}}(\omega) \subseteq \mathfrak{W}$$

records every failed guard. An instance is admissible exactly when this set is empty.

The ordinary partial prediction map is then extended to a total diagnostic evaluator

$$\mathcal{E}_{\mathcal{P}} : \Omega \longrightarrow \mathcal{Y} \sqcup \mathcal{P}^+(\mathfrak{W}),$$

which returns either a projected output or a nonempty set of failure witnesses.

This construction distinguishes four logically different outcomes:

1. admissible prediction;
2. undefined evaluation;
3. contract invalidity or insufficiency;
4. empirical contradiction.

Failure to produce a prediction is not empirical falsification. A lawful empirical contradiction can be declared only after the instance has passed the projection guards and the predicted and observed uncertainty sets are disjoint.

A mismatch between an external observation and a projection does not falsify a Book 0 identity. It falsifies, at most, the particular projection contract, calibration, phase-inference rule, or external interpretation under test.

The essay also proves that restriction of a lawful validity domain preserves lawfulness, whereas extension to a larger domain requires a new admissibility proof. It introduces repair locality: a failure witness identifies the contract field that must be revised without silently altering unrelated fields.

The result is a complete operational airlock between structural mathematics and external claims.

IDENTIFICATION DIVISION

PROGRAM-ID

S1-II-ADMISSIBILITY-AND-FAILURE

PURPOSE

To define the procedures by which a proposed projection is:

- admitted;
- rejected;
- diagnosed;
- repaired;
- empirically tested;
- or declared underdetermined.

PRINCIPAL QUESTION

Given a projection contract $\mathcal{P}$ and an external instance ω , what result is logically justified?

NON-PURPOSE

This essay does not:

- choose a particular physical application;
- define a universal statistical test;
- guarantee that every failure can be repaired;
- prove that a successful projection is unique;
- convert an undefined expression into an empirical refutation;

- permit silent extrapolation beyond a declared domain;
- alter any theorem of Book 0.

ENVIRONMENT DIVISION

1. Inherited Projection Contract

From S1.I, a projection contract is

$$\mathcal{P} = (\Omega, \Omega_{\mathcal{P}}, \Phi, \mathbf{F}, \Gamma_{\theta}, U_{\mathcal{P}}, W_{\mathcal{P}}).$$

Here:

- Ω is the external instance space;
- $\Omega_{\mathcal{P}} \subseteq \Omega$ is the declared validity domain;
- Φ is the phase-inference map;
- $\mathbf{F}$ is the selected Book 0 structural profile;
- Γ_{θ} is the readout map;
- θ contains all calibration parameters;
- $U_{\mathcal{P}}$ assigns projection uncertainty;
- $W_{\mathcal{P}}$ reports failure.

For an admissible instance,

$$\hat{y}_{\mathcal{P}}(\omega) = \Gamma_{\theta}(E_{\mathbf{F}}(\Phi(\omega))).$$

1.1 Interior and boundary-aware forms

An interior contract uses

$$\Phi : \Omega_{\mathcal{P}} \rightarrow \mathcal{D}.$$

A boundary-aware contract uses

$$\widehat{\Phi} : \Omega_{\mathcal{P}} \rightarrow \widehat{\mathcal{D}}.$$

The two forms may not be silently interchanged.

1.2 No new structural axiom

S1.II introduces no new statement about the Book 0 operators.

Its definitions govern only the logical and operational status of external projections.

DATA DIVISION

2. Evaluation Outcomes

Definition 2.1 — Admitted instance

An external instance

$$\omega \in \Omega$$

is admitted by $\mathcal{P}$ when every required projection guard succeeds.

Definition 2.2 — Rejected instance

An instance is rejected by $\mathcal{P}$ when at least one required guard fails.

Rejection is contract-relative. The same instance may be admissible under another contract.

Definition 2.3 — Undefined evaluation

An evaluation is undefined when one of the required mathematical functions is not defined at the value supplied to it.

Examples include:

- an inferred phase at a forbidden wall;
- a zero structural denominator;
- a readout outside its declared input domain.

Undefinedness is a mathematical status, not an empirical discrepancy.

Definition 2.4 — Contract insufficiency

A contract is insufficient for an instance when it lacks information required to decide or evaluate that instance.

Examples include:

- no phase can be inferred;

- two branches remain equally admissible;
- calibration is missing;
- uncertainty is required but unavailable.

Definition 2.5 — Empirical contradiction

An empirical contradiction occurs when:

1. the instance is admitted;
2. a lawful prediction exists;
3. a lawful observation record exists;
4. the predicted and observed admissible-output sets are disjoint.

This definition is formalized in Section 11.

Definition 2.6 — Underdetermined outcome

An outcome is underdetermined when available information permits more than one contract-consistent conclusion and the contract supplies no selection rule.

Underdetermination is neither successful prediction nor empirical contradiction.

3. Guard System

Definition 3.1 — Guard

A guard is a Boolean condition

$$G_j(\mathcal{P}, \omega) \in \{\text{true}, \text{false}\}$$

that must be satisfied before a prediction is admitted.

Definition 3.2 — Contract-completeness guard

$$G_0(\mathcal{P})$$

is true exactly when all required contract fields are present and internally typed.

At minimum, the contract must declare:

- Ω ;
- $\Omega_{\mathcal{P}}$;

- Φ or $\widehat{\Phi}$;
- $\mathbf{F}$;
- Γ_θ ;
- θ ;
- $U_{\mathcal{P}}$;
- $W_{\mathcal{P}}$.

Definition 3.3 — Instance-domain guard

$$G_1(\mathcal{P}, \omega)$$

is true exactly when

$$\omega \in \Omega_{\mathcal{P}}.$$

Definition 3.4 — Phase-inference guard

$$G_2(\mathcal{P}, \omega)$$

is true exactly when the contract's declared procedure returns a phase value for ω .

Definition 3.5 — Phase-uniqueness guard

$$G_3(\mathcal{P}, \omega)$$

is true exactly when the phase-inference procedure returns one contract-admissible phase state, or when the contract explicitly permits a set-valued phase result and supplies a corresponding readout rule.

If two or more unresolved phase states remain and no set-valued rule is declared, then G_3 is false.

Definition 3.6 — Structural-domain guard

For an interior contract,

$$G_4(\mathcal{P}, \omega)$$

is true exactly when

$$\Phi(\omega) \in \mathcal{D}_{\mathbf{F}}.$$

For a boundary-aware contract, it is true exactly when:

1. the inferred split-boundary point is declared;
2. every selected structural quantity has a proved or explicitly assumed extension to that point.

Definition 3.7 — Structural-evaluation guard

$$G_5(\mathcal{P}, \omega)$$

is true exactly when every component of

$$E_{\mathbf{F}}(\Phi(\omega))$$

can be evaluated without division by zero, undefined branching, or an undeclared singular convention.

Definition 3.8 — Calibration guard

$$G_6(\mathcal{P}, \omega)$$

is true exactly when all calibration parameters required for the instance are:

- present;
- assigned;
- dimensionally compatible;
- and within their declared validity conditions.

Definition 3.9 — Readout-domain guard

$$G_7(\mathcal{P}, \omega)$$

is true exactly when the evaluated structural state belongs to the declared domain of Γ_{θ} .

Definition 3.10 — Uncertainty guard

$$G_8(\mathcal{P}, \omega)$$

is true exactly when the uncertainty description required by the contract can be produced.

A purely exact mathematical illustration may declare that no empirical uncertainty is being asserted. An empirical contract may not silently omit required uncertainty.

Definition 3.11 — Observation guard

When empirical comparison is attempted,

$$G_9(\mathcal{P}, \omega, \mathcal{O})$$

is true exactly when the observation record $\mathcal{O}$ is valid under the declared observation protocol.

4. Admissibility Predicate**Definition 4.1 — Instance admissibility**

Define

$$A_{\mathcal{P}}(\omega) = \bigwedge_{j=0}^8 G_j(\mathcal{P}, \omega).$$

Thus,

$$A_{\mathcal{P}}(\omega) = \text{true}$$

exactly when all non-observational projection guards succeed.

Definition 4.2 — Empirical-test admissibility

For an observation record $\mathcal{O}$, define

$$A_{\mathcal{P}}^{\text{test}}(\omega, \mathcal{O}) = A_{\mathcal{P}}(\omega) \wedge G_9(\mathcal{P}, \omega, \mathcal{O}).$$

Definition 4.3 — Maximal operational admissibility predicate

The instance-domain guard G_1 tests membership in the already declared set $\Omega_{\mathcal{P}}$. It must therefore not be used to define the maximal domain against which that declaration is assessed.

Define the non-domain operational predicate

$$A_{\mathcal{P}}^{\text{max}}(\omega) = G_0(\mathcal{P}) \wedge \bigwedge_{j=2}^8 G_j(\mathcal{P}, \omega).$$

The corresponding maximal operational admissibility envelope is

$$\Omega_{\mathcal{P}}^{\text{max}} = \{\omega \in \Omega : A_{\mathcal{P}}^{\text{max}}(\omega) = \text{true}\}.$$

The ordinary admissibility predicate factors as

$$A_{\mathcal{P}}(\omega) = G_1(\mathcal{P}, \omega) \wedge A_{\mathcal{P}}^{\max}(\omega).$$

Theorem 4.4 — Declared-domain lawfulness and maximality

A declared validity domain is operationally lawful when

$$\Omega_{\mathcal{P}} \subseteq \Omega_{\mathcal{P}}^{\max}.$$

It is maximal relative to the declared contract fields and guards when

$$\Omega_{\mathcal{P}} = \Omega_{\mathcal{P}}^{\max}.$$

Proof. If $\omega \in \Omega_{\mathcal{P}}$, then $G_1(\mathcal{P}, \omega)$ is true by definition. Lawful prediction additionally requires every non-domain guard to succeed, which is exactly the condition $\omega \in \Omega_{\mathcal{P}}^{\max}$. Hence lawfulness requires $\Omega_{\mathcal{P}} \subseteq \Omega_{\mathcal{P}}^{\max}$.

If equality holds, every instance admitted by the non-domain guards is included in the declared scope, so the declaration is maximal relative to those guards. If the inclusion is strict, the contract is conservatively scoped but remains lawful. $\square$

Clarification 4.5 — Conservative domains

A contract may intentionally choose a conservative validity domain

$$\Omega_{\mathcal{P}} \subsetneq \Omega_{\mathcal{P}}^{\max}.$$

Such a restriction is lawful when explicitly stated. The contract must not claim that the restricted domain is maximal.

5. Failure-Witness Sets

Definition 5.1 — Witness assignment by guard

To each guard G_j , assign a witness code

$$w_j \in \mathfrak{W}.$$

A recommended base vocabulary is:

j	w_j
0	CONTRACT-INCOMPLETE
1	INSTANCE-OUTSIDE-VALIDITY-DOMAIN
2	PHASE-NOT-INFERABLE
3	PHASE-OR-BRANCH-AMBIGUOUS
4	PHASE-OUTSIDE-STRUCTURAL-DOMAIN
5	STRUCTURAL-EVALUATION-UNDEFINED
6	CALIBRATION-MISSING-OR-INVALID
7	READOUT-OUTSIDE-DOMAIN
8	UNCERTAINTY-NOT-AVAILABLE
9	OBSERVATION-RECORD-INVALID

Definition 5.2 — Failure-witness set

For an external instance ω , define

$$\mathcal{W}_{\mathcal{P}}(\omega) = \{w_j : G_j(\mathcal{P}, \omega) = \text{false}, 0 \leq j \leq 8\}.$$

For an empirical test record,

$$\mathcal{W}_{\mathcal{P}}^{\text{test}}(\omega, \mathcal{O}) = \mathcal{W}_{\mathcal{P}}(\omega) \cup \{w_9 : G_9(\mathcal{P}, \omega, \mathcal{O}) = \text{false}\}.$$

Theorem 5.3 — Empty-witness criterion

$$A_{\mathcal{P}}(\omega) = \text{true} \iff \mathcal{W}_{\mathcal{P}}(\omega) = \emptyset.$$

Proof. By Definition 4.1, $A_{\mathcal{P}}(\omega)$ is true exactly when every guard $G_0, \dots, G_8$ is true.

By Definition 5.2, a witness is included exactly when its corresponding guard is false.

Therefore, every guard is true exactly when no witness is included.

□

Corollary 5.4 — Rejection criterion

$$A_{\mathcal{P}}(\omega) = \text{false} \iff \mathcal{W}_{\mathcal{P}}(\omega) \neq \emptyset.$$

Definition 5.5 — Primary witness

A contract may declare a precedence relation

$\prec$

on $\mathfrak{W}$.

When

$$\mathcal{W}_{\mathcal{P}}(\omega) \neq \emptyset,$$

a primary witness is a $\prec$ -minimal element of the witness set.

Clarification 5.6 — Primary witness need not be unique

If the precedence relation is partial rather than total, more than one minimal witness may exist.

The complete witness set should be preserved even when a primary witness is displayed.

6. Totalized Diagnostic Evaluation

Definition 6.1 — Disjoint outcome space

Let

$$\mathcal{Y} \sqcup \mathcal{P}^+(\mathfrak{W})$$

denote the disjoint union of:

- the output space $\mathcal{Y}$;
- the nonempty subsets of the witness space.

The disjoint union ensures that an output value cannot be confused with a failure code.

Definition 6.2 — Totalized evaluator

Define

$$\boxed{\mathcal{E}_{\mathcal{P}} : \Omega \rightarrow \mathcal{Y} \sqcup \mathcal{P}^+(\mathfrak{W})}$$

by

$$\mathcal{E}_{\mathcal{P}}(\omega) = \begin{cases} \hat{y}_{\mathcal{P}}(\omega), & \mathcal{W}_{\mathcal{P}}(\omega) = \emptyset, \\ \mathcal{W}_{\mathcal{P}}(\omega), & \mathcal{W}_{\mathcal{P}}(\omega) \neq \emptyset. \end{cases}$$

Theorem 6.3 — Totality

For every

$$\omega \in \Omega,$$

the evaluator $\mathcal{E}_{\mathcal{P}}(\omega)$ returns exactly one of:

1. an output in $\mathcal{Y}$;
2. a nonempty failure-witness set.

Proof. By classical logic, either

$$\mathcal{W}_{\mathcal{P}}(\omega) = \emptyset$$

or

$$\mathcal{W}_{\mathcal{P}}(\omega) \neq \emptyset.$$

The two conditions are mutually exclusive.

Definition 6.2 assigns one result in each case.

Therefore, the evaluator is total and single-valued.

□

Corollary 6.4 — No silent failure

A complete implementation of the contract cannot return neither an output nor a witness.

Corollary 6.5 — No output with unresolved failure

A complete implementation cannot lawfully return a projected output while retaining a nonempty unresolved witness set, unless the contract explicitly classifies that witness as a warning rather than a failed guard.

7. Failure Taxonomy

Definition 7.1 — Contract-definition failure

A contract-definition failure concerns the specification itself rather than one particular external instance.

Examples include:

- missing readout map;
- undeclared units;

- incomplete parameter list;
- no failure behavior;
- contradictory domain declarations.

Definition 7.2 — Instance-admission failure

An instance-admission failure occurs when the contract is complete but the particular instance lies outside its lawful applicability.

Example:

$$\omega \notin \Omega_p.$$

Definition 7.3 — Inference failure

An inference failure occurs when phase or branch cannot be determined under the contract.

Definition 7.4 — Structural failure

A structural failure occurs when the inferred state cannot be lawfully evaluated by the selected Book 0 expressions.

Definition 7.5 — Readout failure

A readout failure occurs when the structural state exists but cannot be converted to an output under the declared calibration and readout map.

Definition 7.6 — Uncertainty failure

An uncertainty failure occurs when the contract requires an uncertainty statement but none can be lawfully produced.

Definition 7.7 — Observation failure

An observation failure occurs when the empirical record is missing, malformed, outside protocol, or otherwise unsuitable for comparison.

Definition 7.8 — Empirical contradiction witness

An empirical contradiction witness records a valid mismatch after every admission and observation guard has passed.

Recommended code:

PREDICTION-OBSERVATION-DISJOINT.

Theorem 7.9 — Failure-class separation

Contract-definition failure, undefined evaluation, and empirical contradiction are logically distinct.

Proof. A contract-definition failure concerns whether a projection has been completely specified.

Undefined evaluation concerns whether declared mathematical functions possess a value at the supplied arguments.

Empirical contradiction concerns disagreement between a lawfully produced prediction and a lawfully obtained observation.

Each may occur without either of the others.

For example:

- a contract may be incomplete before any instance is evaluated;
- a complete contract may infer a forbidden phase, producing undefined evaluation;
- a complete and evaluable contract may produce a prediction contradicted by observation.

Therefore, the classes are distinct.

□

8. Undefinedness Is Not Falsification**Theorem 8.1 — Non-evaluation theorem**

If

$$\mathcal{W}_{\mathcal{P}}(\omega) \neq \emptyset,$$

then the instance does not produce a lawful projected output under $\mathcal{P}$.

Consequently, that instance cannot by itself empirically contradict the contract's prediction.

Proof. A nonempty witness set implies, by Corollary 5.4, that the instance is not admitted.

By Definition 6.2, the evaluator returns witnesses rather than a projected output.

Without a projected output, there is no predicted value or predicted admissible-output set to compare with an observation.

Therefore, the instance demonstrates non-applicability or insufficiency, not empirical contradiction.

□

Corollary 8.2 — Boundary failure is not a failed prediction

If an interior contract returns

$$\text{PHASE-OUTSIDE-STRUCTURAL-DOMAIN}$$

because the inferred phase lies at

$$x = \frac{k\pi}{2},$$

the result is a domain failure.

It is not evidence that a numerical prediction was wrong.

Corollary 8.3 — Ambiguity is not contradiction

If two unresolved phases remain admissible and the contract has no branch-selection rule, the result is underdetermination.

It is not a contradiction between the model and the observation.

9. Failure Soundness and Relative Completeness

Definition 9.1 — Sound witness

A witness is sound when its presence implies that the associated guard has genuinely failed.

Definition 9.2 — Guard-complete witness system

A witness system is guard-complete when every failed guard produces at least one witness.

Theorem 9.3 — Soundness of the canonical witness system

The witness system of Section 5 is sound.

Proof. By Definition 5.2, w_j is included only when

$$G_j = \text{false}.$$

Therefore, the presence of w_j directly certifies failure of its corresponding guard.

□

Theorem 9.4 — Relative completeness

The witness system is complete relative to the declared guard set $G_0, \dots, G_8$.

Proof. If a declared guard fails, its corresponding witness is included by Definition 5.2.

Therefore, no failure among the declared guards is left unwitnessed.

□

Clarification 9.5 — Relative completeness is not universal completeness

A contract may omit a guard that later proves necessary.

The witness system cannot diagnose a condition the contract has not represented.

Thus guard completeness is always relative to the declared audit structure.

10. Prediction and Observation Sets

A single predicted number is often insufficient to express calibration and uncertainty. We therefore compare sets of admissible outputs.

Definition 10.1 — Predicted output set

For an admitted instance ω , let

$$P_{\mathcal{P}}(\omega) \subseteq \mathcal{Y}$$

be the set of outputs permitted by the projection prediction and its declared uncertainty.

For an exact deterministic prediction,

$$P_{\mathcal{P}}(\omega) = \{\hat{y}_{\mathcal{P}}(\omega)\}.$$

Definition 10.2 — Observation record

An observation record is a tuple

$$\mathcal{O} = (\omega, \mathcal{M}, y_{\text{obs}}, U_{\text{obs}})$$

where:

- ω identifies the observed instance;
- $\mathcal{M}$ is the declared measurement procedure;

- y_{obs} is the reported observation;
- U_{obs} is its uncertainty description.

Definition 10.3 — Observed output set

A lawful observation record determines a set

$$O_{\mathcal{O}} \subseteq \mathcal{Y}$$

of outputs compatible with the observation and its declared uncertainty.

For an exact observation,

$$O_{\mathcal{O}} = \{y_{\text{obs}}\}.$$

Definition 10.4 — Empirical compatibility

The prediction and observation are compatible when

$$P_{\mathcal{P}}(\omega) \cap O_{\mathcal{O}} \neq \emptyset.$$

Definition 10.5 — Record-level empirical contradiction

The prediction and observation contradict one another when

$$P_{\mathcal{P}}(\omega) \cap O_{\mathcal{O}} = \emptyset.$$

11. Falsification Conditions

Theorem 11.1 — Record-level contradiction theorem

A record-level empirical contradiction is lawfully established exactly when:

1. $A_{\mathcal{P}}(\omega) = \text{true}$;
2. $G_{\mathcal{G}}(\mathcal{P}, \omega, \mathcal{O}) = \text{true}$;
3. both $P_{\mathcal{P}}(\omega)$ and $O_{\mathcal{O}}$ are defined;
4. their intersection is empty.

Proof. Conditions 1–3 establish that both the prediction and observation are lawful objects in the common output space.

Condition 4 states that no output is compatible with both.

Therefore, the prediction and observation cannot simultaneously be correct under the declared contract and observation protocol.

Conversely, if any of Conditions 1–3 fails, the comparison is not lawfully formed. If Condition 4 fails, at least one common output remains possible.

Thus all four conditions are necessary and sufficient.

□

Corollary 11.2 — Exact deterministic case

If both prediction and observation are exact singletons, contradiction occurs exactly when

$$\hat{y}_{\mathcal{P}}(\omega) \neq y_{\text{obs}}.$$

Definition 11.3 — Contract-level rejection rule

A contract-level empirical decision rule is a declared function

$$\Delta : \mathcal{R} \rightarrow \{\text{retain, revise, reject, undetermined}\}.$$

where $\mathcal{R}$ is a permitted collection of observation records and contradiction results.

Clarification 11.4 — One contradiction need not reject every contract

Whether one record-level contradiction rejects an entire contract depends on the declared decision rule.

The contradiction may instead indicate:

- a bad observation;
- a calibration failure;
- a phase-inference error;
- an incorrect validity domain;
- an omitted variable;
- or failure of the complete projection hypothesis.

The decision rule must state how these possibilities are treated.

12. What Empirical Contradiction Can Falsify

Theorem 12.1 — Projection-specific falsification

Suppose a lawful observation contradicts the output of $\mathcal{P}$.

Then at least one empirical or projection-layer claim used by $\mathcal{P}$ is false, inadequate, or incomplete.

The contradiction does not, by itself, falsify an exact Book 0 identity.

Proof. The prediction has the form

$$\hat{y}_{\mathcal{P}} = \Gamma_{\theta} \circ E_{\mathbf{F}} \circ \Phi.$$

A contradiction can arise from failure of:

- the phase-inference map Φ ;
- relevance of the selected structural profile $\mathbf{F}$;
- calibration θ ;
- readout Γ_{θ} ;
- validity-domain declaration;
- uncertainty model;
- observation protocol;
- or an external assumption coupling these components.

The Book 0 identity constrains the structural expressions internally on their mathematical domain.

An external mismatch does not exhibit a counterexample to that mathematical identity unless it supplies an admissible $x \in \mathcal{D}$ for which the two sides of the identity are unequal.

Therefore, the contradiction targets the projection or external model, not automatically the structural theorem.

□

Corollary 12.2 — Structural identity versus relevance hypothesis

A structural identity may remain true while the hypothesis that it models a particular external system is false.

Corollary 12.3 — Failed physical interpretation

If a projection interpreting Flatwave as a particular conserved physical quantity is contradicted, the contradiction may falsify that interpretation without affecting

$$\text{Flatwave} = \text{sgn}(\sin 2x).$$

13. Successful Prediction and Its Limits

Theorem 13.1 — Compatibility is weaker than confirmation of uniqueness

If

$$P_{\mathcal{P}}(\omega) \cap O_{\mathcal{O}} \neq \emptyset,$$

then the projection is compatible with that observation record.

It does not follow that:

- the projection is uniquely correct;
- the phase inference is unique;
- the calibration is uniquely determined;
- competing projections are false;
- the structural profile is necessary.

Proof. Compatibility means only that at least one output is permitted by both prediction and observation.

S1.I proved that reparameterized internal descriptions may produce the same output.

A noninjective readout may erase structural differences.

Multiple projections may have overlapping predicted sets.

Therefore, compatibility does not establish uniqueness or necessity.

□

Corollary 13.2 — Repeated success strengthens evidence but not logical necessity

A larger collection of successful tests may reduce the set of viable alternatives.

It does not transform empirical adequacy into a mathematical proof of unique ontology.

14. Domain Restriction

Definition 14.1 — Restricted contract

Let

$$\Omega_{\mathcal{P}'} \subseteq \Omega_{\mathcal{P}}.$$

Define the restricted contract $\mathcal{P}'$ by retaining the same:

- phase-inference rule;
- structural profile;
- readout;
- calibration;
- uncertainty formalism;

but limiting evaluation to $\Omega_{\mathcal{P}'}$.

Theorem 14.2 — Restriction preserves instance lawfulness

If $\mathcal{P}$ is lawful on $\Omega_{\mathcal{P}}$, then its restriction $\mathcal{P}'$ is lawful on every retained instance in $\Omega_{\mathcal{P}'}$.

Proof. Every retained instance already belonged to the lawful validity domain of $\mathcal{P}$.

All maps and guards are unchanged on those instances.

Therefore, every guard that succeeded under $\mathcal{P}$ continues to succeed under $\mathcal{P}'$.

□

Corollary 14.3 — Restriction reduces coverage

A restricted contract may become more conservative, but it makes fewer predictions.

Improved reliability due solely to removing difficult cases must not be reported as increased explanatory range.

15. Domain Extension

Definition 15.1 — Extended contract

An extension proposes a validity domain

$$\Omega_{\mathcal{P}^+} \supsetneq \Omega_{\mathcal{P}}.$$

Theorem 15.2 — Extension burden

Lawfulness on $\Omega_{\mathcal{P}}$ does not imply lawfulness on the added region

$$\Omega_{\mathcal{P}^+} \setminus \Omega_{\mathcal{P}}.$$

Every added instance must pass the complete guard sequence.

Proof. The original lawfulness proof concerns only instances in $\Omega_{\mathcal{P}}$.

No conclusion has been established for instances outside that set.

The added instances may require:

- new phase branches;
- extrapolated calibration;
- new structural domains;
- boundary handling;
- different uncertainty;
- or a new readout.

Therefore, the old proof does not cover the new region.

□

Corollary 15.3 — No silent extrapolation

A contract may not extend its prediction by merely evaluating the same formula outside its declared validity domain.

Formula evaluability is not sufficient evidence of projection validity.

Corollary 15.4 — Extension may create a new contract

When an extension changes inference, calibration, branch handling, or uncertainty, it should be assigned a new contract identifier.

16. Calibration Failure and Drift

Definition 16.1 — Calibration provenance

Calibration provenance records how each component of θ was obtained.

Possible statuses include:

- fixed by convention;
- directly measured;
- estimated from a training set;

- inferred jointly with the model;
- imported from another experiment.

Definition 16.2 — Calibration domain

Let

$$\Omega_\theta \subseteq \Omega$$

be the set on which the calibration is declared valid.

Theorem 16.3 — Calibration-domain guard

A contract using calibration θ is not lawful for

$$\omega \notin \Omega_\theta$$

unless an extrapolation rule is explicitly declared and tested.

Proof. The readout depends on θ .

The validity of that parameter has been established only on Ω_θ .

Outside this domain, the required calibration guard lacks support.

Therefore, the instance must be rejected or treated under a declared extrapolation contract.

□

Definition 16.4 — Calibration drift

Calibration drift occurs when a parameter treated as fixed changes with external conditions not represented by the contract.

Corollary 16.5 — Drift may mimic structural failure

An empirical contradiction produced by calibration drift does not prove that the selected Book 0 profile is structurally unsuitable.

The calibration and structural hypotheses must be tested separately where possible.

17. Branch Ambiguity

Definition 17.1 — Candidate phase set

For an external instance ω , let

$$C_{\Phi}(\omega) \subseteq \mathcal{D}$$

be the set of phase values compatible with the inference data before branch selection.

Definition 17.2 — Unique inference

Phase inference is unique when

$$|C_{\Phi}(\omega)| = 1.$$

Definition 17.3 — Branch ambiguity

Branch ambiguity occurs when

$$|C_{\Phi}(\omega)| > 1$$

and no declared rule selects one candidate or combines them.

Theorem 17.4 — Ambiguous phase produces ambiguous structural state

If there exist

$$x_1, x_2 \in C_{\Phi}(\omega)$$

such that

$$E_{\mathbf{F}}(x_1) \neq E_{\mathbf{F}}(x_2),$$

then the structural state is not uniquely determined by the instance.

Proof. Both x_1 and x_2 satisfy the inference data.

They produce distinct structural states.

Without a selection or aggregation rule, the contract does not determine which state to use.

□

Corollary 17.5 — Flatwave branch ambiguity

If two candidate phases lie in quadrants of opposite sign, then

$$\text{Flatwave}(x_1) = -\text{Flatwave}(x_2).$$

A contract cannot assign a unique Flatwave sign until the quadrant ambiguity is resolved.

18. Failure Locality and Contract Repair

Definition 18.1 — Contract field map

Associate each witness with the contract field or fields it directly concerns.

For example:

Witness	Primary field
PHASE-NOT-INFERABLE	Φ
PHASE-OR-BRANCH-AMBIGUOUS	Φ
STRUCTURAL-EVALUATION-UNDEFINED	$\mathbf{F}, \mathcal{D}_{\mathbf{F}}$
CALIBRATION-MISSING-OR-INVALID	θ
READOUT-OUTSIDE-DOMAIN	Γ_{θ}
UNCERTAINTY-NOT-AVAILABLE	$U_{\mathcal{P}}$

Definition 18.2 — Local repair

A repair is local when it modifies only the contract fields implicated by the witness, together with any fields that depend on them.

Theorem 18.3 — Repair noninterference condition

Suppose a repaired contract $\mathcal{P}'$ changes a field C .

Every theorem or validation result depending on C must be rechecked.

Results independent of C remain unchanged.

Proof. A derived result can change only if one of its premises or definitions changes.

If the result depends on C , changing C may alter the result and requires reevaluation.

If its dependency graph excludes C , the repair does not alter any of its premises.

Therefore, the result remains valid.

□

Corollary 18.4 — Calibration repair does not rewrite Book 0

Changing θ requires rechecking projected outputs.

It does not require reproving the structural identities of Book 0.

Corollary 18.5 — Phase-rule repair requires broad retesting

Changing Φ may alter every downstream structural state and projected output.

All instance-level results depending on phase inference must therefore be recomputed.

19. Audit Record

Definition 19.1 — Evaluation record

For each attempted instance, define an audit record

$$\mathcal{A}_{\mathcal{P}}(\omega) = (\text{id}_{\mathcal{P}}, \omega, \mathbf{G}, \Phi(\omega), E_{\mathbf{F}}(\Phi(\omega)), \theta, \hat{y}_{\mathcal{P}}(\omega), U_{\mathcal{P}}(\omega), \mathcal{W}_{\mathcal{P}}(\omega)),$$

with undefined entries explicitly marked when a prior guard fails.

Here

$$\mathbf{G} = (G_0, \dots, G_8)$$

is the guard vector.

Definition 19.2 — Contract identifier

Every materially distinct contract should have a distinct identifier or version.

A material change includes:

- validity-domain change;
- phase-inference change;
- structural-profile change;
- calibration change;
- readout change;
- uncertainty-model change;
- boundary-rule change.

Theorem 19.3 — Reproducibility requirement

A projected result is reproducible from its audit record when the record contains every non-derived input required by the contract.

Proof. The prediction is a deterministic composition of declared maps once all non-derived inputs and parameters are supplied.

If the audit record contains these values and the contract version, another evaluator can repeat the same composition.

If any required external input or parameter is absent, exact reproduction may be impossible.

□

PROCEDURE DIVISION

20. Canonical Evaluation Procedure

The following procedure expresses the projection airlock in a COBOL-like form.

EVALUATE-PROJECTION.

PERFORM CHECK-CONTRACT-COMPLETENESS.

IF CONTRACT-INCOMPLETE

RETURN FAILURE-WITNESS-SET

END-IF.

PERFORM CHECK-INSTANCE-DOMAIN.

IF INSTANCE-OUTSIDE-VALIDITY-DOMAIN

RETURN FAILURE-WITNESS-SET

END-IF.

PERFORM INFER-PHASE.

IF PHASE-NOT-INFERABLE

RETURN FAILURE-WITNESS-SET

END-IF.

PERFORM RESOLVE-BRANCH.

IF PHASE-OR-BRANCH-AMBIGUOUS

RETURN FAILURE-WITNESS-SET

END-IF.

PERFORM CHECK-STRUCTURAL-DOMAIN.

IF PHASE-OUTSIDE-STRUCTURAL-DOMAIN

RETURN FAILURE-WITNESS-SET

END-IF.

PERFORM EVALUATE-STRUCTURAL-PROFILE.

IF STRUCTURAL-EVALUATION-UNDEFINED

RETURN FAILURE-WITNESS-SET

END-IF.

PERFORM CHECK-CALIBRATION.

IF CALIBRATION-MISSING-OR-INVALID

RETURN FAILURE-WITNESS-SET

END-IF.

PERFORM CHECK-READOUT-DOMAIN.

IF READOUT-OUTSIDE-DOMAIN

RETURN FAILURE-WITNESS-SET

END-IF.

PERFORM APPLY-READOUT.

PERFORM ASSIGN-UNCERTAINTY.

IF UNCERTAINTY-NOT-AVAILABLE

RETURN FAILURE-WITNESS-SET

END-IF.

RETURN PROJECTED-OUTPUT

PROJECTED-UNCERTAINTY

COMPLETE-AUDIT-RECORD.

20.1 Empirical comparison procedure

COMPARE-WITH-OBSERVATION.

PERFORM EVALUATE-PROJECTION.

IF FAILURE-WITNESS-SET IS NONEMPTY

RETURN NOT-TESTABLE-UNDER-THIS-CONTRACT

END-IF.

PERFORM VALIDATE-OBSERVATION-RECORD.

IF OBSERVATION-RECORD-INVALID

RETURN OBSERVATION-FAILURE

END-IF.

FORM PREDICTED-OUTPUT-SET.

FORM OBSERVED-OUTPUT-SET.

IF INTERSECTION IS EMPTY

RETURN PREDICTION-OBSERVATION-DISJOINT

ELSE

RETURN EMPIRICALLY-COMPATIBLE

END-IF.

21. Theorem Ledger

Theorem II.1 — Empty-witness criterion

$$A_{\mathcal{P}}(\omega) \iff \mathcal{W}_{\mathcal{P}}(\omega) = \emptyset.$$

Theorem II.2 — Totalized evaluation

Every instance returns either:

- an output;
- or a nonempty witness set.

Theorem II.3 — Non-evaluation theorem

An inadmissible instance cannot produce an empirical contradiction because it produces no lawful prediction.

Theorem II.4 — Witness soundness

Every returned canonical witness corresponds to a genuinely failed declared guard.

Theorem II.5 — Relative witness completeness

Every failed declared guard produces a witness.

Theorem II.6 — Record-level contradiction

A lawful empirical contradiction occurs exactly when the admitted prediction set and valid observation set are disjoint.

Theorem II.7 — Projection-specific falsification

External contradiction targets the projection contract or its interpretation, not automatically the Book 0 identity.

Theorem II.8 — Restriction preservation

Restricting a lawful contract to a smaller validity domain preserves lawfulness on retained instances.

Theorem II.9 — Extension burden

Extending the validity domain requires a new guard proof on every added region.

Theorem II.10 — Ambiguous-phase theorem

Distinct admissible phase candidates that produce distinct structural states prevent unique projection unless a selection or aggregation rule is declared.

Theorem II.11 — Repair noninterference

Only results depending on a modified contract field require revalidation.

22. Negative Theorems**Negative Theorem 22.1 — No prediction from undefinedness**

An undefined structural expression is not a numerical prediction.

Negative Theorem 22.2 — No falsification without admission

An instance outside the contract's validity domain cannot falsify a prediction the contract did not make.

Negative Theorem 22.3 — No silent branch choice

A projection may not select one of several admissible phases without a declared selection rule.

Negative Theorem 22.4 — No extrapolation by formula alone

The ability to evaluate a formula outside its calibration or validity domain does not establish projection validity there.

Negative Theorem 22.5 — No structural refutation by external mismatch alone

An external mismatch does not refute a Book 0 identity unless it supplies an actual mathematical counterexample on the theorem's domain.

Negative Theorem 22.6 — No unique explanation from compatibility

Empirical compatibility does not prove that the tested projection is the only compatible projection.

Negative Theorem 22.7 — No witness suppression

A failed guard may not be hidden merely because a downstream numerical value can still be calculated.

23. Terminal Theorem

Theorem 23.1 — Projection Airlock Theorem

For every projection contract $\mathcal{P}$ and external instance ω , exactly one of the following operational results is justified:

1. a lawful projected output with declared uncertainty;
2. a nonempty failure-witness set;
3. after a lawful observation comparison, empirical compatibility;
4. after a lawful observation comparison, record-level empirical contradiction.

These outcomes are mutually distinguishable.

Undefined evaluation, insufficient contract specification, and empirical contradiction may not be substituted for one another.

Proof. The totalized evaluator separates lawful output from guard failure.

The observation guard separates evaluable instances from valid empirical tests.

The intersection rule separates empirical compatibility from empirical contradiction.

The outcome categories are defined by mutually exclusive conditions:

- empty versus nonempty projection witness sets;
- valid versus invalid observation records;
- nonempty versus empty prediction–observation intersections.

Therefore, each evaluated case has a determinate logical status.

□

24. Closing Declaration

S1.I established the form of a projection contract:

$$\hat{y}_{\mathcal{P}} = \Gamma_{\theta} \circ E_{\mathbf{F}} \circ \Phi.$$

S1.II establishes the admission procedure governing its use.

A projection is admitted only when every declared guard succeeds:

$$A_{\mathcal{P}}(\omega) = \bigwedge_{j=0}^s G_j(\mathcal{P}, \omega).$$

The declared scope must satisfy

$$\Omega_{\mathcal{P}} \subseteq \Omega_{\mathcal{P}}^{\max}, \quad \Omega_{\mathcal{P}}^{\max} = \{\omega \in \Omega : A_{\mathcal{P}}^{\max}(\omega) = \text{true}\}.$$

Failure is reported explicitly:

$$\mathcal{W}_{\mathcal{P}}(\omega) = \{w_j : G_j(\mathcal{P}, \omega) = \text{false}\}.$$

The evaluator returns either a prediction or witnesses:

$$\mathcal{E}_{\mathcal{P}}(\omega) \in \mathcal{Y} \sqcup \mathcal{P}^+(\mathfrak{W}).$$

Empirical contradiction requires lawful comparison:

$$P_{\mathcal{P}}(\omega) \cap O_{\mathcal{O}} = \emptyset.$$

The resulting discipline is:

No output before admission;
 no admission without passed guards;
 no rejection without a witness;
 no falsification without a lawful prediction and observation;
 no extrapolation without a new domain proof.

END OF S1.II

S1.III

Measurement Protocols, Error, and Uncertainty

Book 1 - Projection and Measurement

Abstract

S1.I defined the projection contract

$$\hat{y}_{\mathcal{P}} = \Gamma_{\theta} \circ E_{\mathbf{F}} \circ \Phi,$$

and S1.II established the guards, failure witnesses, and empirical comparison conditions governing its use.

This essay defines the observation side of that comparison.

A measurement is not merely a reported number. It is a protocol specifying:

- the measurand;
- the external instances to which the measurement applies;
- sampling and preparation;
- instrument response;
- data reduction;
- units;
- quality guards;
- uncertainty;
- and failure behavior.

A measurement protocol is represented by

$$\mathcal{M} = (\Omega_{\mathcal{M}}, \mu, S, I, A, \mathcal{U}_{\mathcal{M}}, \mathcal{Q}_{\mathcal{M}}, W_{\mathcal{M}}).$$

The realized observation is produced by

$$y_{\text{obs}} = (A \circ I \circ S)(\omega),$$

when all measurement guards succeed.

The essay distinguishes measurement error from model residual. If $\mu(\omega)$ is the measurand value and $\hat{y}_{\mathcal{P}}(\omega)$ is the projected prediction, then

$$y_{\text{obs}} - \hat{y}_{\mathcal{P}} = \underbrace{(y_{\text{obs}} - \mu(\omega))}_{\text{measurement error}} + \underbrace{(\mu(\omega) - \hat{y}_{\mathcal{P}})}_{\text{model discrepancy}} .$$

A residual therefore cannot be assigned wholly to measurement error or wholly to the model without additional evidence.

Uncertainty is represented generally as a set, distribution, covariance structure, or other declared description of values compatible with the measurement. Exact propagation through the projection layer is defined by image sets and pushforward measures.

Within a single open quadrant, continuous structural functions propagate bounded phase uncertainty continuously. At Flatwave boundaries, however, arbitrarily small unresolved phase uncertainty can produce the two-valued result

$$\{-1, +1\}.$$

Near primitive poles, small phase uncertainty can also produce arbitrarily large output uncertainty. Linearized uncertainty propagation is therefore forbidden across a discontinuity or uncontrolled singularity.

Repeated measurement may reduce random dispersion under declared independence assumptions. It does not remove systematic bias. Correlation between repeated measurements limits the benefit of averaging.

The final result is a formal separation among:

- the quantity intended to be measured;
- the realized observation;
- measurement error;
- uncertainty;
- model discrepancy;
- and prediction residual.

IDENTIFICATION DIVISION

PROGRAM-ID

S1-III-MEASUREMENT-ERROR-UNCERTAINTY

PURPOSE

To define the observation protocol required for lawful comparison between projected predictions and external measurements.

PRINCIPAL QUESTION

What must be declared before a reported value may be used as evidence for or against a projection contract?

NON-PURPOSE

This essay does not:

- prescribe one universal instrument;
- require one universal statistical philosophy;
- assume that all uncertainty is probabilistic;
- equate uncertainty with error;
- equate precision with accuracy;
- infer a true value merely from repeated measurements;
- select the calibration model of S1.IV;
- determine whether one projection is superior to another;
- alter any theorem of Book 0.

ENVIRONMENT DIVISION

1. Inherited Projection Environment

From S1.I, a projection contract is

$$\mathcal{P} = (\Omega, \Omega_{\mathcal{P}}, \Phi, \mathbf{F}, \Gamma_{\theta}, U_{\mathcal{P}}, W_{\mathcal{P}}).$$

Its prediction map is

$$\hat{y}_{\mathcal{P}}(\omega) = \Gamma_{\theta}(E_{\mathbf{F}}(\Phi(\omega))).$$

From S1.II, empirical comparison is lawful only when:

1. the instance passes every projection guard;
2. the observation record passes every measurement guard;

3. the prediction and observation belong to a common output space;
4. their uncertainty descriptions are declared.

1.1 Projection and measurement are distinct

The projection contract answers:

What does the model predict?

The measurement protocol answers:

What was observed, and under what conditions?

Neither answer may be substituted for the other.

METHODOLOGICAL AXIOM DIVISION

Axiom M.1 — Observation Declaration

Every empirical claim used to test a projection must be supported by an explicitly declared measurement protocol.

A reported number without its measurand, units, method, applicability conditions, uncertainty, and failure status is not a complete observation record.

Axiom M.2 — Prediction–Observation Separation

The observation record must declare every way in which prediction, model choice, or expected outcome influenced:

- sampling;
- exclusion;
- preprocessing;
- instrument settings;
- data reduction;
- uncertainty estimation;
- or reporting.

A prediction-dependent procedure is not automatically invalid, but the dependence may not remain hidden.

DATA DIVISION

2. Measurands

Definition 2.1 — Measurement instance space

Let

$$\Omega_{\mathcal{M}} \subseteq \Omega$$

be the external instances for which a measurement protocol is declared.

Definition 2.2 — Measurand space

Let

$$\mathcal{Y}$$

be the declared space in which the target quantity takes values.

The measurand space must include its relevant structure, such as:

- scalar or vector form;
- units;
- coordinate convention;
- reference frame;
- category definitions;
- permissible range.

Definition 2.3 — Measurand

A measurand is a declared target map

$$\mu : \Omega_{\mathcal{M}} \rightarrow \mathcal{Y}.$$

The value

$$\mu(\omega)$$

is the quantity the protocol intends to determine for instance ω .

Clarification 2.4 — Measurand value need not be known

Declaring the measurand does not imply that its value is exactly known.

The measurand identifies the target of measurement.

The measurement result is an estimate, report, or constraint concerning that target.

Example 2.5 — Incomplete measurand

The phrase “measure the temperature” is incomplete when the following remain unspecified:

- which object or region;
- at what time;
- under which equilibrium assumptions;
- on which temperature scale;
- with what spatial or temporal averaging.

Different specifications define different measurands.

3. Measurement Protocol**Definition 3.1 — Sampling map**

A sampling map is

$$S : \Omega_{\mathcal{M}} \rightarrow \mathcal{X},$$

where $\mathcal{X}$ is the space of samples, signals, specimens, or prepared inputs supplied to the instrument.

The sampling map includes any declared:

- selection;
- timing;
- spatial window;
- averaging;
- preparation;
- filtering;
- or transformation occurring before instrument response.

Definition 3.2 — Instrument-response map

An instrument-response map is

$$I : \mathcal{X} \rightarrow \mathcal{R},$$

where $\mathcal{R}$ is the raw-record space.

The instrument map may be deterministic, stochastic, or partially unknown.

Definition 3.3 — Analysis map

An analysis map is

$$A : \mathcal{R} \rightarrow \mathcal{Y}_{\text{rep}},$$

where $\mathcal{Y}_{\text{rep}}$ is the reported-value space.

The analysis map includes:

- digitization;
- preprocessing;
- baseline subtraction;
- transformation;
- fitting;
- aggregation;
- classification;
- rounding;
- and reporting conventions.

Definition 3.4 — Realized observation

For a valid measurement instance,

$$y_{\text{obs}}(\omega) = A(I(S(\omega))).$$

This value is the reported observation, not necessarily the measurand value.

Definition 3.5 — Measurement uncertainty assignment

The measurement uncertainty assignment is

$$\mathcal{U}_{\mathcal{M}} : \Omega_{\mathcal{M}} \rightarrow \mathfrak{U}(\mathcal{Y}).$$

Definition 3.6 — Measurement quality guards

Let

$$\mathcal{Q}_{\mathcal{M}} = (Q_0, \dots, Q_n)$$

be the declared collection of guards required for a valid observation.

Definition 3.7 — Measurement failure map

$$W_{\mathcal{M}} : \Omega \setminus \Omega_{\mathcal{M}}^* \rightarrow \mathfrak{W}_{\mathcal{M}},$$

where $\Omega_{\mathcal{M}}^*$ is the set of instances passing every measurement guard.

Definition 3.8 — Measurement protocol

A measurement protocol is the tuple

$$\mathcal{M} = (\Omega_{\mathcal{M}}, \mu, S, I, A, \mathcal{U}_{\mathcal{M}}, \mathcal{Q}_{\mathcal{M}}, W_{\mathcal{M}}).$$

4. Measurement Guards

A protocol should declare guards appropriate to its instrument and measurand.

The following form a recommended minimum.

Definition 4.1 — Protocol-completeness guard

The measurand, sampling rule, instrument, analysis, units, uncertainty method, and failure behavior must be declared.

Recommended witness:

MEASUREMENT-PROTOCOL-INCOMPLETE.

Definition 4.2 — Instance-applicability guard

The instance must lie within the protocol's declared measurement domain.

Recommended witness:

INSTANCE-OUTSIDE-MEASUREMENT-DOMAIN.

Definition 4.3 — Sampling-validity guard

The obtained sample or signal must be suitable for the declared measurand.

Recommended witness:

SAMPLE-NOT-REPRESENTATIVE.

Definition 4.4 — Instrument-range guard

The instrument response must lie inside its declared operating range.

Recommended witnesses include:

- INSTRUMENT-SATURATED;
- SIGNAL-BELOW-DETECTION-LIMIT;
- INSTRUMENT-OUTSIDE-RATED-RANGE.

Definition 4.5 — Resolution guard

The instrument resolution must be sufficient for the reported number of digits or classification.

Recommended witness:

REPORTED-PRECISION-EXCEEDS-RESOLUTION.

Definition 4.6 — Calibration-status guard

The instrument must possess a valid calibration status for the measurement conditions.

Recommended witness:

INSTRUMENT-CALIBRATION-INVALID.

The full theory of calibration is deferred to S1.IV.

Definition 4.7 — Analysis-validity guard

The analysis procedure must be defined for the raw record and must not conceal failed preprocessing assumptions.

Recommended witness:

ANALYSIS-PROCEDURE-INVALID.

Definition 4.8 — Unit-consistency guard

The reported value, uncertainty, and projection output must use compatible units or a declared conversion.

Recommended witness:

MEASUREMENT-UNIT-MISMATCH.

Definition 4.9 — Uncertainty-completeness guard

An empirical result must include the uncertainty representation required by its protocol.

Recommended witness:

MEASUREMENT-UNCERTAINTY-MISSING.

Definition 4.10 — Independence-disclosure guard

Any prediction-dependent sampling, filtering, exclusion, or analysis must be disclosed.

Recommended witness:

PREDICTION-DEPENDENCE-UNDISCLOSED.

5. Observation Record

Definition 5.1 — Complete observation record

A complete observation record is

$$\mathcal{O} = (\text{id}_{\mathcal{M}}, \omega, \mu, y_{\text{obs}}, \mathcal{U}_{\mathcal{M}}(\omega), \mathbf{Q}, \mathbf{m}),$$

where:

- $\text{id}_{\mathcal{M}}$ identifies the protocol version;
- ω identifies the measured instance;
- μ identifies the measurand;
- y_{obs} is the reported observation;
- $\mathcal{U}_{\mathcal{M}}(\omega)$ is its uncertainty description;

- $\mathbf{Q}$ is the measurement-guard vector;
- $\mathbf{m}$ contains required metadata.

Definition 5.2 — Measurement witness set

$$\mathcal{W}_{\mathcal{M}}(\omega) = \{w_j : Q_j(\mathcal{M}, \omega) = \text{false}\}.$$

Theorem 5.3 — Valid-observation criterion

An observation record is valid exactly when

$$\mathcal{W}_{\mathcal{M}}(\omega) = \emptyset.$$

Proof. The record is valid precisely when every declared measurement guard succeeds.

A witness is included precisely when its guard fails.

Therefore, every guard succeeds exactly when the witness set is empty.

□

Corollary 5.4 — A failed measurement is not an observation against the model

If

$$\mathcal{W}_{\mathcal{M}}(\omega) \neq \emptyset,$$

the record cannot be used as a lawful contradiction of the projection until the failure is resolved or the protocol is replaced.

6. Error

Definition 6.1 — Measurement error

When subtraction is defined in $\mathcal{Y}$, the measurement error is

$$e_{\mathcal{M}}(\omega) = y_{\text{obs}}(\omega) - \mu(\omega).$$

Clarification 6.2 — Error is usually not directly known

The exact measurement error requires knowledge of the measurand value.

If $\mu(\omega)$ is unknown, the exact error is also unknown.

An uncertainty estimate is not the same thing as the unknown error.

Definition 6.3 — Signed error

For a scalar measurand,

$$e_{\mathcal{M}} > 0$$

indicates an observation above the measurand, while

$$e_{\mathcal{M}} < 0$$

indicates an observation below it.

Definition 6.4 — Absolute error

$$|e_{\mathcal{M}}| = |y_{\text{obs}} - \mu|.$$

Definition 6.5 — Relative error

Where

$$\mu(\omega) \neq 0,$$

define

$$e_{\text{rel}} = \frac{y_{\text{obs}} - \mu}{\mu}.$$

Relative error is undefined when the reference measurand is zero unless another normalization is declared.

7. Bias and Random Error

The following definitions require a declared probability model for repeated outcomes of the measurement process.

Definition 7.1 — Expected observation

Suppose the realized observation is represented by a random variable Y with finite expectation.

Then

$$\mathbb{E}[Y]$$

is the expected reported value under the declared repetition model.

Definition 7.2 — Measurement bias

$$b = \mathbb{E}[Y] - \mu.$$

Definition 7.3 — Random error component

$$\eta = Y - \mathbb{E}[Y].$$

Then

$$\mathbb{E}[\eta] = 0.$$

Theorem 7.4 — Error decomposition

$$Y - \mu = b + \eta.$$

Proof. Add and subtract $\mathbb{E}[Y]$:

$$Y - \mu = Y - \mathbb{E}[Y] + \mathbb{E}[Y] - \mu.$$

By definition,

$$Y - \mathbb{E}[Y] = \eta$$

and

$$\mathbb{E}[Y] - \mu = b.$$

Therefore,

$$Y - \mu = b + \eta.$$

□

Clarification 7.5 — Systematic and random are model-relative

An effect may appear random under one description and systematic under a richer description that includes an omitted condition.

The classification depends on the declared repetition and conditioning model.

8. Accuracy and Precision

Definition 8.1 — Accuracy

Accuracy concerns closeness of a measurement result, or expected measurement result, to the measurand.

A scalar measure of inaccuracy may be based on:

$$|y_{\text{obs}} - \mu|$$

or

$$|\mathbb{E}[Y] - \mu|.$$

Definition 8.2 — Precision

Precision concerns the dispersion of repeated measurement results under declared repeatability or reproducibility conditions.

Possible precision measures include:

- variance;
- standard deviation;
- interquartile range;
- repeatability interval;
- resolution.

Theorem 8.3 — Precision does not imply accuracy

Arbitrarily precise measurements may remain inaccurate.

Proof. Let every repeated measurement return the same value

$$Y = \mu + c$$

for a fixed nonzero constant c .

Then

$$\text{Var}(Y) = 0,$$

so the measurement is perfectly precise under the repetition model.

However,

$$Y - \mu = c \neq 0.$$

Therefore, perfect precision does not imply accuracy.

□

Theorem 8.4 — Accuracy does not imply high precision

A measurement process may be unbiased but highly dispersed.

Proof. Let Y satisfy

$$\mathbb{E}[Y] = \mu$$

while

$$\text{Var}(Y)$$

is large.

Then the bias is zero, but individual results may be far from the measurand.

Therefore, zero bias does not imply high precision.

□

9. Resolution, Sensitivity, and Detection

Definition 9.1 — Resolution

Resolution is the smallest declared change in input or measurand that the instrument or reporting system can distinguish under specified conditions.

Definition 9.2 — Sensitivity

For a differentiable instrument-response function $r(u)$, local sensitivity is

$$\left[\frac{dr}{du} \right]$$

Sensitivity measures response change per input change.

It is not identical to resolution or accuracy.

Definition 9.3 — Detection limit

A detection limit is a declared threshold below which the protocol does not reliably distinguish signal from background under its stated decision rule.

Definition 9.4 — Quantization

If an instrument reports values only on a discrete grid of spacing Δ , then the reporting process introduces quantization.

Under nearest-grid rounding, the quantization error satisfies

$$|e_q| \leq \frac{\Delta}{2}.$$

This bound does not include other sources of error.

10. Measurement Uncertainty**Definition 10.1 — Uncertainty description**

A measurement uncertainty description is a declared representation of values, distributions, or parameter states compatible with the measurement information.

It may be:

- a bounded set;
- an interval;
- a covariance matrix;
- a probability distribution;
- a confidence region;
- a credible region;
- an ensemble;
- a qualitative classification.

Definition 10.2 — Observed admissible-value set

For a set-valued uncertainty representation, define

$$O_o \subseteq \mathcal{Y}$$

as the set of measurand values compatible with the observation under the declared protocol.

Definition 10.3 — Interval uncertainty

For a scalar observation, an interval representation has the form

$$O_{\mathcal{O}} = [y_{\text{obs}} - u_-, y_{\text{obs}} + u_+].$$

The symmetric case is

$$[y_{\text{obs}} - u, y_{\text{obs}} + u].$$

Definition 10.4 — Probabilistic uncertainty

A probabilistic uncertainty description assigns a probability measure

$$\boxed{\nu_{\mathcal{O}}}$$

to the measurand or to the measurement process under declared assumptions.

The interpretation of $\nu_{\mathcal{O}}$ must be stated.

Clarification 10.5 — Uncertainty is not automatically probability

Incomplete information may be represented by a set without assigning relative probabilities to the values inside it.

Clarification 10.6 — Confidence and credibility are different declarations

A confidence interval is defined by the long-run coverage properties of a procedure under a sampling model.

A credible interval is defined by probability under a declared posterior distribution.

Neither term should be used without its supporting interpretation.

11. Measurement Error and Model Residual**Definition 11.1 — Projected prediction**

For an admitted instance,

$$\hat{y}_{\mathcal{P}}(\omega)$$

is the output of the projection contract.

Definition 11.2 — Residual

Where subtraction is defined,

$$r(\omega) = y_{\text{obs}}(\omega) - \hat{y}_{\mathcal{P}}(\omega).$$

Definition 11.3 — Model discrepancy

$$d_{\mathcal{P}}(\omega) = \mu(\omega) - \hat{y}_{\mathcal{P}}(\omega).$$

Theorem 11.4 — Residual decomposition

$$r = e_{\mathcal{M}} + d_{\mathcal{P}}.$$

Proof. By definition,

$$r = y_{\text{obs}} - \hat{y}_{\mathcal{P}}.$$

Add and subtract μ :

$$r = y_{\text{obs}} - \mu + \mu - \hat{y}_{\mathcal{P}}.$$

The first difference is measurement error:

$$e_{\mathcal{M}} = y_{\text{obs}} - \mu.$$

The second is model discrepancy:

$$d_{\mathcal{P}} = \mu - \hat{y}_{\mathcal{P}}.$$

Therefore,

$$r = e_{\mathcal{M}} + d_{\mathcal{P}}.$$

□

Corollary 11.5 — A residual is not measurement error

Unless

$$d_{\mathcal{P}} = 0$$

has been independently established,

$$r \neq e_{\mathcal{M}}$$

in general.

Corollary 11.6 — A residual is not model discrepancy

Unless

$$e_{\mathcal{M}} = 0$$

has been independently established,

$$r \neq d_{\mathcal{P}}$$

in general.

Negative Theorem 11.7 — Residual attribution is underdetermined

A nonzero residual alone does not determine how much arises from measurement error and how much arises from projection discrepancy.

12. Repeated Measurement

Suppose repeated observations satisfy

$$Y_i = \mu + b + \eta_i,$$

where

$$\mathbb{E}[\eta_i] = 0.$$

Definition 12.1 — Sample mean

$$\bar{Y}_n = \frac{1}{n} \sum_{i=1}^n Y_i.$$

Theorem 12.2 — Averaging does not remove bias

$$\mathbb{E}[\bar{Y}_n] = \mu + b.$$

Proof. By linearity of expectation,

$$\mathbb{E}[\bar{Y}_n] = \frac{1}{n} \sum_{i=1}^n \mathbb{E}[Y_i].$$

Each expectation is

$$\mu + b.$$

Therefore,

$$\mathbb{E}[\bar{Y}_n] = \mu + b.$$

□

Corollary 12.3 — Infinite repetition does not correct fixed bias

Increasing n does not cause the expected mean to approach μ unless $b = 0$.

Theorem 12.4 — Independent equal-variance averaging

If the random errors are independent and each has variance σ^2 , then

$$\boxed{\text{Var}(\bar{Y}_n) = \frac{\sigma^2}{n}.}$$

Proof.

$$\text{Var}(\bar{Y}_n) = \frac{1}{n^2} \text{Var} \left(\sum_{i=1}^n \eta_i \right).$$

Under independence,

$$\text{Var} \left(\sum_{i=1}^n \eta_i \right) = n\sigma^2.$$

Therefore,

$$\text{Var}(\bar{Y}_n) = \frac{n\sigma^2}{n^2} = \frac{\sigma^2}{n}.$$

□

Theorem 12.5 — Equal-correlation averaging

Suppose each random error has variance σ^2 , and every distinct pair has correlation ρ . Then

$$\text{Var}(\bar{Y}_n) = \frac{\sigma^2}{n} (1 + (n-1)\rho).$$

Proof. The variance of the sum is

$$n\sigma^2 + 2\binom{n}{2}\rho\sigma^2.$$

Thus,

$$\text{Var}\left(\sum_{i=1}^n \eta_i\right) = n\sigma^2 + n(n-1)\rho\sigma^2.$$

Divide by n^2 :

$$\text{Var}(\bar{Y}_n) = \frac{\sigma^2}{n} (1 + (n-1)\rho).$$

□

Corollary 12.6 — Correlation limits averaging benefit

When $\rho > 0$, repeated measurements contain less independent information than the formula σ^2/n would imply.

When $\rho = 1$,

$$\text{Var}(\bar{Y}_n) = \sigma^2.$$

Averaging provides no dispersion reduction.

13. Set-Valued Phase Inference

Measurement uncertainty may affect the phase-inference map.

Definition 13.1 — Candidate external-state set

Let

$$X_{\mathcal{O}} \subseteq \Omega$$

be the external states compatible with an observation record.

Definition 13.2 — Candidate phase set

$$C_{\Phi}(\mathcal{O}) = \{\Phi(\omega) : \omega \in X_{\mathcal{O}}\}.$$

When phase inference itself is set-valued, C_{Φ} includes every contract-admissible inferred phase.

Definition 13.3 — Candidate structural set

For a structural profile $\mathbf{F}$,

$$C_{\mathbf{F}}(\mathcal{O}) = E_{\mathbf{F}}(C_{\Phi}(\mathcal{O})).$$

Definition 13.4 — Candidate projected-output set

$$P_{\mathcal{P}}(\mathcal{O}) = \Gamma_{\theta}(C_{\mathbf{F}}(\mathcal{O})),$$

with calibration uncertainty included where required.

Theorem 13.5 — Exact set propagation

Every output generated by a candidate phase in C_{Φ} belongs to $P_{\mathcal{P}}$, and every member of $P_{\mathcal{P}}$ arises from at least one candidate phase.

Proof. The set

$$C_{\mathbf{F}} = E_{\mathbf{F}}(C_{\Phi})$$

is the image of the candidate phase set under structural evaluation.

The set

$$P_{\mathcal{P}} = \Gamma_{\theta}(C_{\mathbf{F}})$$

is its image under the readout.

This is precisely the definition of an image set.

□

14. Continuous Uncertainty Propagation

Theorem 14.1 — Continuous-image compactness

Let

$$X \subset Q_k$$

be compact, and let

$$F : X \rightarrow \mathbb{R}$$

be continuous.

Then

$$F(X)$$

is compact and therefore bounded.

Proof. The continuous image of a compact set is compact.

Every compact subset of $\mathbb{R}$ is bounded.

□

Theorem 14.2 — Monotone interval propagation

Let

$$X = [a, b] \subset Q_k$$

and suppose F is monotone increasing on X . Then

$$F(X) = [F(a), F(b)].$$

If F is monotone decreasing, then

$$F(X) = [F(b), F(a)].$$

Proof. A continuous monotone function on an interval attains its extrema at the endpoints and assumes every intermediate value.

□

Theorem 14.3 — Lipschitz uncertainty bound

Suppose

$$|F(x) - F(y)| \leq L|x - y|$$

for all $x, y \in X$.

Then

$$\text{diam}(F(X)) \leq L \text{diam}(X).$$

Proof. For any $x, y \in X$,

$$|F(x) - F(y)| \leq L|x - y| \leq L \text{diam}(X).$$

Taking the supremum over $x, y \in X$ gives the result.

□

Corollary 14.4 — Derivative-based bound

If F is differentiable on an interval X and

$$|F'(x)| \leq L$$

throughout X , then

$$\text{diam}(F(X)) \leq L \text{diam}(X).$$

This follows from the mean-value theorem.

15. Probabilistic Propagation**Definition 15.1 — Phase probability measure**

Let

$$\nu_\Phi$$

be a declared probability measure over candidate phases.

Definition 15.2 — Pushforward measure

For a measurable structural function F , define

$$F_{\#}\nu_{\Phi}(B) = \nu_{\Phi}(F^{-1}(B))$$

for measurable output sets B .

Theorem 15.3 — Exact probabilistic propagation

The probability distribution of

$$Y = F(X)$$

for $X \sim \nu_{\Phi}$ is

$$\nu_Y = F_{\#}\nu_{\Phi}.$$

This is exact and does not require linearization.

Definition 15.4 — Readout pushforward

With readout Γ_{θ} ,

$$\nu_{\text{out}} = (\Gamma_{\theta} \circ F)_{\#}\nu_{\Phi}.$$

16. Linearized Uncertainty Propagation**Definition 16.1 — First-order propagation**

For a differentiable scalar function F near a nominal phase x_0 ,

$$F(x) \approx F(x_0) + F'(x_0)(x - x_0).$$

If the phase uncertainty has variance σ_x^2 , the first-order propagated variance is

$$\sigma_F^2 \approx F'(x_0)^2 \sigma_x^2.$$

Definition 16.2 — Multivariate first-order propagation

For a differentiable map

$$g : \mathbb{R}^n \rightarrow \mathbb{R}^m$$

with Jacobian J , the first-order covariance approximation is

$$\Sigma_y \approx J \Sigma_x J^\top.$$

Theorem 16.3 — Exactness for affine maps

If

$$g(x) = Ax + b,$$

then

$$\Sigma_y = A \Sigma_x A^\top$$

exactly.

Clarification 16.4 — Nonlinear propagation is approximate

For nonlinear g , the Jacobian covariance formula neglects higher-order terms.

Its adequacy must be checked against:

- uncertainty width;
- curvature;
- singularity distance;
- branch structure;
- discontinuities.

17. Flatwave Boundary Uncertainty

Theorem 17.1 — Quadrant-stable Flatwave

If the candidate phase set satisfies

$$C_\Phi \subset Q_k,$$

then

$$\text{Flatwave}(C_\Phi) = \{\varepsilon_k\}.$$

Proof. Flatwave is constant on Q_k :

$$\text{Flatwave}(x) = \varepsilon_k.$$

Therefore, every candidate phase produces the same value.

□

Theorem 17.2 — Boundary-straddling Flatwave uncertainty

Suppose C_Φ contains at least one phase in Q_{k-1} and at least one phase in Q_k .

Then

$$\{-1, +1\} \subseteq \text{Flatwave}(C_\Phi).$$

If no other Flatwave values are permitted, then

$$\text{Flatwave}(C_\Phi) = \{-1, +1\}.$$

Proof. Adjacent quadrants have opposite signs:

$$\varepsilon_{k-1} = -\varepsilon_k.$$

One candidate phase therefore produces +1, and the other produces −1.

Flatwave assumes only those two values on its interior domain.

□

Corollary 17.3 — Arbitrarily narrow phase uncertainty may produce maximal sign uncertainty

For every $\delta > 0$, a candidate phase interval of width less than δ can straddle an excluded boundary and produce both Flatwave signs.

Thus small phase width does not imply small Flatwave uncertainty near a boundary.

Negative Theorem 17.4 — No linearized Flatwave propagation across a wall

Because Flatwave is discontinuous across an identified boundary, derivative-based uncertainty propagation cannot represent boundary-straddling uncertainty.

Inside a quadrant,

$$\text{Flatwave}' = 0,$$

but across the wall the output changes by 2.

18. Primitive Singular Amplification

Theorem 18.1 — No uniform primitive bound near a pole

Let b be a boundary at which a primitive F has asymptotic form

$$F(b+h) \sim \frac{2}{h}$$

as $h \rightarrow 0^+$.

Then F is unbounded on every interval

$$(b, b + \delta)$$

with $\delta > 0$.

Proof. Since

$$F(b+h) \sim \frac{2}{h}$$

and $2/h \rightarrow +\infty$, the function exceeds every finite bound for sufficiently small h .

□

Corollary 18.2 — Small phase uncertainty may produce arbitrarily large primitive uncertainty

A candidate phase set approaching a primitive pole may have arbitrarily small width while its structural image has arbitrarily large diameter.

Theorem 18.3 — No uniform Lipschitz constant at a primitive pole

No finite L provides a Lipschitz bound for the divergent primitive on every sufficiently small interval approaching the pole.

Proof. A Lipschitz function on a bounded interval has bounded variation and finite values throughout that interval.

The primitive diverges at the boundary approach.

Therefore, no finite uniform Lipschitz constant exists on intervals whose closure reaches the pole.

□

Corollary 18.4 — Linear propagation requires a singularity-distance guard

A projection using primitive values must declare a minimum distance from the relevant excluded boundary or use an exact nonlinear propagation procedure.

19. Saw Uncertainty**Theorem 19.1 — Saw boundedness**

For every admissible phase,

$$\boxed{-1 < \text{saw}_r(x) < 1}$$

and

$$\boxed{-1 < \text{saw}_x(x) < 1.}$$

Their quadrant-relative endpoint extensions lie in $[-1, 1]$.

Proof. From Book 0,

$$\text{saw}_r = \varepsilon\lambda,$$

and

$$\text{saw}_x = \varepsilon(1 - \lambda),$$

with

$$0 < \lambda < 1.$$

Therefore, both magnitudes are strictly less than 1 in the interior.

□

Corollary 19.2 — Saw uncertainty cannot diverge

Unlike primitive uncertainty near a pole, saw-value uncertainty remains bounded.

Clarification 19.3 — Bounded does not mean continuous across identified seams

Each individual saw has unequal one-sided limits at a quadrant boundary.

A phase set straddling the boundary may therefore produce separated bounded values.

20. Uncertainty Budgets

Definition 20.1 — Uncertainty source

An uncertainty source is a declared contribution arising from one stage of measurement or projection.

Possible sources include:

- sampling uncertainty;
- instrument noise;
- instrument bias uncertainty;
- preprocessing uncertainty;
- phase-inference uncertainty;
- branch ambiguity;
- calibration uncertainty;
- readout approximation;
- numerical approximation;
- model discrepancy allowance.

Definition 20.2 — Uncertainty budget

An uncertainty budget is a ledger

$$\mathcal{B} = \{(B_i, \text{magnitude}_i, \text{dependence}_i, \text{method}_i)\}.$$

recording each included source and how it enters the final uncertainty description.

Definition 20.3 — Independence declaration

Sources may be combined by independent-error formulas only when their independence or negligible dependence is justified.

Theorem 20.4 — Variance addition under zero covariance

For random variables $X_1, \dots, X_n$,

$$\text{Var} \left(\sum_i X_i \right) = \sum_i \text{Var}(X_i)$$

exactly when the total covariance contribution is zero:

$$\sum_{i \neq j} \text{Cov}(X_i, X_j) = 0.$$

Proof. The general variance identity is

$$\text{Var} \left(\sum_i X_i \right) = \sum_i \text{Var}(X_i) + \sum_{i \neq j} \text{Cov}(X_i, X_j).$$

The reduced formula holds exactly when the covariance sum vanishes.

□

Negative Theorem 20.5 — No root-sum-square without dependence assumptions

The formula

$$u_{\text{total}} = \sqrt{u_1^2 + \dots + u_n^2}$$

is not universally valid.

It requires an interpretation under which the corresponding contributions are uncorrelated standard uncertainties or satisfy an equivalent declared condition.

Negative Theorem 20.6 — No double counting

The same uncertainty contribution may not be included once in the measurement interval and again as an independent projection term unless the duplication is explicitly intended and justified.

21. Missing, Censored, and Selected Data

Definition 21.1 — Missing observation

A value is missing when no report is available for an instance expected under the protocol.

Definition 21.2 — Censored observation

A value is censored when only a constraint is reported, such as

$$y < L$$

or

$$y > U.$$

Definition 21.3 — Truncated protocol

A protocol is truncated when instances outside a selection range are excluded from the recorded sample.

Theorem 21.4 — A censored value is a set-valued observation

The report

$$y < L$$

corresponds to an observed admissible set of the form

$$O_{\mathcal{O}} \subseteq (-\infty, L),$$

subject to the declared protocol and uncertainty.

It may not be replaced by the single value L without an added convention.

Negative Theorem 21.5 — Missingness is not zero

An absent observation may not be assigned the numerical value 0 unless zero is the explicit coding convention and is kept distinct from an actual measured zero.

Negative Theorem 21.6 — Selection may alter empirical distributions

A sample selected according to the observed or predicted value need not represent the distribution of the full instance population.

The selection rule must be included in the measurement protocol.

22. Compatibility with Projection Testing**Definition 22.1 — Predicted admissible-output set**

From S1.II,

$$P_{\mathcal{P}}(\omega) \subseteq \mathcal{Y}$$

is the output set compatible with the projection and its uncertainty.

Definition 22.2 — Observed admissible-output set

From the measurement protocol,

$$O_{\mathcal{O}} \subseteq \mathcal{Y}$$

is the output set compatible with the observation and its uncertainty.

Theorem 22.3 — Compatibility condition

The prediction and observation are compatible when

$$P_{\mathcal{P}}(\omega) \cap O_{\mathcal{O}} \neq \emptyset.$$

Theorem 22.4 — Contradiction condition

They are contradictory at the record level when

$$P_{\mathcal{P}}(\omega) \cap O_{\mathcal{O}} = \emptyset.$$

Clarification 22.5 — Larger uncertainty weakens discrimination

Enlarging either admissible set cannot convert a nonempty intersection into an empty one.

Therefore, wider uncertainty generally makes contradiction harder to establish.

Clarification 22.6 — Artificially narrow uncertainty inflates apparent contradiction

Understating uncertainty may create a disjoint-set result that disappears under a complete uncertainty budget.

PROCEDURE DIVISION

23. Canonical Measurement Procedure

PERFORM-MEASUREMENT.

PERFORM CHECK-MEASUREMENT-PROTOCOL.

IF MEASUREMENT-PROTOCOL-INCOMPLETE

RETURN MEASUREMENT-WITNESS-SET

END-IF.

PERFORM CHECK-INSTANCE-APPLICABILITY.

IF INSTANCE-OUTSIDE-MEASUREMENT-DOMAIN

RETURN MEASUREMENT-WITNESS-SET
END-IF.
PERFORM OBTAIN-SAMPLE.
IF SAMPLE-NOT-REPRESENTATIVE
RETURN MEASUREMENT-WITNESS-SET
END-IF.
PERFORM ACQUIRE-INSTRUMENT-RECORD.
IF INSTRUMENT-SATURATED
RETURN MEASUREMENT-WITNESS-SET
END-IF.
IF SIGNAL-BELOW-DETECTION-LIMIT
RETURN CENSORED-OBSERVATION
WITH DECLARED LIMIT
END-IF.
PERFORM APPLY-ANALYSIS-MAP.
IF ANALYSIS-PROCEDURE-INVALID
RETURN MEASUREMENT-WITNESS-SET
END-IF.
PERFORM CHECK-UNITS.
PERFORM CHECK-RESOLUTION.
PERFORM BUILD-UNCERTAINTY-DESCRIPTION.
PERFORM DISCLOSE-PREDICTION-DEPENDENCE.
IF ANY REQUIRED GUARD FAILED
RETURN MEASUREMENT-WITNESS-SET
ELSE
RETURN COMPLETE-OBSERVATION-RECORD
END-IF.

23.1 Uncertainty propagation procedure

PROPAGATE-UNCERTAINTY.
FORM CANDIDATE-EXTERNAL-STATE-SET.
FORM CANDIDATE-PHASE-SET.
IF CANDIDATE-PHASE-SET CROSSES FORBIDDEN BOUNDARY

DO NOT USE SINGLE-BRANCH LINEARIZATION
 END-IF.
 IF STRUCTURAL-FUNCTION HAS UNCONTROLLED POLE
 USE EXACT SET OR DISTRIBUTIONAL PROPAGATION
 OR RETURN SINGULARITY-PROXIMITY-WITNESS
 END-IF.
 APPLY STRUCTURAL-IMAGE MAP.
 APPLY CALIBRATION AND READOUT MAP.
 RECORD CORRELATIONS AND ASSUMPTIONS.
 RETURN PREDICTED-OUTPUT-SET
 OR OUTPUT-DISTRIBUTION.

24. Theorem Ledger

Theorem III.1 — Valid-observation criterion

$$\mathcal{W}_{\mathcal{M}}(\omega) = \emptyset$$

if and only if every declared measurement guard succeeds.

Theorem III.2 — Error decomposition

$$Y - \mu = b + \eta.$$

Theorem III.3 — Precision does not imply accuracy

Zero dispersion is compatible with nonzero bias.

Theorem III.4 — Residual decomposition

$$y_{\text{obs}} - \hat{y}_{\mathcal{P}} = (y_{\text{obs}} - \mu) + (\mu - \hat{y}_{\mathcal{P}}).$$

Theorem III.5 — Averaging does not remove bias

$$\mathbb{E}[\bar{Y}_n] = \mu + b.$$

Theorem III.6 — Independent averaging

$$\text{Var}(\bar{Y}_n) = \frac{\sigma^2}{n}$$

under independence and equal variance.

Theorem III.7 — Correlated averaging

$$\text{Var}(\bar{Y}_n) = \frac{\sigma^2}{n} (1 + (n-1)\rho)$$

under equal pairwise correlation.

Theorem III.8 — Exact set propagation

$$P_{\mathcal{P}} = \Gamma_{\theta} (E_{\mathbf{F}}(C_{\Phi})) .$$

Theorem III.9 — Lipschitz propagation bound

$$\text{diam}(F(X)) \leq L \text{diam}(X).$$

Theorem III.10 — Pushforward propagation

$$\nu_{\text{out}} = (\Gamma_{\theta} \circ E_{\mathbf{F}})_{\#} \nu_{\Phi}.$$

Theorem III.11 — Flatwave quadrant stability

If

$$C_{\Phi} \subset Q_k,$$

then

$$\text{Flatwave}(C_{\Phi}) = \{\varepsilon_k\}.$$

Theorem III.12 — Flatwave boundary ambiguity

If candidate phases occupy adjacent quadrants, then

$$\text{Flatwave}(C_{\Phi}) = \{-1, +1\}.$$

Theorem III.13 — Primitive singular amplification

Arbitrarily small phase uncertainty near a primitive pole may produce arbitrarily large primitive-value uncertainty.

Theorem III.14 — Saw boundedness

Saw-value uncertainty remains inside $[-1, 1]$.

Theorem III.15 — Record-level compatibility

$$P_{\mathcal{P}}(\omega) \cap O_{\mathcal{O}} \neq \emptyset$$

is the general compatibility condition.

25. Negative Theorems**Negative Theorem 25.1 — No number without a measurand**

A numerical report without a declared measurand is not a complete measurement result.

Negative Theorem 25.2 — No error without a reference value

Exact measurement error cannot be calculated without a measurand value or accepted reference value.

Negative Theorem 25.3 — No uncertainty–error identity

An uncertainty interval is not the unknown realized error.

Negative Theorem 25.4 — No precision–accuracy identity

Small dispersion does not prove closeness to the measurand.

Negative Theorem 25.5 — No residual attribution without further evidence

A residual does not uniquely identify measurement error or model discrepancy.

Negative Theorem 25.6 — No independent-error formula without dependence analysis

Covariance may not be discarded without justification.

Negative Theorem 25.7 — No linearization across a Flatwave wall

A zero interior derivative does not describe the two-valued uncertainty created by a boundary-straddling phase set.

Negative Theorem 25.8 — No linearization through an uncontrolled pole

Derivative-based propagation is invalid when the required derivative bound does not exist on the uncertainty domain.

Negative Theorem 25.9 — No failed measurement as model falsification

A record with unresolved measurement witnesses cannot serve as a lawful contradiction.

Negative Theorem 25.10 — No hidden prediction-dependent filtering

Data selection influenced by model expectations must be disclosed.

26. Terminal Theorem**Theorem 26.1 — Measurement–Projection Separation Theorem**

A lawful empirical comparison requires two independently auditable constructions:

projection contract $\mathcal{P}$

and

measurement protocol $\mathcal{M}$.

The projection contract produces

$$P_{\mathcal{P}}(\omega),$$

the set or distribution of outputs permitted by the model.

The measurement protocol produces

$$O_{\mathcal{O}},$$

the set or distribution of measurand values permitted by the observation.

The comparison is meaningful only when:

1. both constructions pass their guards;

2. both refer to the same measurand;
3. both use compatible units and reference conventions;
4. uncertainty and dependence have been declared.

A residual between central values does not, by itself, identify either measurement error or model discrepancy.

Proof. The projection result depends on phase inference, structural evaluation, calibration, and readout.

The observation result depends on sampling, instrument response, analysis, and measurement uncertainty.

These are logically distinct processes.

The residual decomposition proves that their difference contains contributions from both measurement error and model discrepancy.

Therefore, lawful comparison requires separate audit and compatible definitions before intersection, residual, or falsification statements are made.

□

27. Closing Declaration

S1.I established the projection contract.

S1.II established admissibility, witnesses, and falsification conditions.

S1.III establishes the observation protocol.

A measurement protocol is

$$\mathcal{M} = (\Omega_{\mathcal{M}}, \mu, S, I, A, \mathcal{U}_{\mathcal{M}}, \mathcal{Q}_{\mathcal{M}}, W_{\mathcal{M}}).$$

Its realized report is

$$y_{\text{obs}}(\omega) = (A \circ I \circ S)(\omega).$$

Measurement error is

$$e_{\mathcal{M}} = y_{\text{obs}} - \mu.$$

Model discrepancy is

$$d_{\mathcal{P}} = \mu - \hat{y}_{\mathcal{P}}.$$

The residual is their sum:

$$r = e_{\mathcal{M}} + d_{\mathcal{P}}.$$

Uncertainty propagates by exact image sets or pushforward measures:

$$P_{\mathcal{P}} = \Gamma_{\theta}(E_{\mathbf{F}}(C_{\Phi})).$$

Inside one quadrant, continuous propagation is available.

Across a Flatwave wall,

$$\text{Flatwave}(C_{\Phi}) = \{-1, +1\}.$$

Near a primitive pole, arbitrarily small phase uncertainty may be amplified without finite bound.

The resulting measurement discipline is:

No observation without a measurand;
 no error without a reference;
 no uncertainty without a declared representation;
 no precision claim without repetition conditions;
 no residual attribution without separating
 measurement error from model discrepancy;
 no linear propagation across a discontinuity or pole.

END OF S1.III

S1.IV

Calibration and Dimensional Readout

Book 1 - Projection and Measurement

Abstract

Book 0 established a dimensionless structural calculus. S1.I defined the projection contract

$$\hat{y}_{\mathcal{P}} = \Gamma_{\theta} \circ E_{\mathbf{F}} \circ \Phi,$$

S1.II established admissibility and failure witnesses, and S1.III separated measurement error, uncertainty, and model discrepancy.

This essay formalizes the readout and calibration layer.

Calibration is the process by which external reference information is used to select the parameters

$$\theta \in \Theta$$

of a readout map

$$\Gamma_{\theta} : \mathcal{S}_{\mathcal{P}} \rightarrow \mathcal{Y}.$$

Because the Book 0 structural variables are dimensionless, every dimensional scale, offset, unit, and reference value appearing in the projected output must enter through the readout, the calibration data, or another explicitly declared external quantity.

A calibration is represented by

$$\hat{\theta} = K(\mathcal{C}),$$

where $\mathcal{C}$ is the calibration record and K is the declared calibration procedure.

For a scalar affine readout,

$$y = a + bs,$$

two distinct exact structural anchors determine a unique pair (a, b) . If the structural anchors coincide, the calibration is either inconsistent or underdetermined.

For a multivariable linear readout,

$$y = b + \beta^T \mathbf{s},$$

parameter identifiability requires full column rank of the design matrix. A numerically optimized parameter is not automatically identifiable; multiple parameter values may produce the same outputs.

Calibration uncertainty is propagated jointly with phase, measurement, and readout uncertainty:

$$P_{\mathcal{P}}(\omega) = \{\Gamma_{\theta}(\mathbf{s}) : \theta \in C_{\theta}, \mathbf{s} \in C_{\mathbf{F}}(\omega)\}.$$

The essay distinguishes fitting data from validation data. Reusing the same information to select parameters and to claim independent confirmation does not provide an independent test.

Calibration is valid only on its declared domain. Evaluation outside the calibrated structural or external region is extrapolation and requires an explicit guard. Parameter drift may invalidate a formerly lawful calibration without altering the Book 0 identities.

The resulting principle is:

Book 0 supplies form; calibration supplies scale.

IDENTIFICATION DIVISION

PROGRAM-ID

S1-IV-CALIBRATION-AND-DIMENSIONAL-READOUT

PURPOSE

To define:

- dimensional readout;
- calibration anchors;
- parameter estimation;
- parameter identifiability;
- calibration uncertainty;
- unit transformation;
- interpolation and extrapolation;
- calibration drift;
- and calibration failure witnesses.

PRINCIPAL QUESTION

How may dimensionless structural quantities be converted into externally meaningful, dimensionally declared outputs without hiding scale choices or empirical assumptions?

NON-PURPOSE

This essay does not:

- select a particular physical constant;
- calibrate the R-operators to hydrogen, gravity, cosmology, or another specific system;
- prove that a selected external anchor is fundamental;
- infer units from dimensionless identities alone;
- guarantee that a fitted model is correct;
- establish model superiority;
- replace independent validation with parameter fitting;
- remove Book 0 boundary restrictions.

ENVIRONMENT DIVISION

1. Inherited Projection Structure

A lawful projection contract has the form

$$\mathcal{P} = (\Omega, \Omega_{\mathcal{P}}, \Phi, \mathbf{F}, \Gamma_{\theta}, U_{\mathcal{P}}, W_{\mathcal{P}}).$$

For an admitted instance ω ,

$$\mathbf{s}(\omega) = E_{\mathbf{F}}(\Phi(\omega))$$

is its dimensionless structural state.

The predicted output is

$$\boxed{\hat{y}_{\mathcal{P}}(\omega) = \Gamma_{\theta}(\mathbf{s}(\omega)).}$$

1.1 Structural state space

Let

$$\mathcal{S}_{\mathcal{P}} \subseteq \mathbb{R}^m$$

be the set of structural states admitted by the contract.

Every coordinate of $\mathbf{s} \in \mathcal{S}_{\mathcal{P}}$ is dimensionless unless a later contract explicitly introduces a dimension-bearing external coordinate.

1.2 Measurement environment

A valid calibration record must be built from observation records satisfying the measurement requirements of S1.III.

A failed, censored, ambiguous, or dimensionally incompatible observation may not be used as an exact calibration anchor without an explicit treatment rule.

METHODOLOGICAL AXIOM DIVISION

Axiom C.1 — Explicit Scale Provenance

Every dimensional scale, offset, unit, conversion constant, and reference value used by a readout must have declared provenance.

The provenance must state whether the quantity is:

- fixed by unit convention;
- measured externally;
- selected as a reference;
- estimated from calibration data;
- fitted jointly with other parameters;
- imported from another model;
- or assumed.

No dimensional constant may be presented as though it were generated by the dimensionless structural calculus.

Axiom C.2 — Calibration–Validation Separation

Every datum used to select, constrain, or tune the calibration parameters must be identified as calibration information.

A claim of independent validation must use information not already exhausted in selecting those parameters, or must explicitly account for the dependence created by reuse.

DATA DIVISION

2. Dimensions and Units

Definition 2.1 — Dimension set

Let

$$\mathcal{D}$$

be the set of declared quantity dimensions.

Examples may include:

$$L, \quad T, \quad M, \quad Q,$$

representing length, time, mass, charge, or other base dimensions.

The dimensionless element is denoted

$$1.$$

Definition 2.2 — Dimension assignment

For a quantity q , write

$$[q] \in \mathcal{D}$$

for its dimension.

For every Book 0 structural variable s ,

$$\boxed{[s] = 1.}$$

Definition 2.3 — Unit representation

A unit representation assigns a numerical coordinate to a dimensional quantity relative to a declared unit.

Changing units changes the numerical representation but not the underlying quantity.

Definition 2.4 — Dimensional compatibility

Two quantities may be added or compared numerically only when they have compatible dimensions and reference conventions.

Thus,

$$q_1 + q_2$$

requires

$$[q_1] = [q_2].$$

Theorem 2.5 — Dimensional isolation

Let

$$\mathbf{s} \in \mathcal{S}_{\mathcal{P}}$$

be dimensionless.

If

$$y = \Gamma_{\theta}(\mathbf{s})$$

has nontrivial dimension $[y] \neq \mathbf{1}$, then at least one component of θ , the readout definition, or an external input to the readout must carry that dimension.

Proof. The structural state consists only of dimensionless numbers.

Algebraic combinations of dimensionless numbers remain dimensionless unless combined with a dimension-bearing quantity or assigned a dimension through an external convention.

Therefore, any nontrivial output dimension must enter through the readout layer or another declared external element.

□

Corollary 2.6 — No unit from Flatwave alone

Since

$$\text{Flatwave} \in \{-1, +1\},$$

Flatwave does not by itself determine seconds, metres, joules, hertz, or another unit.

Corollary 2.7 — No dimensional scale from λ

Since

$$0 \leq \lambda \leq 1$$

on its closed-quadrant extension, λ provides a normalized coordinate but not a dimensional magnitude.

3. Readout Maps

Definition 3.1 — Parametric readout

A parametric readout is a family of maps

$$\Gamma : \Theta \times \mathcal{S}_{\mathcal{P}} \rightarrow \mathcal{Y}$$

written

$$\Gamma_{\theta}(\mathbf{s}) = \Gamma(\theta, \mathbf{s}).$$

Definition 3.2 — Readout parameter

The parameter

$$\theta \in \Theta$$

contains every free or externally supplied quantity required to convert structural values into the output.

Definition 3.3 — Fixed parameter

A parameter is fixed when its value is specified before the calibration data under consideration are analyzed.

Definition 3.4 — Estimated parameter

A parameter is estimated when calibration information is used to determine or constrain its value.

Definition 3.5 — Fitted parameter

A parameter is fitted when it is selected by optimizing a declared objective function over calibration data.

Every fitted parameter is estimated, but not every estimate must arise from optimization.

Definition 3.6 — Readout domain

The readout domain is the set

$$\mathcal{S}_\theta \subseteq \mathcal{S}_\mathcal{P}$$

on which Γ_θ is declared valid.

It may be smaller than the complete structural state space.

4. Calibration Records**Definition 4.1 — Calibration instance set**

Let

$$\Omega_{\mathcal{C}} \subseteq \Omega_{\mathcal{P}} \cap \Omega_{\mathcal{M}}$$

be the external instances used for calibration.

Definition 4.2 — Structural calibration state

For each calibration instance ω_i ,

$$\boxed{\mathbf{s}_i = E_{\mathbf{F}}(\Phi(\omega_i)).}$$

Definition 4.3 — Calibration observation

Let

$$\mathcal{O}_i$$

be a valid observation record for the same measurand the readout is intended to predict.

Its admissible observed-value set is

$$\mathcal{O}_i \subseteq \mathcal{Y}.$$

Definition 4.4 — Exact calibration anchor

An exact calibration anchor is a pair

$$\boxed{(\mathbf{s}_i, y_i)}$$

for which the structural state and output value are both treated as exact under the declared calibration procedure.

Exact anchors are idealizations unless their uncertainty is demonstrably negligible for the intended purpose.

Definition 4.5 — Uncertain calibration anchor

An uncertain anchor is a pair

$$(S_i, O_i),$$

where

$$S_i \subseteq \mathcal{S}_{\mathcal{P}}$$

and

$$O_i \subseteq \mathcal{Y}$$

represent admissible structural and observed values.

Definition 4.6 — Calibration record

The calibration record is

$$\mathcal{C} = \{(\omega_i, S_i, O_i, \mathcal{O}_i)\}_{i=1}^n.$$

Definition 4.7 — Calibration procedure

A calibration procedure is a map or algorithm

$$K : \mathfrak{C} \rightarrow \mathfrak{T},$$

where $\mathfrak{C}$ is the class of admissible calibration records and $\mathfrak{T}$ is a class of:

- parameter values;
- parameter sets;
- or parameter distributions.

Definition 4.8 — Calibrated parameter result

The calibrated result is

$$\widehat{\Theta} = K(\mathcal{C}).$$

It may be:

- a single estimate $\hat{\theta}$;
- a confidence region;
- a credible distribution;
- an admissible parameter set;
- or a nonidentifiability report.

5. Calibration Lawfulness**Definition 5.1 — Calibration completeness**

A calibration is complete when it declares:

1. the readout family Γ_{θ} ;
2. the parameter space Θ ;
3. the calibration instances;
4. the structural states or structural uncertainty sets;
5. the observation records;
6. the objective, constraint, or anchor equations;
7. the treatment of uncertainty and dependence;
8. the calibration domain;
9. the parameter result;
10. failure behavior.

Definition 5.2 — Calibration admissibility

A calibration record is admissible when:

- all observation records are valid;
- all structural evaluations are lawful;
- units and measurands are compatible;

- prediction-dependent selection is disclosed;
- the calibration procedure is defined for the record;
- parameter constraints are declared.

Definition 5.3 — Calibration witness set

Recommended witnesses include:

- CALIBRATION-SPECIFICATION-INCOMPLETE;
- CALIBRATION-DATA-INVALID;
- CALIBRATION-MEASURAND-MISMATCH;
- CALIBRATION-UNIT-MISMATCH;
- CALIBRATION-ANCHORS-DEGENERATE;
- PARAMETERS-NOT-IDENTIFIABLE;
- CALIBRATION-OBJECTIVE-UNDEFINED;
- CALIBRATION-UNCERTAINTY-MISSING;
- CALIBRATION-DOMAIN-UNDECLARED;
- CALIBRATION-VALIDATION-LEAKAGE;
- CALIBRATION-DRIFT-DETECTED;
- EXTRAPOLATION-UNDECLARED.

6. Scalar Affine Calibration

Definition 6.1 — Scalar affine readout

Let

$$s \in \mathbb{R}$$

be dimensionless.

Define

$$\Gamma_{a,b}(s) = a + bs.$$

For dimensional consistency,

$$[a] = [b] = [y].$$

Theorem 6.2 — Unique two-anchor affine calibration

Let

$$(s_1, y_1)$$

and

$$(s_2, y_2)$$

be exact anchors with

$$s_1 \neq s_2.$$

Then there exists exactly one affine readout

$$y = a + bs$$

passing through both anchors.

Its parameters are

$$b = \frac{y_2 - y_1}{s_2 - s_1}$$

and

$$a = \frac{s_2 y_1 - s_1 y_2}{s_2 - s_1}.$$

Proof. The anchor equations are

$$y_1 = a + bs_1$$

and

$$y_2 = a + bs_2.$$

Subtracting,

$$y_2 - y_1 = b(s_2 - s_1).$$

Since

$$s_2 - s_1 \neq 0,$$

the slope is uniquely determined:

$$b = \frac{y_2 - y_1}{s_2 - s_1}.$$

Substituting into either anchor equation uniquely determines a .

□

Corollary 6.3 — Offset-free calibration

If the readout is constrained to pass through zero,

$$y = bs,$$

then one nonzero exact anchor determines

$$b = \frac{y}{s}.$$

Theorem 6.4 — Degenerate equal-state anchors

Suppose

$$s_1 = s_2.$$

Then:

1. if $y_1 \neq y_2$, no single-valued affine readout can satisfy both anchors;
2. if $y_1 = y_2$, infinitely many affine readouts satisfy both anchors.

Proof. Both anchor equations have the same structural input.

A function must assign one output to that input.

If the outputs differ, the requirements are inconsistent.

If the outputs agree, the single equation

$$y_1 = a + bs_1$$

leaves one free parameter.

□

Corollary 6.5 — Distinct outputs require structurally distinguishable anchors

Calibration cannot assign different deterministic outputs to two calibration instances that the selected structural profile and readout treat as the same state.

7. Normalized Transfer Readout**Definition 7.1 — Two-endpoint transfer readout**

For a declared lower anchor y_0 and upper anchor y_1 , define

$$\Gamma_{y_0, y_1}(\lambda) = y_0 + (y_1 - y_0)\lambda.$$

Theorem 7.2 — Endpoint correspondence

On the closed-quadrant extension,

$$\lambda = 0 \implies \Gamma(\lambda) = y_0,$$

and

$$\lambda = 1 \implies \Gamma(\lambda) = y_1.$$

Corollary 7.3 — Interior boundedness

If

$$y_0 < y_1$$

and

$$0 < \lambda < 1,$$

then

$$y_0 < \Gamma(\lambda) < y_1.$$

Clarification 7.4 — Endpoint anchors are external choices

The structural calculus supplies the normalized endpoints 0 and 1.

It does not supply the dimensional values y_0 and y_1 .

Clarification 7.5 — Boundary-aware requirement

If y_0 and y_1 are interpreted as values at the exact quadrant boundaries, the contract must use the split-boundary extensions of S0.V.

The primitive interior formulas may not be evaluated directly at those boundaries.

8. Polynomial and Nonlinear Readouts

Definition 8.1 — Polynomial readout

For a dimensionless scalar s ,

$$y = \sum_{k=0}^d a_k s^k.$$

Theorem 8.2 — Polynomial coefficient dimensions

Every nonzero coefficient must satisfy

$$[a_k] = [y].$$

Proof. Since s is dimensionless,

$$[s^k] = \mathbf{1}.$$

Therefore,

$$[a_k s^k] = [a_k].$$

All terms being added must have the same dimension as y .

□

Definition 8.3 — Nonlinear readout

A nonlinear readout is any declared map

$$\Gamma_\theta(\mathbf{s})$$

that is not affine in $\mathbf{s}$.

Examples include:

- rational maps;
- exponentials;
- thresholds;
- logistic functions;
- piecewise maps;
- saturation functions.

The use of such a readout is a projection choice. It is not a new theorem of Book 0.

Negative Theorem 8.4 — A flexible readout can conceal structural weakness

If the readout class is sufficiently flexible, it may reproduce many external datasets independently of the distinctive structure of $\mathbf{F}$.

Therefore, goodness of fit must be interpreted relative to:

- parameter count;
- readout flexibility;
- competing models;
- and out-of-sample performance.

9. Linear Multivariable Calibration

Definition 9.1 — Linear structural readout

Let

$$\mathbf{s} \in \mathbb{R}^m.$$

Define the scalar readout

$$y = b + \beta^\top \mathbf{s}.$$

The parameter vector is

$$\theta = (b, \beta_1, \dots, \beta_m)^\top.$$

Definition 9.2 — Design matrix

For calibration states $\mathbf{s}_1, \dots, \mathbf{s}_n$, define

$$X = \begin{pmatrix} 1 & \mathbf{s}_1^\top \\ \vdots & \vdots \\ 1 & \mathbf{s}_n^\top \end{pmatrix}.$$

The observed-output vector is

$$\mathbf{y} = (y_1, \dots, y_n)^\top.$$

The calibration equations are

$$X\theta = \mathbf{y}.$$

Theorem 9.3 — Exact linear identifiability

The parameter vector θ is uniquely determined by exact calibration equations only if

$$\text{rank}(X) = m + 1.$$

When the system is consistent and X has full column rank, the solution is unique.

Proof. If X does not have full column rank, there exists a nonzero vector v such that

$$Xv = 0.$$

If θ is one solution, then

$$X(\theta + v) = X\theta + Xv = \mathbf{y}.$$

Thus the solution is not unique.

If X has full column rank, its null space is trivial. Hence no two distinct parameter vectors can produce the same exact calibration outputs.

□

Corollary 9.4 — Duplicate structural information does not add rank

Repeating calibration instances with identical structural states may improve precision under a measurement model, but it does not increase the algebraic rank of X .

Corollary 9.5 — Collinear structural coordinates cause parameter degeneracy

If one selected structural coordinate is a linear combination of the others throughout the calibration record, its coefficient cannot be independently identified by the linear readout.

10. Weighted Linear Calibration

Definition 10.1 — Weighted quadratic objective

Let W be a symmetric positive-semidefinite matrix.

Define

$$J(\theta) = (\mathbf{y} - X\theta)^\top W (\mathbf{y} - X\theta).$$

Theorem 10.2 — Unique weighted least-squares estimate

If

$$X^\top W X$$

is positive definite, then J has the unique minimizer

$$\hat{\theta} = (X^\top W X)^{-1} X^\top W \mathbf{y}.$$

Proof. Differentiating,

$$\nabla J(\theta) = -2X^\top W (\mathbf{y} - X\theta).$$

Setting the gradient to zero gives

$$X^\top W X \theta = X^\top W \mathbf{y}.$$

Positive definiteness makes $X^\top W X$ invertible and the quadratic objective strictly convex.

Therefore, the stationary point is unique and is the global minimizer.

□

Clarification 10.3 — Weight interpretation must be declared

Weights may represent:

- inverse variances;
- relative importance;
- sampling corrections;
- robust-loss approximations;
- or another convention.

Different weight interpretations produce different calibrations.

Negative Theorem 10.4 — Numerical uniqueness is not physical necessity

A unique minimizer of a chosen objective proves only that the optimization problem has one solution.

It does not prove that:

- the readout family is correct;
- the weights are correct;
- the structural profile is externally relevant;
- or the parameter has fundamental physical status.

11. Nonlinear Parameter Identifiability

Definition 11.1 — Calibration prediction vector

For calibration instances $\omega_1, \dots, \omega_n$, define

$$\mathbf{g}(\theta) = \begin{pmatrix} \Gamma_{\theta}(\mathbf{s}_1) \\ \vdots \\ \Gamma_{\theta}(\mathbf{s}_n) \end{pmatrix}.$$

Definition 11.2 — Global identifiability

The parameter is globally identifiable on Θ when

$$\mathbf{g}(\theta_1) = \mathbf{g}(\theta_2) \implies \theta_1 = \theta_2.$$

Definition 11.3 — Observationally equivalent parameters

Parameters θ_1 and θ_2 are observationally equivalent on the calibration record when

$$\mathbf{g}(\theta_1) = \mathbf{g}(\theta_2).$$

Definition 11.4 — Sensitivity matrix

When differentiable, define

$$J_{\theta} = \frac{\partial \mathbf{g}}{\partial \theta}.$$

Theorem 11.5 — Rank-deficiency obstruction

If J_θ has a nontrivial null direction at θ^* , then there exists a first-order parameter perturbation that produces no first-order change in the calibration predictions.

Proof. If

$$J_\theta v = 0$$

for nonzero v , then

$$\mathbf{g}(\theta^* + tv) = \mathbf{g}(\theta^*) + tJ_\theta v + o(t).$$

Thus,

$$\mathbf{g}(\theta^* + tv) = \mathbf{g}(\theta^*) + o(t).$$

The perturbation is invisible to first order.

□

Corollary 11.6 — Full rank is a local diagnostic

Full column rank supports local parameter distinguishability.

It does not, by itself, prove global uniqueness across the complete parameter space.

Negative Theorem 11.7 — Parameter estimation does not imply parameter identification

An optimization routine may return one parameter value even when many values produce indistinguishable predictions.

12. Calibration Uncertainty**Definition 12.1 — Parameter admissibility set**

Let

$$C_\theta \subseteq \Theta$$

be the set of parameter values compatible with the calibration record and the declared calibration procedure.

Definition 12.2 — Parameter distribution

Under a probabilistic calibration, let

$$\nu_\theta$$

be the declared distribution over parameter values.

Definition 12.3 — Joint structural–calibration output set

Let

$$C_{\mathbf{F}}(\omega) \subseteq \mathcal{S}_{\mathcal{P}}$$

be the structural-state uncertainty set.

Define

$$P_{\mathcal{P}}(\omega) = \{\Gamma_\theta(\mathbf{s}) : \theta \in C_\theta, \mathbf{s} \in C_{\mathbf{F}}(\omega)\}.$$

Theorem 12.4 — Exact joint set propagation

The complete predicted-output set is the image of

$$C_\theta \times C_{\mathbf{F}}(\omega)$$

under the map

$$(\theta, \mathbf{s}) \mapsto \Gamma_\theta(\mathbf{s}).$$

Proof. The displayed set contains exactly the outputs generated by every admissible parameter and structural-state pair.

This is the definition of the image of the Cartesian product under the readout map.

□

Definition 12.5 — Joint probabilistic propagation

Let

$$\nu_{\theta, \mathbf{s}}$$

be the declared joint distribution of calibration parameters and structural state.

Then

$$\nu_y = \Gamma_{\#} \nu_{\theta, \mathbf{s}}.$$

Clarification 12.6 — Independence may not be assumed silently

If the same observations influence both phase inference and calibration, then θ and $\mathbf{s}$ may be statistically dependent.

Replacing the joint distribution by

$$\nu_{\theta} \otimes \nu_{\mathbf{s}}$$

requires an independence justification.

13. Calibration Contribution to Residuals

Definition 13.1 — Reference calibration

Suppose θ^* denotes the parameter value that would produce the correct readout within the selected readout family.

This is a conceptual reference and need not be known.

Definition 13.2 — Calibration discrepancy

For structural state $\mathbf{s}$, define

$$c_{\theta}(\mathbf{s}) = \Gamma_{\theta^*}(\mathbf{s}) - \Gamma_{\hat{\theta}}(\mathbf{s}).$$

Definition 13.3 — Structural-model discrepancy

Define

$$d_{\text{str}} = \mu - \Gamma_{\theta^*}(\mathbf{s}).$$

Theorem 13.4 — Three-part residual decomposition

$$y_{\text{obs}} - \Gamma_{\hat{\theta}}(\mathbf{s}) = e_{\mathcal{M}} + d_{\text{str}} + c_{\theta}.$$

Proof. Add and subtract μ and $\Gamma_{\theta^*}(\mathbf{s})$:

$$y_{\text{obs}} - \Gamma_{\hat{\theta}}(\mathbf{s}) = (y_{\text{obs}} - \mu) + (\mu - \Gamma_{\theta^*}(\mathbf{s})) + (\Gamma_{\theta^*}(\mathbf{s}) - \Gamma_{\hat{\theta}}(\mathbf{s})).$$

The three terms are measurement error, structural-model discrepancy, and calibration discrepancy.

□

Corollary 13.5 — Calibration error may mimic structural failure

A nonzero residual does not by itself distinguish incorrect structural modeling from incorrect calibration.

14. Calibration Domain

Definition 14.1 — External calibration domain

Let

$$\Omega_\theta \subseteq \Omega_{\mathcal{P}}$$

be the external region on which the calibration is declared valid.

Definition 14.2 — Structural calibration domain

Let

$$\mathcal{S}_\theta \subseteq \mathcal{S}_{\mathcal{P}}$$

be the corresponding region of structural states covered by the calibration declaration.

Definition 14.3 — Calibration range

For a scalar structural coordinate, the calibration range is the interval or union of intervals represented by the calibration states.

For a multivariable state, it is the declared region containing the calibration states.

Definition 14.4 — Interpolation

In a Euclidean structural state space, a prediction is interpolative when its structural state lies inside the declared interpolation region, often the convex hull of the calibration states.

Definition 14.5 — Extrapolation

A prediction is extrapolative when its structural state lies outside the declared calibration region.

Theorem 14.6 — Calibration lawfulness does not automatically extend

A calibration validated on $\mathcal{S}_\theta$ is not thereby validated on

$$\mathcal{S}_{\mathcal{P}} \setminus \mathcal{S}_\theta.$$

Proof. The calibration record constrains the readout only on the represented calibration region.

Different readout functions may agree on that region and diverge outside it.

Therefore, agreement within the calibration domain does not determine or validate behavior outside it.

□

Corollary 14.7 — Formula evaluation is not calibration validation

The numerical ability to evaluate $\Gamma_{\hat{\theta}}(\mathbf{s})$ outside $\mathcal{S}_\theta$ does not establish that the output is reliable there.

Required witness

Undeclared extrapolation must return

EXTRAPOLATION-UNDECLARED.

15. Singular and Boundary Calibration Guards**Theorem 15.1 — Primitive-pole calibration instability**

Suppose a readout uses a primitive operator that diverges at a boundary.

Calibration based on states approaching that boundary may be arbitrarily sensitive to small phase error.

Proof. S1.III proved that primitive values are unbounded on every neighborhood of the relevant pole.

Therefore, arbitrarily small phase perturbations may produce arbitrarily large changes in the structural input to the readout.

Unless the readout suppresses that divergence in a controlled and declared way, calibration sensitivity is unbounded.

□

Corollary 15.2 — Minimum boundary distance

A primitive-based calibration should declare a minimum admissible distance from excluded boundaries or use an exact singular uncertainty treatment.

Theorem 15.3 — Flatwave calibration cannot resolve interior phase

Within a fixed open quadrant,

$$\text{Flatwave}(x) = \varepsilon$$

is constant.

Therefore, a readout using Flatwave alone cannot distinguish two phases within the same quadrant.

Proof. Equal structural inputs produce equal deterministic readout outputs.

Since every phase in the quadrant has the same Flatwave value, no Flatwave-only readout can distinguish them.

□

Corollary 15.4 — Flatwave supplies a sign channel, not a continuous calibration coordinate

A continuously varying external quantity cannot be calibrated uniquely from Flatwave alone within one quadrant.

Theorem 15.5 — Saw-based calibration remains bounded

A readout based continuously on the saw coordinates receives structural inputs bounded in $[-1, 1]$.

This avoids primitive divergence but does not remove seam discontinuity or branch ambiguity.

16. Unit Transformations**Definition 16.1 — Affine unit conversion**

Let two numerical unit systems be related by

$$u(y) = cy + d,$$

with $c \neq 0$.

Pure scale conversions have $d = 0$.

Offset units may have $d \neq 0$.

Definition 16.2 — Converted readout

Define

$$\Gamma_{\theta}^{(u)} = u \circ \Gamma_{\theta}.$$

Theorem 16.3 — Unit-covariant prediction

If

$$\hat{y} = \Gamma_{\theta}(\mathbf{s}),$$

then the numerical prediction in the converted unit is

$$\hat{y}^{(u)} = u(\hat{y}).$$

Proof. By definition,

$$\hat{y}^{(u)} = \Gamma_{\theta}^{(u)}(\mathbf{s}) = u(\Gamma_{\theta}(\mathbf{s})) = u(\hat{y}).$$

□

Corollary 16.4 — Affine parameter transformation

For

$$\Gamma_{a,b}(s) = a + bs,$$

the converted readout is

$$u(\Gamma_{a,b}(s)) = (ca + d) + (cb)s.$$

Thus,

$$a^{(u)} = ca + d, \quad b^{(u)} = cb.$$

Clarification 16.5 — Numerical parameter values depend on units

A dimensional calibration parameter may change numerically under unit conversion while representing the same underlying quantity.

Negative Theorem 16.6 — Unit-dependent numerical equality is not physical identity

Two parameters with the same numerical value in different unit systems need not represent the same dimensional quantity.

17. Calibration Drift

Definition 17.1 — Context-dependent parameter

Let c denote external conditions such as:

- time;
- temperature;
- instrument state;
- population;
- environment;
- manufacturing batch;
- reference frame.

A drifting parameter has form

$$\theta = \theta(c).$$

Definition 17.2 — Fixed-calibration assumption

A contract using one constant $\hat{\theta}$ assumes

$$\theta(c) = \hat{\theta}$$

throughout its declared calibration domain.

Definition 17.3 — Calibration drift

Calibration drift occurs when the fixed-calibration assumption fails materially.

Theorem 17.4 — Drift invalidates fixed readout without altering structural identities

If the true readout parameter changes from θ_1 to θ_2 while the contract continues to use θ_1 , projected outputs may become inaccurate even though every Book 0 structural identity remains exact.

Proof. The structural state $\mathbf{s}$ and its identities do not depend on the external calibration parameter. The output does:

$$\hat{y} = \Gamma_{\theta_1}(\mathbf{s}).$$

If the appropriate readout is instead

$$\Gamma_{\theta_2}(\mathbf{s}),$$

the output discrepancy arises in the readout layer.

The Book 0 identities are unchanged.

□

Definition 17.5 — Drift detection rule

A calibration contract should declare:

- monitoring variables;
- acceptable parameter variation;
- recalibration threshold;
- and failure action.

Required witness

Detected material drift should return

CALIBRATION-DRIFT-DETECTED.

18. Calibration Transfer

Definition 18.1 — Source calibration domain

Let a calibration be established on domain

$$\Omega_A.$$

Definition 18.2 — Target domain

Let

$$\Omega_B$$

be a distinct external domain to which the same parameters are proposed to apply.

Definition 18.3 — Calibration transfer assumption

Calibration transfer assumes that the same readout parameters remain valid on Ω_B .

Theorem 18.4 — Transfer is an additional hypothesis

Calibration validity on Ω_A does not imply validity on Ω_B .

Proof. The calibration record contains evidence from Ω_A , not necessarily from Ω_B .

Conditions affecting the readout may differ between the domains.

Therefore, parameter invariance across the two domains is an additional claim requiring justification or testing.

□

Corollary 18.5 — Imported constants require provenance

A parameter imported from another experiment or field must state:

- its source;
- original calibration domain;
- uncertainty;
- unit convention;
- and transfer justification.

19. Calibration and Validation Data

Definition 19.1 — Calibration set

Let

$$\mathcal{C}_{\text{fit}}$$

contain the records used to select $\hat{\theta}$.

Definition 19.2 — Validation set

Let

$$\mathcal{C}_{\text{val}}$$

contain records used to evaluate the calibrated projection after parameter selection.

Definition 19.3 — Independent validation

Validation is independent when its information did not influence:

- parameter choice;
- model form;
- structural-profile selection;
- preprocessing choices;
- stopping rules;
- or reporting thresholds.

Theorem 19.4 — Training fit is not independent validation

Agreement with

$$\mathcal{C}_{\text{fit}}$$

does not constitute an independent test of the calibration selected from that same information.

Proof. The parameter was chosen, at least in part, to improve agreement with the fitting data.

The observed agreement is therefore statistically and procedurally dependent on the parameter-selection process.

It may demonstrate optimization success, but not independent predictive performance.

□

Corollary 19.5 — Data reuse must be disclosed

If the same data are used for both calibration and evaluation, the dependence must be represented in the uncertainty or validation procedure.

Definition 19.6 — Validation leakage

Validation leakage occurs when information from the nominal validation set influences model or parameter selection without being accounted for.

Required witness

CALIBRATION-VALIDATION-LEAKAGE.

20. Overfitting and Readout Complexity**Definition 20.1 — Calibration fit error**

Let

$$L_{\text{fit}}(\hat{\theta})$$

be the declared loss on the calibration data.

Definition 20.2 — Validation error

Let

$$L_{\text{val}}(\hat{\theta})$$

be the corresponding loss on independent validation data.

Definition 20.3 — Overfitting

Overfitting occurs when a readout captures features specific to the calibration record that do not generalize to the declared target domain.

Negative Theorem 20.4 — Perfect fit does not prove correct projection

A sufficiently flexible readout may interpolate a finite calibration dataset exactly while making poor predictions elsewhere.

Proof. For finitely many distinct scalar structural states, a polynomial of sufficiently high degree can interpolate arbitrary output values.

Exact agreement on those points therefore does not uniquely determine behavior between or beyond them.

□

Corollary 20.5 — Parameter count must be reported

The number of fitted parameters and the flexibility of the readout class are part of the calibration record.

Corollary 20.6 — Simpler readouts require fewer calibration constraints

A low-parameter readout is generally easier to identify and audit, though simplicity alone does not establish correctness.

21. Calibration Failure Locality

Definition 21.1 — Calibration-field dependency

The readout output depends on:

$$\Gamma, \theta, \mathcal{C}, K, S_\theta.$$

Theorem 21.2 — Calibration repair locality

Changing calibration parameters requires recalculating projected outputs and their calibration uncertainty.

It does not require reproving Book 0 identities.

Proof. Book 0 identities depend on the structural definitions and phase domain.

Calibration parameters appear only after structural evaluation.

Therefore, changing them affects the readout layer but not the structural theorem layer.

□

Corollary 21.3 — Readout-form revision requires broader reevaluation

Changing the readout family Γ may alter:

- parameter identifiability;
- dimensional consistency;
- calibration domain;
- extrapolation behavior;
- uncertainty propagation;
- and validation results.

All dependent claims must be rechecked.

PROCEDURE DIVISION

22. Canonical Calibration Procedure

```
CALIBRATE-READOUT.  
PERFORM CHECK-CALIBRATION-SPECIFICATION.  
IF CALIBRATION-SPECIFICATION-INCOMPLETE  
RETURN CALIBRATION-WITNESS-SET  
END-IF.  
PERFORM VALIDATE-CALIBRATION-OBSERVATIONS.  
IF CALIBRATION-DATA-INVALID  
RETURN CALIBRATION-WITNESS-SET  
END-IF.  
PERFORM VERIFY-MEASURAND-AND-UNIT-COMPATIBILITY.  
IF CALIBRATION-MEASURAND-MISMATCH  
RETURN CALIBRATION-WITNESS-SET  
END-IF.  
IF CALIBRATION-UNIT-MISMATCH  
RETURN CALIBRATION-WITNESS-SET  
END-IF.  
PERFORM EVALUATE-CALIBRATION-STRUCTURAL-STATES.  
IF STRUCTURAL-EVALUATION-UNDEFINED  
RETURN CALIBRATION-WITNESS-SET  
END-IF.  
PERFORM CHECK-ANCHOR-DEGENERACY.  
PERFORM CHECK-PARAMETER-IDENTIFIABILITY.  
IF CALIBRATION-ANCHORS-DEGENERATE  
RETURN CALIBRATION-WITNESS-SET  
END-IF.  
IF PARAMETERS-NOT-IDENTIFIABLE  
RETURN PARAMETER-SET  
OR IDENTIFIABILITY-WITNESS  
END-IF.  
PERFORM APPLY-CALIBRATION-PROCEDURE.
```

PERFORM ESTIMATE-CALIBRATION-UNCERTAINTY.
PERFORM DECLARE-CALIBRATION-DOMAIN.
PERFORM CHECK-CALIBRATION-VALIDATION-SEPARATION.
IF CALIBRATION-UNCERTAINTY-MISSING
RETURN CALIBRATION-WITNESS-SET
END-IF.
IF CALIBRATION-DOMAIN-UNDECLARED
RETURN CALIBRATION-WITNESS-SET
END-IF.
RETURN CALIBRATED-PARAMETER-RESULT
CALIBRATION-UNCERTAINTY
CALIBRATION-DOMAIN
COMPLETE-AUDIT-RECORD.

22.1 Readout procedure

APPLY-DIMENSIONAL-READOUT.
PERFORM CHECK-INSTANCE-STRUCTURAL-STATE.
PERFORM CHECK-CALIBRATION-STATUS.
PERFORM CHECK-CALIBRATION-DOMAIN.
PERFORM CHECK-UNIT-CONVENTION.
PERFORM CHECK-DRIFT-STATUS.
IF STRUCTURAL-STATE OUTSIDE CALIBRATION-DOMAIN
IF EXTRAPOLATION-RULE IS NOT DECLARED
RETURN EXTRAPOLATION-UNDECLARED
END-IF
END-IF.
IF CALIBRATION-DRIFT-DETECTED
RETURN CALIBRATION-DRIFT-DETECTED
END-IF.
APPLY GAMMA-THETA TO STRUCTURAL-STATE.
PROPAGATE JOINT STRUCTURAL
AND CALIBRATION UNCERTAINTY.
RETURN DIMENSIONAL-PREDICTION

OUTPUT-UNIT

PREDICTED-OUTPUT-SET

CALIBRATION-PROVENANCE.

23. Theorem Ledger

Theorem IV.1 — Dimensional isolation

Nontrivial output dimensions must enter through the calibration or readout layer.

Theorem IV.2 — Unique two-anchor affine calibration

Two exact anchors with distinct structural coordinates determine one affine scalar readout.

Theorem IV.3 — Degenerate-anchor classification

Equal structural anchors with unequal outputs are inconsistent; equal outputs are underdetermining.

Theorem IV.4 — Polynomial coefficient dimensions

For dimensionless structural input, every polynomial coefficient carries the output dimension.

Theorem IV.5 — Linear rank identifiability

A linear multivariable readout requires full column rank for unique parameter determination.

Theorem IV.6 — Weighted least-squares uniqueness

If

$$X^T W X$$

is positive definite, the weighted quadratic calibration has a unique minimizer.

Theorem IV.7 — Nonlinear rank obstruction

A null direction of the parameter sensitivity matrix is invisible to first order.

Theorem IV.8 — Exact joint uncertainty propagation

$$P_{\mathcal{P}}(\omega) = \{\Gamma_{\theta}(\mathbf{s}) : \theta \in C_{\theta}, \mathbf{s} \in C_{\mathbf{F}}(\omega)\}.$$

Theorem IV.9 — Three-part residual decomposition

$$y_{\text{obs}} - \Gamma_{\hat{\theta}}(\mathbf{s}) = e_{\mathcal{M}} + d_{\text{str}} + c_{\theta}.$$

Theorem IV.10 — Calibration-domain restriction

Calibration validity does not automatically extend beyond its declared structural or external domain.

Theorem IV.11 — Primitive-pole instability

Primitive-based calibration may become arbitrarily sensitive near excluded boundaries.

Theorem IV.12 — Flatwave nonresolution

Flatwave alone cannot resolve phase location within one open quadrant.

Theorem IV.13 — Unit covariance

Changing units transforms the numerical readout and parameters while preserving the underlying projected quantity.

Theorem IV.14 — Drift separation

Calibration drift may invalidate projected outputs without affecting Book 0 identities.

Theorem IV.15 — Calibration-transfer burden

Parameter validity in one domain does not imply validity in another.

Theorem IV.16 — Fit-validation separation

Agreement with data used to select parameters is not an independent validation result.

24. Negative Theorems**Negative Theorem 24.1 — No dimensional scale from structural form alone**

A dimensionless identity cannot select a physical unit or numerical scale without external input.

Negative Theorem 24.2 — No unique calibration from degenerate anchors

Identical structural anchors do not uniquely determine a nontrivial readout.

Negative Theorem 24.3 — No identifiability from optimizer output alone

A numerical routine returning one estimate does not prove that alternative parameter values are distinguishable.

Negative Theorem 24.4 — No physical necessity from best fit alone

The parameter minimizing a declared loss is not thereby a fundamental constant.

Negative Theorem 24.5 — No independent confirmation from reused data

Calibration data cannot simultaneously function as unqualified independent validation data.

Negative Theorem 24.6 — No extrapolation by analytic continuation alone

A readout formula remaining finite outside its calibration domain does not establish external validity there.

Negative Theorem 24.7 — No fixed calibration under undeclared drift

A parameter known to vary with external conditions may not be treated as constant without restricting or modeling those conditions.

Negative Theorem 24.8 — No unit-free comparison of dimensional parameters

Numerical parameter values in incompatible unit systems may not be compared without conversion.

Negative Theorem 24.9 — No residual attribution to structure alone

A residual may contain measurement error, structural discrepancy, and calibration discrepancy.

Negative Theorem 24.10 — No continuous phase information from Flatwave alone

A two-valued quadrant sign cannot uniquely encode a continuous within-quadrant coordinate.

25. Terminal Theorem**Theorem 25.1 — Dimensional Readout Separation Theorem**

Let

$$\mathbf{s}(\omega) = E_{\mathbf{F}}(\Phi(\omega))$$

be a dimensionless structural state.

Every lawful dimensional prediction has the form

$$\hat{y}_{\mathcal{P}}(\omega) = \Gamma_{\hat{\theta}}(\mathbf{s}(\omega)),$$

where the calibrated parameter result

$$\widehat{\Theta} = K(\mathcal{C})$$

is obtained from explicitly declared external information.

The Book 0 structure determines neither:

- the output dimension;
- the unit;
- the scale;
- the offset;
- the calibration anchors;
- the parameter values;
- nor the calibration domain.

These belong to the readout and calibration contract.

Proof. The structural state is dimensionless by Book 0 and S1.I.

The output dimension and numerical scale enter only when the readout map and calibration parameters are applied.

The parameter result is selected from external calibration information by K .

None of those external choices is contained in the structural definitions.

Therefore, structural form and dimensional calibration are logically distinct.

□

26. Closing Declaration

S1.I established the projection contract.

S1.II established admission and falsification.

S1.III established measurement and uncertainty.

S1.IV establishes calibration and dimensional readout.

The calibrated parameter result is

$$\widehat{\Theta} = K(\mathcal{C}).$$

The projected output is

$$\hat{y}_{\mathcal{P}} = \Gamma_{\hat{\theta}} \circ E_{\mathbf{F}} \circ \Phi.$$

For an affine scalar readout,

$$y = a + bs,$$

two distinct exact anchors determine

$$b = \frac{y_2 - y_1}{s_2 - s_1}$$

and

$$a = \frac{s_2 y_1 - s_1 y_2}{s_2 - s_1}.$$

For a linear multivariable readout, unique parameter determination requires

$$\text{rank}(X) = m + 1.$$

Calibration uncertainty combines with structural uncertainty through

$$P_{\mathcal{P}}(\omega) = \{\Gamma_{\theta}(\mathbf{s}) : \theta \in C_{\theta}, \mathbf{s} \in C_{\mathbf{F}}(\omega)\}.$$

The complete residual may contain

$$\text{measurement error} + \text{structural discrepancy} + \text{calibration discrepancy}.$$

The resulting calibration discipline is:

$$\begin{array}{l} \text{No unit without declared provenance;} \\ \text{no scale without an external anchor;} \\ \text{no fitted parameter without an objective;} \\ \text{no unique parameter without identifiability;} \\ \text{no extrapolation without a domain declaration;} \\ \text{no independent validation from calibration data alone;} \\ \text{no fixed readout after material drift.} \end{array}$$

Book 0 supplies form; calibration supplies scale.

END OF S1.IV

S1.V

Projection Equivalence and Model Comparison

Book 1 - Projection and Measurement

Abstract

S1.I defined projection contracts. S1.II established admissibility and failure witnesses. S1.III formalized measurement, error, and uncertainty. S1.IV established calibration and dimensional readout.

This essay determines when two projections are genuinely different, when they are merely different descriptions of the same prediction, and what is required to prefer one model over another.

Several levels of equivalence are distinguished:

1. representational equivalence, in which two contracts differ only by invertible reparameterization;
2. structural equivalence, in which their internal structural states correspond through an invertible map;
3. predictive equivalence, in which they produce the same output sets or distributions;
4. observational equivalence, in which a declared measurement protocol cannot distinguish their outputs;
5. finite-record indistinguishability, in which they agree only on the observations presently available.

These relations are not interchangeable.

An invertible change of phase coordinate or structural variable does not create a new empirical model when the readout is transformed correspondingly. Conversely, two structurally different projections may remain observationally equivalent because calibration, readout, uncertainty, or instrument resolution erases their differences.

Within a fixed open quadrant, the transfer coordinate

$$\lambda$$

is strictly increasing and therefore identifies phase uniquely. Globally, however,

$$\lambda\left(x + \frac{\pi}{2}\right) = \lambda(x),$$

so λ alone cannot identify the quadrant. Flatwave supplies quadrant sign but cannot identify position within the quadrant. The pair

$$(\text{Flatwave}, \lambda)$$

distinguishes positive from negative quadrants and within-quadrant position, but still does not distinguish phases separated by π . Additional orientation or quadrant information is required for global phase identification.

A comparison contract is introduced:

$$\mathcal{K} = (\{\mathcal{P}_j\}_{j=1}^r, \Omega_{\mathcal{K}}, \mathcal{M}, \mathcal{D}_{\text{fit}}, \mathcal{D}_{\text{val}}, L, C, \Delta),$$

where L is a declared predictive loss or score, C is a declared complexity measure, and Δ is a declared decision rule.

For nested model classes, the more flexible class cannot have worse minimum fitting loss on the same calibration data. This does not imply better validation performance. A perfect fit may result from excessive readout flexibility, parameter nonidentifiability, or data reuse.

No universal comparison metric is imposed. Deterministic error, interval compatibility, likelihood, proper scoring rules, residual diagnostics, and other methods may be used only when their assumptions are declared.

The final result is:

equivalent predictions cannot be distinguished by prediction-only evidence.

A model may be preferred only relative to a common domain, observation protocol, uncertainty model, validation record, scoring rule, and complexity criterion. Even then, comparative preference does not prove unique ontology.

IDENTIFICATION DIVISION

PROGRAM-ID

S1-V-PROJECTION-EQUIVALENCE-AND-MODEL-COMPARISON

PURPOSE

To formalize:

- equivalence between projection contracts;
- reparameterization;
- identifiability;
- observational indistinguishability;

- model nesting;
- predictive scoring;
- complexity;
- validation;
- comparative preference;
- and the limits of model selection.

PRINCIPAL QUESTIONS

1. When do two apparently different projections make the same claim?
2. When can observations distinguish them?
3. What evidence justifies preferring one over another?
4. What does comparative success fail to prove?

NON-PURPOSE

This essay does not:

- select one physical projection;
- prescribe one universal statistical method;
- claim that the simplest model is always true;
- infer ontology from predictive success;
- equate parameter count with complete model complexity;
- permit comparison across incompatible measurands or units;
- convert finite-data agreement into universal equivalence;
- alter the structural theorems of Book 0.

ENVIRONMENT DIVISION

1. Inherited Projection Contract

For $j = 1, \dots, r$, let

$$\mathcal{P}_j = (\Omega_j, \Omega_{\mathcal{P}_j}, \Phi_j, \mathbf{F}_j, \Gamma_{j, \theta_j}, U_j, W_j)$$

be lawful projection contracts.

Their predictions may be represented as:

- point outputs

$$\hat{y}_j(\omega);$$

- admissible output sets

$$P_j(\omega) \subseteq \mathcal{Y};$$

- or probability measures

$$\nu_j(\cdot \mid \omega).$$

Definition 1.1 — Common external domain

The common external domain is

$$\Omega_{\cap} = \bigcap_{j=1}^r \Omega_{\mathcal{P}_j}.$$

A comparative domain must satisfy

$$\Omega_{\mathcal{K}} \subseteq \Omega_{\cap}.$$

Definition 1.2 — Common measurand requirement

Two projections are directly comparable only when their outputs refer to the same declared measurand or are connected by a lawful conversion.

Definition 1.3 — Common output representation

Before comparison, all predictions must be expressed in a common:

- output space;
- unit system;
- reference frame;
- coordinate convention;
- and uncertainty interpretation.

A numerical comparison without this normalization is not lawful.

METHODOLOGICAL AXIOM DIVISION

Axiom E.1 — Comparison Declaration

Every claim that one projection is equivalent, superior, simpler, better supported, or falsified relative to another must be mediated by an explicit comparison contract.

Axiom E.2 — Equivalence-Level Declaration

Every equivalence claim must identify its level.

The statement “the models are equivalent” is incomplete unless it specifies whether the equivalence is:

- representational;
- structural;
- predictive;
- observational;
- or restricted to a finite record.

Axiom E.3 — Common-Evidence Requirement

Comparative preference must be based on evidence and scoring procedures applied under compatible conditions.

A model may not be credited for one domain while its competitor is judged on a different, more difficult domain without explicit adjustment.

DATA DIVISION

2. Projection Representations

Definition 2.1 — Complete prediction map

For deterministic comparison, define

$$H_j = \Gamma_{j, \theta_j} \circ E_{\mathbf{F}_j} \circ \Phi_j.$$

Thus,

$$H_j : \Omega_{\mathcal{P}_j} \rightarrow \mathcal{Y}.$$

Definition 2.2 — Structural-state map

Define

$$S_j = E_{\mathbf{F}_j} \circ \Phi_j.$$

Thus,

$$S_j : \Omega_{\mathcal{P}_j} \rightarrow \mathcal{S}_j.$$

Definition 2.3 — Prediction family

When parameters remain free, define

$$\mathfrak{H}_j = \{H_{j,\theta} : \theta \in \Theta_j\}.$$

The projection family, rather than one fitted parameter value, is the relevant object when comparing model flexibility.

3. Representational Equivalence**Definition 3.1 — External-coordinate reparameterization**

Let

$$h : \Omega_{\mathcal{P}_1} \rightarrow \Omega_{\mathcal{P}_2}$$

be a bijection preserving the external instances under a change of representation.

Definition 3.2 — Phase reparameterization

Let

$$r : \Phi_1(\Omega_{\mathcal{K}}) \rightarrow \Phi_2(\Omega_{\mathcal{K}})$$

be a bijection.

Definition 3.3 — Structural reparameterization

Let

$$g : S_1(\Omega_{\mathcal{K}}) \rightarrow S_2(\Omega_{\mathcal{K}})$$

be a bijection.

Definition 3.4 — Readout compensation

The readouts compensate for g when

$$\Gamma_{2,\theta_2} \circ g = \Gamma_{1,\theta_1}$$

on the relevant structural image.

Definition 3.5 — Representational equivalence

Two projections are representationally equivalent on $\Omega_{\mathcal{K}}$ when they differ only by invertible coordinate or parameter transformations that leave the complete prediction unchanged.

Theorem 3.6 — Reparameterization invariance

Suppose

$$S_2 = g \circ S_1$$

and

$$\Gamma_{2,\theta_2} \circ g = \Gamma_{1,\theta_1}.$$

Then

$$H_2(\omega) = H_1(\omega)$$

for every

$$\omega \in \Omega_{\mathcal{K}}.$$

Proof. For each ω ,

$$H_2(\omega) = \Gamma_{2,\theta_2}(S_2(\omega)).$$

Using

$$S_2 = g \circ S_1,$$

we obtain

$$H_2(\omega) = \Gamma_{2,\theta_2}(g(S_1(\omega))).$$

By readout compensation,

$$\Gamma_{2,\theta_2} \circ g = \Gamma_{1,\theta_1}.$$

Therefore,

$$H_2(\omega) = \Gamma_{1,\theta_1}(S_1(\omega)) = H_1(\omega).$$

□

Corollary 3.7 — Coordinate change does not create new predictive content

An invertible change of internal coordinates, together with the corresponding transformed readout, does not produce a new empirical prediction.

Corollary 3.8 — Parameter names have no independent empirical force

Renaming, rescaling, or bijectively transforming parameters does not create a distinct empirical model when the prediction family remains unchanged.

4. Structural Equivalence

Definition 4.1 — Structural equivalence

Two projections are structurally equivalent on $\Omega_{\mathcal{K}}$ when there exists a bijection

$$g : S_1(\Omega_{\mathcal{K}}) \rightarrow S_2(\Omega_{\mathcal{K}})$$

such that

$$\boxed{S_2 = g \circ S_1.}$$

Structural equivalence concerns the internal state descriptions before readout.

Definition 4.2 — Exact structural identity

Two projections are exactly structurally identical when

$$S_1(\omega) = S_2(\omega)$$

for every ω in the comparison domain.

Clarification 4.3 — Structural equivalence need not imply identical notation

One model may describe the same state with:

- a primitive operator;
- its reciprocal;
- a half-angle coordinate;
- a rational generator;
- or another invertible state coordinate.

Clarification 4.4 — Structural equivalence alone does not imply predictive equivalence

If the readouts do not compensate for the structural transformation, equivalent structural states may be assigned different outputs.

5. Predictive Equivalence**Definition 5.1 — Point-predictive equivalence**

Two deterministic projections are point-predictively equivalent on $\Omega_{\mathcal{K}}$ when

$$\boxed{H_1(\omega) = H_2(\omega)}$$

for every

$$\omega \in \Omega_{\mathcal{K}}.$$

Definition 5.2 — Set-predictive equivalence

Two uncertain projections are set-predictively equivalent when

$$\boxed{P_1(\omega) = P_2(\omega)}$$

for every ω in the comparison domain.

Definition 5.3 — Distributional predictive equivalence

Two probabilistic projections are distributionally equivalent when

$$\boxed{\nu_1(\cdot \mid \omega) = \nu_2(\cdot \mid \omega)}$$

for every $\omega \in \Omega_{\mathcal{K}}$.

Theorem 5.4 — Predictive equivalence is an equivalence relation

On a fixed comparison domain and output representation, predictive equivalence is:

1. reflexive;
2. symmetric;
3. transitive.

Proof. For point predictions, equality of functions has all three properties.

The same is true for equality of output sets and equality of probability measures.

□

Corollary 5.5 — Predictive equivalence classes

Projection contracts may be partitioned into equivalence classes whose members produce identical predictions on the declared comparison domain.

Negative Theorem 5.6 — Distinct derivations need not imply distinct predictions

Two projections may use different structural profiles, calibrations, or narratives and still belong to the same predictive equivalence class.

6. Observational Equivalence

The measurement protocol may erase differences present in the projected outputs.

Definition 6.1 — Measurement channel

Let

$$\mathcal{C}_{\mathcal{M}}$$

be the declared measurement channel mapping an output state y to a distribution or admissible set of possible observation records.

For a probabilistic protocol,

$$\mathcal{C}_{\mathcal{M}}(\cdot | y)$$

is the observation distribution conditional on output y .

Definition 6.2 — Induced observation distribution

For a deterministic prediction $H_j(\omega)$, define

$$\zeta_j(\cdot \mid \omega) = \mathcal{C}_{\mathcal{M}}(\cdot \mid H_j(\omega)).$$

For an uncertain prediction, integrate or propagate through the measurement channel according to the declared uncertainty formalism.

Definition 6.3 — Observational equivalence

Two projections are observationally equivalent under $\mathcal{M}$ when

$$\zeta_1(\cdot \mid \omega) = \zeta_2(\cdot \mid \omega)$$

for every

$$\omega \in \Omega_{\mathcal{K}}.$$

Theorem 6.4 — Predictive equivalence implies observational equivalence

If two projections are predictively equivalent and are passed through the same measurement channel, then they are observationally equivalent.

Proof. Equal predictions provide equal inputs to the same measurement channel.

A function or conditional distribution assigns the same observation law to equal inputs.

□

Negative Theorem 6.5 — Observational equivalence does not imply predictive equivalence

Two distinct outputs may induce the same observation distribution under a noninjective or resolution-limited measurement channel.

Proof. Suppose there exist

$$y_1 \neq y_2$$

such that

$$\mathcal{C}_{\mathcal{M}}(\cdot \mid y_1) = \mathcal{C}_{\mathcal{M}}(\cdot \mid y_2).$$

A projection predicting y_1 and another predicting y_2 are observationally equivalent under that channel despite differing predictions.

□

Example 6.6 — Resolution-limited equivalence

If an instrument rounds every value to the nearest integer, predictions

$$2.1 \quad \text{and} \quad 2.4$$

may produce the same reported result.

The projections are observationally indistinguishable under that instrument even though their exact predictions differ.

7. Finite-Record Indistinguishability

Definition 7.1 — Evaluation record

Let

$$\mathcal{R} = \{(\omega_i, \mathcal{O}_i)\}_{i=1}^n .$$

be a finite valid observation record.

Definition 7.2 — Finite-record indistinguishability

Two projections are indistinguishable on $\mathcal{R}$ when the declared comparison procedure returns equal or unresolved comparative status for every record in $\mathcal{R}$.

For exact deterministic observations, a simple form is

$$H_1(\omega_i) = H_2(\omega_i)$$

for every i .

Theorem 7.3 — Finite agreement does not imply global predictive equivalence

Two functions may agree at every point in a finite comparison record and differ elsewhere.

Proof. Let the sampled external coordinates be

$$t_1, \dots, t_n .$$

Define

$$H_1(t) = 0$$

and

$$H_2(t) = \prod_{i=1}^n (t - t_i).$$

Then

$$H_2(t_i) = 0 = H_1(t_i)$$

at every sampled point, but H_2 is not identically zero.

Therefore, finite agreement does not imply global equality.

□

Corollary 7.4 — Finite-data equivalence is record-relative

A new observation outside the existing record may distinguish previously indistinguishable projections.

8. Identification of Book 0 Phase

Definition 8.1 — Structural phase identifiability

A structural coordinate $F(x)$ identifies phase on a domain S when

$$F|_S$$

is injective.

Theorem 8.2 — Transfer-coordinate identifiability within a quadrant

On every open quadrant Q_k ,

$$\lambda : Q_k \rightarrow (0, 1)$$

is a bijection.

Proof. Book 0 proves that λ is continuous and strictly increasing on every open quadrant.

It also proves the endpoint limits

$$\lambda \rightarrow 0$$

at the entering wall and

$$\lambda \rightarrow 1$$

at the exiting wall.

A continuous strictly increasing function with those limits maps the quadrant bijectively onto $(0, 1)$.

□

Corollary 8.3 — Within-quadrant phase recovery

If the quadrant Q_k is known and λ is known exactly, then the phase $x \in Q_k$ is uniquely determined.

Theorem 8.4 — Global nonidentifiability of λ

$$\lambda\left(x + \frac{\pi}{2}\right) = \lambda(x).$$

Therefore, λ alone does not identify global phase modulo 2π .

Proof. S0.IV proves quarter-turn invariance of λ .

Thus four distinct phase positions separated by quarter-turns may possess the same λ value.

□

Theorem 8.5 — Flatwave cannot identify within-quadrant position

For every $x, y \in Q_k$,

$$\text{Flatwave}(x) = \text{Flatwave}(y) = \varepsilon_k.$$

Therefore, Flatwave is noninjective on every open quadrant.

Theorem 8.6 — Combined (Flatwave, λ) limitation

The pair

$$(\text{Flatwave}(x), \lambda(x))$$

distinguishes:

- positive-sign quadrants from negative-sign quadrants;

- and position within a quadrant.

However,

$$(\text{Flatwave}(x + \pi), \lambda(x + \pi)) = (\text{Flatwave}(x), \lambda(x)).$$

Therefore, the pair does not distinguish phases separated by a half-turn.

Proof. Flatwave is π -periodic, and λ is already $\pi/2$ -periodic.

Thus both coordinates repeat after π .

□

Corollary 8.7 — Global phase identification requires additional information

A complete phase identification modulo 2π requires at least one additional orientation, quadrant, or half-turn label beyond $(\text{Flatwave}, \lambda)$.

9. Parameter Identifiability

Definition 9.1 — Parameter prediction map

For projection family $\mathfrak{H}$, define

$$\Psi : \Theta \rightarrow \mathcal{Y}^{\Omega_{\mathcal{X}}}$$

by

$$\Psi(\theta) = H_{\theta}|_{\Omega_{\mathcal{X}}}.$$

Definition 9.2 — Global parameter identifiability

Parameters are globally identifiable when

$$\Psi(\theta_1) = \Psi(\theta_2) \implies \theta_1 = \theta_2.$$

Definition 9.3 — Parameter equivalence class

Define

$$\theta_1 \sim \theta_2$$

when

$$\Psi(\theta_1) = \Psi(\theta_2).$$

The equivalence class is

$$[\theta] = \{\vartheta \in \Theta : \Psi(\vartheta) = \Psi(\theta)\}.$$

Theorem 9.4 — Nonidentifiable parameters cannot be distinguished by output evidence

If

$$\theta_1 \sim \theta_2,$$

then no observation depending only on the projected outputs over $\Omega_{\mathcal{K}}$ can distinguish θ_1 from θ_2 .

Proof. The parameters produce identical predictions at every comparison instance.

Passing identical predictions through the same observation protocol produces identical observation laws.

Therefore, output evidence on the comparison domain cannot distinguish them.

□

Corollary 9.5 — One fitted value need not be uniquely supported

An optimizer returning $\hat{\theta}$ does not prove that

$$[\hat{\theta}] = \{\hat{\theta}\}.$$

10. Model Classes and Nesting

Definition 10.1 — Model class

A model class is a set of prediction maps

$$\mathfrak{H} = \{H_\theta : \theta \in \Theta\}.$$

Definition 10.2 — Nested model class

Model class $\mathfrak{H}_0$ is nested in $\mathfrak{H}_1$ when

$$\mathfrak{H}_0 \subseteq \mathfrak{H}_1.$$

Definition 10.3 — Strict nesting

The nesting is strict when

$$\mathfrak{H}_0 \subsetneq \mathfrak{H}_1.$$

Theorem 10.4 — Training-loss monotonicity under nesting

Let $L_{\text{fit}}(H)$ be the same fitting loss applied to both model classes.

If

$$\mathfrak{H}_0 \subseteq \mathfrak{H}_1,$$

then

$$\boxed{\inf_{H \in \mathfrak{H}_1} L_{\text{fit}}(H) \leq \inf_{H \in \mathfrak{H}_0} L_{\text{fit}}(H).}$$

Proof. The optimization domain for $\mathfrak{H}_1$ contains every candidate in $\mathfrak{H}_0$, possibly with additional candidates.

The infimum over a larger set cannot exceed the infimum over its subset.

□

Corollary 10.5 — Better fitting loss may be purely a flexibility effect

A more flexible nested class is guaranteed the opportunity to match or improve the best fitting loss, even when its additional structure has no external validity.

Negative Theorem 10.6 — Training improvement does not imply validation improvement

The inequality of Theorem 10.4 applies to the fitting record.

It imposes no corresponding inequality on independent validation loss.

11. Comparison Contract

Definition 11.1 — Comparison contract

A model-comparison contract is

$$\boxed{\mathcal{K} = (\{\mathcal{P}_j\}_{j=1}^r, \Omega_{\mathcal{K}}, \mathcal{M}, \mathcal{D}_{\text{fit}}, \mathcal{D}_{\text{val}}, L, C, \Delta, W_{\mathcal{K}}),}$$

where:

- $\{\mathcal{P}_j\}$ is the candidate projection set;
- $\Omega_{\mathcal{K}}$ is the common comparison domain;
- $\mathcal{M}$ is the common observation protocol or lawful protocol conversion;
- $\mathcal{D}_{\text{fit}}$ is the calibration or fitting record;
- $\mathcal{D}_{\text{val}}$ is the validation record;
- L is the declared loss or score;
- C is the declared complexity measure;
- Δ is the decision rule;
- $W_{\mathcal{K}}$ is the comparison failure map.

Definition 11.2 — Complete comparison

A comparison is complete when it declares:

1. all candidate contracts and versions;
2. a common validity domain;
3. a common measurand;
4. common units and conventions;
5. the fitting and validation records;
6. uncertainty treatment;
7. scoring or loss;
8. complexity treatment;
9. model-search scope;
10. the decision rule;
11. failure behavior.

Recommended failure witnesses

- COMPARISON-CONTRACT-INCOMPLETE;
- COMMON-VALIDITY-DOMAIN-EMPTY;
- MEASURAND-MISMATCH;
- OUTPUT-UNIT-MISMATCH;
- OBSERVATION-PROTOCOL-INCOMPATIBLE;

- UNCERTAINTY-INTERPRETATION-INCOMPATIBLE;
- COMPARISON-SCORE-UNDECLARED;
- COMPLEXITY-MEASURE-UNDECLARED;
- VALIDATION-RECORD-NOT-INDEPENDENT;
- MODEL-SEARCH-SCOPE-UNDISCLOSED;
- CANDIDATE-MODELS-OBSERVATIONALLY-EQUIVALENT;
- PARAMETER-NONIDENTIFIABILITY-UNRESOLVED;
- COMPARISON-EXTRAPOLATION-MISMATCH;
- COMPARISON-INCONCLUSIVE.

12. Deterministic Comparison

Definition 12.1 — Pointwise residual

For a scalar observation y_i ,

$$r_{j,i} = y_i - H_j(\omega_i).$$

Definition 12.2 — Absolute-error loss

$$L_{\text{AE}}(H_j) = \sum_{i=1}^n w_i |r_{j,i}|.$$

Definition 12.3 — Squared-error loss

$$L_{\text{SE}}(H_j) = \sum_{i=1}^n w_i r_{j,i}^2.$$

The weights and their interpretation must be declared.

Definition 12.4 — Maximum-error loss

$$L_{\infty}(H_j) = \max_i |r_{j,i}|.$$

Clarification 12.5 — Loss choice changes preference

A projection minimizing squared error need not minimize absolute error or maximum error.

Therefore, “best fit” is incomplete without the loss definition.

Theorem 12.6 — Equivalent predictions receive equal prediction-only loss

If

$$H_1(\omega_i) = H_2(\omega_i)$$

for every record i , then every loss depending only on those predictions and observations assigns equal values:

$$L(H_1) = L(H_2).$$

Proof. The complete list of loss inputs is identical for the two projections.

Therefore, the loss outputs are identical.

□

13. Set-Valued Comparison**Definition 13.1 — Compatibility indicator**

For prediction set $P_j(\omega_i)$ and observed set O_i , define

$$c_{j,i} = \begin{cases} 1, & P_j(\omega_i) \cap O_i \neq \emptyset, \\ 0, & P_j(\omega_i) \cap O_i = \emptyset. \end{cases}$$

Definition 13.2 — Contradiction count

$$N_{\text{contra}}(j) = \sum_{i=1}^n (1 - c_{j,i}).$$

Clarification 13.3 — Compatibility count ignores sharpness

A projection making extremely wide predictions may achieve compatibility by admitting nearly every result.

Compatibility alone therefore does not reward informative narrowness.

Definition 13.4 — Set width or volume

When meaningful, define a width or volume measure

$$V(P_j(\omega)).$$

A comparison may jointly evaluate:

- empirical coverage;
- and predictive sharpness.

Negative Theorem 13.5 — Universal prediction is not strong prediction

A projection returning the entire output space

$$P_j(\omega) = \mathcal{Y}$$

is compatible with every observation but excludes nothing.

It provides no discriminatory prediction.

14. Probabilistic Comparison

Definition 14.1 — Predictive probability distribution

Let

$$\nu_j(\cdot \mid \omega)$$

be the declared predictive distribution for projection j .

Definition 14.2 — Scoring rule

A scoring rule assigns a numerical score

$$S(\nu_j, y_{\text{obs}})$$

to a predictive distribution and realized observation.

The direction of preference—larger or smaller—must be declared.

Definition 14.3 — Aggregate predictive score

$$L_j = \sum_{i=1}^n S(\nu_j(\cdot \mid \omega_i), y_i).$$

Clarification 14.4 — Probability models require interpretation

A probability distribution may represent:

- aleatory variation;
- measurement noise;
- uncertainty concerning parameters;
- uncertainty concerning states;
- or a mixture.

The interpretation must be stated.

Theorem 14.5 — Distributional equivalence implies equal probabilistic score

If

$$\nu_1(\cdot \mid \omega_i) = \nu_2(\cdot \mid \omega_i)$$

for every record i , then any prediction-only scoring rule assigns equal aggregate score.

Negative Theorem 14.6 — Likelihood alone does not determine ontology

A greater likelihood under one projection shows stronger support relative to the declared probabilistic models and data.

It does not prove that the internal structural description is uniquely real.

15. Residual Diagnostics

Definition 15.1 — Residual sequence

For projection j ,

$$\mathbf{r}_j = (r_{j,1}, \dots, r_{j,n}).$$

Definition 15.2 — Residual structure

Residual structure includes systematic dependence on:

- phase;
- external covariates;

- time;
- magnitude;
- location;
- instrument;
- calibration batch;
- or previous residuals.

Theorem 15.3 — Zero mean residual does not imply adequate modeling

A residual sequence may have mean zero while retaining systematic structure.

Proof. Consider residuals alternating between $+1$ and -1 .

Their mean may be zero, but they display perfect serial structure.

Therefore, zero average alone does not establish adequacy.

□

Corollary 15.4 — Residual diagnostics must exceed one summary statistic

A complete comparison should examine patterns relevant to the declared model assumptions.

Clarification 15.5 — Residual structure may arise from several layers

Patterns may indicate:

- phase-inference failure;
- readout misspecification;
- calibration drift;
- omitted external variables;
- measurement artifacts;
- dependence;
- or structural inadequacy.

16. Complexity

Definition 16.1 — Nominal parameter count

$$k_{\text{nom}} = \dim(\Theta)$$

when the parameter space has a declared finite dimension.

Definition 16.2 — Identifiable parameter count

The identifiable parameter count is the dimension of the predictionally distinguishable parameter space after quotienting by parameter equivalence.

Symbolically,

$$k_{\text{id}} = \dim(\Theta/\sim)$$

where defined.

Definition 16.3 — Functional complexity

Functional complexity concerns the richness of the prediction family

$$\mathfrak{H}$$

rather than merely the number of symbols used to parameterize it.

Definition 16.4 — Procedural complexity

Procedural complexity includes:

- branch rules;
- preprocessing;
- calibration stages;
- exception handling;
- selected data transformations;
- and post-hoc adjustments.

Definition 16.5 — Declared complexity measure

A comparison contract may use a complexity function

$$C(\mathcal{P}) \geq 0.$$

The measure and its rationale must be stated.

Theorem 16.6 — Parameter count is representation-sensitive under redundancy

The same prediction family may be represented with additional redundant parameters.

Therefore, nominal parameter count alone is not an invariant measure of empirical complexity.

Proof. Let a model depend on parameter a through prediction H_a .

Introduce two parameters b, c with

$$a = b + c.$$

The prediction family is unchanged, but the nominal parameter count increases from one to two.

Because many pairs (b, c) produce the same a , the added parameterization is redundant.

□

Corollary 16.7 — Identifiability must accompany parameter count

Reporting only k_{nom} may overstate the number of empirically distinguishable degrees of freedom.

17. Parsimony**Definition 17.1 — Empirical equivalence tolerance**

Let

$$\epsilon \geq 0$$

be a declared tolerance for considering validation performance practically equivalent.

Definition 17.2 — Conditional parsimony rule

A comparison may prefer $\mathcal{P}_1$ over $\mathcal{P}_2$ by parsimony when:

1. their validation performance differs by no more than ϵ ;
2. both satisfy required adequacy guards;
3. the same complexity measure is applied;
4. and

$$C(\mathcal{P}_1) < C(\mathcal{P}_2).$$

Clarification 17.3 — Parsimony is a decision rule

Parsimony is not a theorem that the simpler model is ontologically true.

It is a rule for choosing among models with sufficiently similar empirical performance.

Negative Theorem 17.4 — Simplicity cannot rescue empirical failure

A simpler projection contradicted by valid observations is not preferable merely because it has fewer parameters.

18. Calibration and Validation Separation**Definition 18.1 — Fitting record**

$$\mathcal{D}_{\text{fit}}$$

is the information used to select:

- model form;
- structural profile;
- readout;
- parameters;
- thresholds;
- preprocessing;
- or comparison criteria.

Definition 18.2 — Validation record

$$\mathcal{D}_{\text{val}}$$

is the information reserved for evaluating decisions made using the fitting record.

Theorem 18.3 — Selection dependence

Performance measured on data used to select among candidate models is statistically and procedurally dependent on the selection process.

Proof. The selected model is chosen in response to the fitting data.

Its measured fitting performance is therefore not the performance of a model specified independently of that data.

□

Definition 18.4 — Model-search scope

The model-search scope is the complete set of:

- projection families considered;
- transformations tried;
- variables screened;
- thresholds tested;
- and stopping rules used.

Negative Theorem 18.5 — Undisclosed search exaggerates evidence

Reporting only the best-performing model without disclosing the broader search omits information required to evaluate selection effects.

Required witness

MODEL-SEARCH-SCOPE-UNDISCLOSED.

19. Cross-Validation and Repeated Assessment**Definition 19.1 — Data partition**

A data partition divides the available record into disjoint or partially overlapping fitting and validation subsets.

Definition 19.2 — Cross-validation procedure

A cross-validation procedure repeatedly fits on one subset and evaluates on another according to a declared partition rule.

Clarification 19.3 — Cross-validation estimates procedural performance

Cross-validation evaluates the complete fitting procedure, including parameter selection, when the procedure is repeated correctly within each training partition.

Negative Theorem 19.4 — Preprocessing leakage invalidates nominal separation

If preprocessing, structural-profile selection, or parameter tuning uses the complete dataset before partitioned evaluation, the validation subsets are not fully independent of the selection process.

20. Model Dominance

Definition 20.1 — Record-wise predictive dominance

Projection $\mathcal{P}_1$ dominates $\mathcal{P}_2$ on a validation record under loss L when

$$L_i(\mathcal{P}_1) \leq L_i(\mathcal{P}_2)$$

for every record i , with strict inequality for at least one.

Definition 20.2 — Aggregate dominance

Projection $\mathcal{P}_1$ has lower aggregate loss when

$$L(\mathcal{P}_1) < L(\mathcal{P}_2).$$

Clarification 20.3 — Aggregate dominance may hide subgroup failure

A model may have lower total loss while performing worse on a scientifically important subset. Comparison domains and subgroup requirements must be declared.

Definition 20.4 — Constraint dominance

Projection $\mathcal{P}_1$ constraint-dominates $\mathcal{P}_2$ when:

1. every output permitted by $\mathcal{P}_1$ is permitted by $\mathcal{P}_2$;
2. $\mathcal{P}_1$ excludes at least one output that $\mathcal{P}_2$ permits;
3. both remain compatible with the validation evidence.

Symbolically,

$$P_1(\omega) \subseteq P_2(\omega)$$

throughout the domain, with strict inclusion somewhere.

Clarification 20.5 — Narrower is better only when calibrated

An unjustifiably narrow prediction set may appear more informative while possessing incorrect coverage.

Constraint dominance requires valid uncertainty construction.

21. Equivalence and Comparison Theorems

Theorem 21.1 — Prediction-only indistinguishability

If two projections are predictively equivalent on $\Omega_{\mathcal{X}}$, then no comparison method using only their predictions and observations from that domain can prefer one on predictive grounds.

Proof. Predictive equivalence gives identical predictions for every comparison instance.

Every prediction-only score, residual, compatibility result, and induced observation distribution is therefore identical.

No predictive difference remains for the comparison method to use.

□

Corollary 21.2 — Preference between predictively equivalent projections requires nonpredictive criteria

Possible criteria include:

- computational cost;
- interpretability;
- portability;
- calibration stability;
- procedural simplicity;
- or compatibility with separately established theory.

These criteria must be declared.

Theorem 21.3 — Observational equivalence blocks protocol-specific discrimination

If two projections are observationally equivalent under protocol $\mathcal{M}$, no amount of repetition using only that unchanged protocol can distinguish them in expectation.

Proof. Every repeated observation has the same declared probability law under both projections.

Therefore, the joint distribution of any finite or countable sequence of repeated observations is the same under both, assuming the same repetition model.

□

Corollary 21.4 — Distinguishing observationally equivalent models requires a new intervention or protocol

The new protocol must access a difference that the former measurement channel erased.

Theorem 21.5 — Better score is comparison-relative

The statement

$$L(\mathcal{P}_1) < L(\mathcal{P}_2)$$

establishes preference only relative to:

- the chosen data;
- the chosen loss;
- the chosen weighting;
- the chosen uncertainty representation;
- the chosen domain;
- and the chosen candidate set.

Proof. Changing any of these elements may change the numerical losses or their interpretation.

Therefore, the preference is conditional on the comparison contract.

□

22. Comparison Decision Rules

Definition 22.1 — Decision set

Let

$$\mathfrak{D}_{\mathcal{K}} = \{\text{equivalent, prefer } \mathcal{P}_j, \text{reject } \mathcal{P}_j, \text{retain_multiple, inconclusive}\}.$$

Definition 22.2 — Comparison decision rule

$$\Delta : \mathcal{Z}_{\mathcal{K}} \rightarrow \mathfrak{D}_{\mathcal{K}},$$

where $\mathcal{Z}_{\mathcal{K}}$ contains the declared scores, diagnostics, uncertainty results, complexity values, and guard statuses.

Definition 22.3 — Predeclared decision rule

A decision rule is predeclared when it is fixed before examining the validation outcomes to which it will be applied.

Clarification 22.4 — Exploratory comparison

A post-hoc rule may be used for exploration, but its result must be labeled exploratory rather than confirmatory.

Theorem 22.5 — Inconclusive comparison

A comparison is inconclusive when:

- candidate predictions overlap within uncertainty;
- observational equivalence has not been broken;
- validation evidence is insufficient;
- scoring rules disagree without a declared priority;
- or required guards fail.

Required witness

COMPARISON-INCONCLUSIVE.

23. Comparison Failure Locality**Definition 23.1 — Equivalence failure**

An equivalence claim fails when the required equality, bijection, or induced observation identity does not hold on the declared domain.

Definition 23.2 — Comparability failure

Comparability fails when the projections do not share:

- a common measurand;
- compatible units;
- a common domain;
- or a lawful observation protocol.

Definition 23.3 — Selection failure

Selection fails when no declared decision rule resolves the available evidence.

Theorem 23.4 — Comparison failure does not invalidate individual contracts

Two individually lawful projections may be impossible to compare directly because their measurands, domains, or protocols differ.

Proof. Lawfulness is assessed relative to each projection's own contract.

Direct comparison imposes additional common-domain and common-output requirements.

Failure of those additional requirements does not imply either projection is internally unlawful.

□

PROCEDURE DIVISION**24. Canonical Equivalence Procedure**

CHECK-PROJECTION-EQUIVALENCE.

DECLARE COMPARISON-DOMAIN.

VERIFY COMMON MEASURAND.

VERIFY COMMON UNITS.

VERIFY COMMON OBSERVATION PROTOCOL.

IF COMMON-VALIDITY-DOMAIN-EMPTY

RETURN COMPARISON-WITNESS-SET

END-IF.

CHECK FOR INVERTIBLE PHASE

OR STRUCTURAL REPARAMETERIZATION.

IF REPARAMETERIZATION EXISTS

AND READOUTS COMPENSATE

RECORD REPRESENTATIONAL-EQUIVALENCE

END-IF.

COMPARE STRUCTURAL-STATE MAPS.

COMPARE PREDICTED OUTPUTS.

COMPARE PREDICTED OUTPUT SETS

OR DISTRIBUTIONS.

PROPAGATE THROUGH MEASUREMENT CHANNEL.

CLASSIFY AS:

STRUCTURALLY-EQUIVALENT,

PREDICTIVELY-EQUIVALENT,

OBSERVATIONALLY-EQUIVALENT,

FINITE-RECORD-INDISTINGUISHABLE,

OR DISTINCT.

RETURN EQUIVALENCE-CLASSIFICATION

DOMAIN

ASSUMPTIONS

WITNESS-SET.

25. Canonical Model-Comparison Procedure

COMPARE-PROJECTION-MODELS.

PERFORM CHECK-COMPARISON-CONTRACT.

IF COMPARISON-CONTRACT-INCOMPLETE

RETURN COMPARISON-WITNESS-SET

END-IF.

PERFORM NORMALIZE-MEASURAND-UNITS

AND OBSERVATION-PROTOCOL.

IF MEASURAND-MISMATCH

OR OUTPUT-UNIT-MISMATCH

OR OBSERVATION-PROTOCOL-INCOMPATIBLE

RETURN COMPARISON-WITNESS-SET

END-IF.

PERFORM CHECK-PROJECTION-EQUIVALENCE.

IF CANDIDATE-MODELS-OBSERVATIONALLY-EQUIVALENT

RETURN NO-PROTOCOL-SPECIFIC-DISCRIMINATION

END-IF.

DECLARE FITTING-RECORD.

DECLARE VALIDATION-RECORD.

DECLARE MODEL-SEARCH-SCOPE.

DECLARE LOSS-OR-SCORE.

DECLARE COMPLEXITY-MEASURE.
 DECLARE DECISION-RULE.
 FIT EACH CANDIDATE
 USING FITTING-RECORD ONLY.
 EVALUATE EACH FITTED CANDIDATE
 ON VALIDATION-RECORD.
 COMPUTE:
 PREDICTIVE SCORES,
 UNCERTAINTY COMPATIBILITY,
 RESIDUAL DIAGNOSTICS,
 CALIBRATION STABILITY,
 COMPLEXITY VALUES,
 AND FAILURE WITNESSES.
 APPLY PREDECLARED DECISION-RULE.
 RETURN:
 EQUIVALENCE STATUS,
 COMPARATIVE SCORES,
 VALIDATION RESULTS,
 COMPLEXITY RESULTS,
 DECISION,
 AND COMPLETE AUDIT RECORD.

26. Theorem Ledger

Theorem V.1 — Reparameterization invariance

Invertible structural reparameterization with compensating readout preserves prediction.

Theorem V.2 — Predictive equivalence relation

Predictive equivalence is reflexive, symmetric, and transitive.

Theorem V.3 — Predictive-to-observational implication

Predictive equivalence under a common measurement channel implies observational equivalence.

Theorem V.4 — Failure of the converse

Observational equivalence may occur despite distinct predictions.

Theorem V.5 — Finite-record limitation

Agreement on a finite record does not prove global predictive equivalence.

Theorem V.6 — Quadrant-local λ identifiability

$$\lambda : Q_k \rightarrow (0, 1)$$

is bijective.

Theorem V.7 — Global λ nonidentifiability

$$\lambda\left(x + \frac{\pi}{2}\right) = \lambda(x).$$

Theorem V.8 — Flatwave nonidentifiability

Flatwave cannot identify within-quadrant phase.

Theorem V.9 — Half-turn ambiguity of $(\text{Flatwave}, \lambda)$

The combined pair repeats after π .

Theorem V.10 — Parameter-equivalence limitation

Predictionally equivalent parameters cannot be distinguished by output evidence on the comparison domain.

Theorem V.11 — Nested fitting monotonicity

A larger nested model class cannot have worse minimum fitting loss on the same record.

Theorem V.12 — Equivalent-prediction score equality

Prediction-only losses assign equal scores to equal predictions.

Theorem V.13 — Training improvement limitation

Improved fitting loss does not imply improved validation loss.

Theorem V.14 — Parameter-count noninvariance under redundancy

Nominal parameter count may change while the prediction family remains unchanged.

Theorem V.15 — Prediction-only indistinguishability

Predictively equivalent models cannot be distinguished on predictive evidence from the equivalence domain.

Theorem V.16 — Observational repetition limitation

Repeated use of the same nondiscriminating measurement protocol does not distinguish observationally equivalent projections.

Theorem V.17 — Comparison-relative preference

A better score establishes preference only relative to the complete comparison contract.

27. Negative Theorems**Negative Theorem 27.1 — No universal meaning of “equivalent”**

An equivalence claim without a declared level is incomplete.

Negative Theorem 27.2 — No new empirical model from notation alone

Changing symbols or invertibly reparameterizing internal states does not create distinct predictions.

Negative Theorem 27.3 — No global equivalence from finite agreement

Agreement on finitely many records does not prove equality outside those records.

Negative Theorem 27.4 — No structural equality from observational equality alone

A measurement protocol may erase genuine structural or predictive differences.

Negative Theorem 27.5 — No global phase from λ alone

Quarter-turn-related phases share the same transfer coordinate.

Negative Theorem 27.6 — No continuous phase from Flatwave alone

Flatwave is constant within each quadrant.

Negative Theorem 27.7 — No unique parameters without identifiability

One optimizer output does not prove one empirically distinguishable parameter state.

Negative Theorem 27.8 — No validation conclusion from fitting loss alone

Nested model flexibility guarantees opportunities for improved training fit.

Negative Theorem 27.9 — No universal best loss function

Different valid comparison goals may require different declared scoring procedures.

Negative Theorem 27.10 — No strong prediction from universal compatibility

A prediction set containing every possible output excludes nothing.

Negative Theorem 27.11 — No universal truth from parsimony

Simplicity is a conditional decision criterion, not proof of ontology.

Negative Theorem 27.12 — No independent validation after hidden data reuse

Validation evidence is compromised when it influenced model selection without adjustment.

Negative Theorem 27.13 — No fair comparison across unequal domains

A model tested only on favorable cases may not be declared superior to one evaluated on a broader domain without domain adjustment.

Negative Theorem 27.14 — No ontology from comparative victory

Outperforming declared competitors does not prove that the winning model is the uniquely real underlying structure.

28. Terminal Theorem

Theorem 28.1 — Projection Comparison Separation Theorem

Let $\mathcal{P}_1, \dots, \mathcal{P}_r$ be projection contracts.

A lawful comparative conclusion requires:

1. a common comparison domain;
2. a common measurand and output representation;
3. compatible observation protocols;
4. declared uncertainty treatment;
5. declared calibration and validation records;
6. a declared score or loss;
7. a declared complexity measure when complexity affects the decision;
8. a declared model-search scope;
9. and a declared decision rule.

If two projections are predictively equivalent on the comparison domain, predictive evidence from that domain cannot distinguish them.

If they are observationally equivalent under the declared measurement protocol, repetition of that same protocol cannot distinguish them.

If one projection receives a better comparative score, the preference is conditional on the comparison contract and does not establish unique structural or physical ontology.

Proof. The first requirements follow from the definitions of lawful comparability and the inherited measurement and calibration disciplines.

Predictive equivalence supplies identical prediction inputs to every prediction-only comparison.

Observational equivalence supplies identical observation laws under the declared protocol.

A comparative score depends on the chosen domain, records, uncertainty representation, score, weighting, calibration, and candidate class.

Therefore, equivalence blocks discrimination at its declared level, while comparative preference remains conditional on the complete comparison design.

□

29. Closing Declaration

Book 1 began by separating structural mathematics from external interpretation.

S1.I established projection contracts:

$$\hat{y}_{\mathcal{P}} = \Gamma_{\theta} \circ E_{\mathbf{F}} \circ \Phi.$$

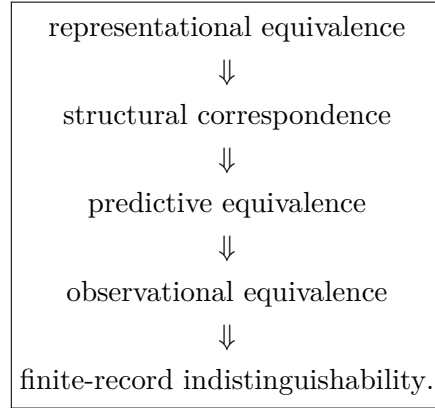
S1.II established admission, failure witnesses, and falsification.

S1.III established measurement protocols, error, and uncertainty.

S1.IV established calibration and dimensional readout.

S1.V establishes equivalence and comparison.

The principal equivalence ladder is:



The downward implications are not all reversible.

Within a known quadrant,

$$\lambda \text{ identifies phase uniquely.}$$

Globally,

$$\lambda\left(x + \frac{\pi}{2}\right) = \lambda(x).$$

Flatwave supplies only quadrant sign:

$$\text{Flatwave} = \text{sgn}(\sin 2x).$$

The pair

$$(\text{Flatwave}, \lambda)$$

still retains half-turn ambiguity.

For model classes,

$$\mathfrak{H}_0 \subseteq \mathfrak{H}_1 \implies \inf_{\mathfrak{H}_1} L_{\text{fit}} \leq \inf_{\mathfrak{H}_0} L_{\text{fit}}.$$

This is a fitting theorem, not a validation theorem.

The resulting comparison discipline is:

No equivalence without a declared level; no comparison without a common domain; no preference without a declared score; no simplicity claim without a complexity measure; no validation after hidden data reuse; no discrimination between equivalent predictions; no ontology from predictive victory alone.
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This completes the projection and measurement framework.

END OF S1.V

END OF BOOK 1
